

the side information over the wireless access network based on the information identifying the network element destination.

20 Claims, 24 Drawing Sheets

Related U.S. Application Data

continuation of application No. 15/977,713, filed on May 11, 2018, now Pat. No. 10,484,858, which is a continuation of application No. 14/952,374, filed on Nov. 25, 2015, now Pat. No. 9,986,413, which is a continuation of application No. 13/718,952, filed on Dec. 18, 2012, now Pat. No. 9,204,282, which is a continuation of application No. 12/695,980, filed on Jan. 28, 2010, now Pat. No. 8,340,634, which is a continuation-in-part of application No. 12/695,019, filed on Jan. 27, 2010, now Pat. No. 8,275,830, and a continuation-in-part of application No. 12/695,021, filed on Jan. 27, 2010, now Pat. No. 8,346,225, which is a continuation-in-part of application No. 12/380,780, filed on Mar. 2, 2009, now Pat. No. 8,839,388, said application No. 12/695,019 is a continuation-in-part of application No. 12/380,778, filed on Mar. 2, 2009, now Pat. No. 8,321,526.

- (60) Provisional application No. 61/264,126, filed on Nov. 24, 2009, provisional application No. 61/252,151, filed on Oct. 15, 2009, provisional application No. 61/252,153, filed on Oct. 15, 2009, provisional application No. 61/270,353, filed on Jul. 6, 2009, provisional application No. 61/207,739, filed on Feb. 13, 2009, provisional application No. 61/207,393, filed on Feb. 10, 2009, provisional application No. 61/206,944, filed on Feb. 4, 2009, provisional application No. 61/206,354, filed on Jan. 28, 2009.

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(56) References Cited

U.S. PATENT DOCUMENTS

5,325,532 A 6/1994 Crosswy et al.
 5,572,528 A 11/1996 Shuen
 5,577,100 A 11/1996 McGregor et al.
 5,594,777 A 1/1997 Makkonen et al.
 5,617,539 A 4/1997 Ludwig et al.
 5,630,159 A 5/1997 Zanchi
 5,633,484 A 5/1997 Zanchi et al.
 5,633,868 A 5/1997 Baldwin et al.
 5,754,953 A 5/1998 Briancon et al.
 5,764,693 A 6/1998 Taylor et al.
 5,774,532 A 6/1998 Gottlieb et al.

5,794,142 A 8/1998 Vantila et al.
 5,814,798 A 9/1998 Zanchi
 5,889,477 A 3/1999 Fastenrath
 5,892,900 A 4/1999 Ginter et al.
 5,903,845 A 5/1999 Buhrmann et al.
 5,915,008 A 6/1999 Dulman
 5,915,226 A 6/1999 Martineau
 5,933,778 A 8/1999 Buhrmann et al.
 5,940,472 A 8/1999 Newman et al.
 5,974,439 A 10/1999 Bollella
 5,983,270 A 11/1999 Abraham et al.
 6,035,281 A 3/2000 Crosskey et al.
 6,038,452 A 3/2000 Strawczynski et al.
 6,038,540 A 3/2000 Krist et al.
 6,047,268 A 4/2000 Bartoli et al.
 6,058,434 A 5/2000 Wilt et al.
 6,061,571 A 5/2000 Tamura
 6,064,878 A 5/2000 Denker et al.
 6,078,953 A 6/2000 Vaid et al.
 6,081,591 A 6/2000 Skoog
 6,098,878 A 8/2000 Dent et al.
 6,104,700 A 8/2000 Haddock et al.
 6,115,823 A 9/2000 Velasco et al.
 6,119,933 A 9/2000 Wong et al.
 6,125,391 A 9/2000 Meltzer et al.
 6,141,565 A 10/2000 Feuerstein et al.
 6,141,686 A 10/2000 Jackowski et al.
 6,148,336 A 11/2000 Thomas et al.
 6,154,738 A 11/2000 Call
 6,157,636 A 12/2000 Voit et al.
 6,185,576 B1 2/2001 McIntosh
 6,198,915 B1 3/2001 McGregor et al.
 6,219,786 B1 4/2001 Cunningham et al.
 6,226,277 B1 5/2001 Chuah
 6,246,870 B1 6/2001 Dent et al.
 6,263,055 B1 7/2001 Garland et al.
 6,292,828 B1 9/2001 Williams
 6,317,584 B1 11/2001 Abu-Amara et al.
 6,370,139 B2 4/2002 Redmond
 6,381,316 B2 4/2002 Joyce et al.
 6,393,014 B1 5/2002 Daly et al.
 6,397,259 B1 5/2002 Lincke et al.
 6,401,113 B2 6/2002 Lazaridis et al.
 6,418,147 B1 7/2002 Wiedeman
 6,421,722 B1 7/2002 Bauer et al.
 6,438,575 B1 8/2002 Khan et al.
 6,445,777 B1 9/2002 Clark
 6,449,479 B1 9/2002 Sanchez
 6,466,984 B1 10/2002 Naveh et al.
 6,470,182 B1 10/2002 Nelson
 6,477,670 B1 11/2002 Ahmadvand
 6,502,131 B1 12/2002 Vaid et al.
 6,505,114 B2 1/2003 Luciani
 6,510,152 B1 1/2003 Gerszberg et al.
 6,522,629 B1 2/2003 Anderson, Sr.
 6,532,235 B1 3/2003 Benson et al.
 6,532,579 B2 3/2003 Sato et al.
 6,535,855 B1 3/2003 Cahill et al.
 6,535,949 B1 3/2003 Parker
 6,539,082 B1 3/2003 Lowe et al.
 6,542,500 B1 4/2003 Gerszberg et al.
 6,542,992 B1 4/2003 Peirce et al.
 6,546,016 B1 4/2003 Gerszberg et al.
 6,556,823 B2 4/2003 Clapton et al.
 6,563,806 B1 5/2003 Yano et al.
 6,570,974 B1 5/2003 Gerszberg et al.
 6,574,321 B1 6/2003 Cox et al.
 6,574,465 B2 6/2003 Marsh et al.
 6,578,076 B1 6/2003 Putzolu
 6,581,092 B1 6/2003 Motoyama
 6,591,098 B1 7/2003 Shieh et al.
 6,598,034 B1 7/2003 Kloth
 6,601,040 B1 7/2003 Kolls
 6,603,969 B1 8/2003 Vuoristo et al.
 6,603,975 B1 8/2003 Inouchi et al.
 6,606,744 B1 8/2003 Mikurak
 6,615,034 B1 9/2003 Alloune et al.
 6,628,934 B2 9/2003 Rosenberg et al.
 6,631,122 B1 10/2003 Arunachalam et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,636,721	B2	10/2003	Threadgill et al.	7,039,027	B2	5/2006	Bridgelall
6,639,975	B1	10/2003	O'Neal et al.	7,039,037	B2	5/2006	Wang et al.
6,640,097	B2	10/2003	Corrigan et al.	7,039,403	B2	5/2006	Wong
6,640,334	B1	10/2003	Rasmussen	7,039,713	B1	5/2006	Van Gunter et al.
6,650,887	B2	11/2003	McGregor et al.	7,042,988	B2	5/2006	Juitt et al.
6,651,101	B1	11/2003	Gai et al.	7,043,225	B1	5/2006	Patel et al.
6,654,786	B1	11/2003	Fox et al.	7,043,226	B2	5/2006	Yamauchi
6,654,814	B1	11/2003	Britton et al.	7,043,268	B2	5/2006	Yukie et al.
6,658,254	B1	12/2003	Purdy et al.	7,047,276	B2	5/2006	Liu et al.
6,662,014	B1	12/2003	Walsh	7,058,022	B1	6/2006	Carolan et al.
6,678,516	B2	1/2004	Nordman et al.	7,058,968	B2	6/2006	Rowland et al.
6,683,853	B1	1/2004	Kannas et al.	7,068,600	B2	6/2006	Cain
6,684,244	B1	1/2004	Goldman et al.	7,069,248	B2	6/2006	Huber
6,690,918	B2	2/2004	Evans et al.	7,082,422	B1	7/2006	Zirngibi et al.
6,694,362	B1	2/2004	Secor et al.	7,084,775	B1	8/2006	Smith
6,697,821	B2	2/2004	Ziff et al.	7,092,696	B1	8/2006	Hosain et al.
6,725,031	B2	4/2004	Watler et al.	7,095,754	B2	8/2006	Benveniste
6,725,256	B1	4/2004	Albal et al.	7,102,620	B2	9/2006	Harries et al.
6,732,176	B1	5/2004	Stewart et al.	7,110,753	B2	9/2006	Campen
6,735,206	B1	5/2004	Oki et al.	7,113,780	B2	9/2006	Mckenna et al.
6,748,195	B1	6/2004	Phillips	7,113,997	B2	9/2006	Jayapalan et al.
6,748,437	B1	6/2004	Mankude et al.	7,120,133	B1	10/2006	Joo et al.
6,751,296	B1	6/2004	Albal et al.	7,133,386	B2	11/2006	Holur et al.
6,754,470	B2	6/2004	Hendrickson et al.	7,133,695	B2	11/2006	Beyda
6,757,717	B1	6/2004	Goldstein	7,136,361	B2	11/2006	Benveniste
6,760,417	B1	7/2004	Wallenius	7,139,569	B2	11/2006	Kato
6,763,000	B1	7/2004	Walsh	7,142,876	B2	11/2006	Trossen et al.
6,763,226	B1	7/2004	McZeal, Jr.	7,149,229	B1	12/2006	Leung
6,765,864	B1	7/2004	Natarajan et al.	7,149,521	B2	12/2006	Sundar et al.
6,765,925	B1	7/2004	Sawyer et al.	7,151,764	B1	12/2006	Heinonen et al.
6,782,412	B2	8/2004	Brophy et al.	7,158,792	B1	1/2007	Cook et al.
6,785,889	B1	8/2004	Williams	7,162,237	B1	1/2007	Silver et al.
6,792,461	B1	9/2004	Hericourt	7,165,040	B2	1/2007	Ehrman et al.
6,829,596	B1	12/2004	Fraze	7,167,078	B2	1/2007	Pourchot
6,829,696	B1	12/2004	Balmer et al.	7,174,156	B1	2/2007	Mangal
6,839,340	B1	1/2005	Voit et al.	7,174,174	B2	2/2007	Boris et al.
6,842,628	B1	1/2005	Arnold et al.	7,177,919	B1	2/2007	Truong et al.
6,873,988	B2	3/2005	Herrmann et al.	7,180,855	B1	2/2007	Lin
6,876,653	B2	4/2005	Ambe et al.	7,181,017	B1	2/2007	Nagel et al.
6,879,825	B1	4/2005	Daly	7,191,248	B2	3/2007	Chattopadhyay et al.
6,882,718	B1	4/2005	Smith	7,197,321	B2	3/2007	Erskine et al.
6,885,997	B1	4/2005	Roberts	7,200,112	B2	4/2007	Sundar et al.
6,901,440	B1	5/2005	Bimm et al.	7,200,551	B1	4/2007	Senez
6,920,455	B1	7/2005	Weschler	7,203,169	B1	4/2007	Okholm et al.
6,922,562	B2	7/2005	Ward et al.	7,203,721	B1	4/2007	Ben-Efraim et al.
6,928,280	B1	8/2005	Xanthos et al.	7,203,752	B2	4/2007	Rice et al.
6,934,249	B1	8/2005	Bertin et al.	7,212,491	B2	5/2007	Koga
6,934,751	B2	8/2005	Jayapalan et al.	7,219,123	B1	5/2007	Flechter et al.
6,947,723	B1	9/2005	Gurnani et al.	7,222,190	B2	5/2007	Klinker et al.
6,947,985	B2	9/2005	Hegli et al.	7,222,304	B2	5/2007	Beaton et al.
6,952,428	B1	10/2005	Necka et al.	7,224,968	B2	5/2007	Dobson et al.
6,957,067	B1	10/2005	Iyer et al.	7,228,354	B2	6/2007	Chambliss et al.
6,959,202	B2	10/2005	Heinonen et al.	7,236,780	B2	6/2007	Benco
6,959,393	B2	10/2005	Hollis et al.	7,242,668	B2	7/2007	Kan et al.
6,965,667	B2	11/2005	Trabandt et al.	7,242,920	B2	7/2007	Morris
6,965,872	B1	11/2005	Grdina	7,245,901	B2	7/2007	McGregor et al.
6,967,958	B2	11/2005	Ono et al.	7,248,570	B2	7/2007	Bahl et al.
6,970,692	B2	11/2005	Tysor	7,251,218	B2	7/2007	Jorgensen
6,970,927	B1	11/2005	Stewart et al.	7,260,382	B1	8/2007	Lamb et al.
6,982,733	B1	1/2006	McNally et al.	7,266,371	B1	9/2007	Amin et al.
6,983,370	B2	1/2006	Eaton et al.	7,269,157	B2	9/2007	Klinker et al.
6,996,062	B1	2/2006	Freed et al.	7,271,765	B2	9/2007	Stilp et al.
6,996,076	B1	2/2006	Forbes et al.	7,272,660	B1	9/2007	Powers et al.
6,996,393	B2	2/2006	Pyhalammii et al.	7,280,816	B2	10/2007	Fratti et al.
6,998,985	B2	2/2006	Reisman et al.	7,280,818	B2	10/2007	Clayton
7,002,920	B1	2/2006	Ayyagari et al.	7,283,561	B1	10/2007	Picher-Dempsey
7,007,295	B1	2/2006	Rose et al.	7,283,963	B1	10/2007	Fitzpatrick et al.
7,013,469	B2	3/2006	Smith et al.	7,286,834	B2	10/2007	Walter
7,017,189	B1	3/2006	DeMello et al.	7,286,848	B2	10/2007	Vireday et al.
7,024,200	B2	4/2006	McKenna et al.	7,289,489	B1	10/2007	Kung et al.
7,024,460	B2	4/2006	Koopmas et al.	7,290,283	B2	10/2007	Copeland, III
7,027,055	B2	4/2006	Anderson et al.	7,310,424	B2	12/2007	Gehring et al.
7,027,408	B2	4/2006	Nabkel et al.	7,313,237	B2	12/2007	Bahl et al.
7,031,733	B2	4/2006	Alminana et al.	7,315,892	B2	1/2008	Freimuth et al.
7,032,072	B1	4/2006	Quinn et al.	7,317,699	B2	1/2008	Godfrey et al.
				7,318,050	B1	1/2008	Musgrave
				7,318,111	B2	1/2008	Zhao
				7,320,029	B2	1/2008	Rinne et al.
				7,322,044	B2	1/2008	Hrastar

(56)

References Cited

U.S. PATENT DOCUMENTS

7,324,447 B1	1/2008	Morford	7,551,921 B2	6/2009	Petermann
7,325,037 B2	1/2008	Lawson	7,551,922 B2	6/2009	Roskowski et al.
7,336,960 B2	2/2008	Zavalkovsky et al.	7,554,983 B1	6/2009	Muppala
7,340,772 B2	3/2008	Panasyuk et al.	7,555,757 B2	6/2009	Smith et al.
7,346,410 B2	3/2008	Uchiyama	7,561,899 B2	7/2009	Lee
7,349,695 B2	3/2008	Oommen et al.	7,562,213 B1	7/2009	Timms
7,353,533 B2	4/2008	Wright et al.	7,564,799 B2	7/2009	Holland et al.
7,356,011 B1	4/2008	Waters et al.	7,565,141 B2	7/2009	Macaluso
7,356,337 B2	4/2008	Florence	7,574,509 B2	8/2009	Nixon et al.
7,366,497 B2	4/2008	Nagata	7,574,731 B2	8/2009	Fascenda
7,366,654 B2	4/2008	Moore	7,577,431 B2	8/2009	Jiang
7,366,934 B1	4/2008	Narayan et al.	7,580,356 B1	8/2009	Mishra et al.
7,369,848 B2	5/2008	Jiang	7,580,857 B2	8/2009	VanFleet et al.
7,369,856 B2	5/2008	Ovadia	7,583,964 B2	9/2009	Wong
7,373,136 B2	5/2008	Wattler et al.	7,584,298 B2	9/2009	Klinker et al.
7,373,179 B2	5/2008	Stine et al.	7,585,217 B2	9/2009	Lutnick et al.
7,379,731 B2	5/2008	Natsuno et al.	7,586,871 B2	9/2009	Hamilton et al.
7,388,950 B2	6/2008	Elsey et al.	7,593,417 B2	9/2009	Wang et al.
7,389,412 B2	6/2008	Sharma et al.	7,593,730 B2	9/2009	Khandelwal et al.
7,391,724 B2	6/2008	Alakoski et al.	7,596,373 B2	9/2009	Mcgregor et al.
7,395,056 B2	7/2008	Petermann	7,599,288 B2	10/2009	Cole et al.
7,395,244 B1	7/2008	Kingsford	7,599,714 B2	10/2009	Kuzminskiy
7,401,338 B1	7/2008	Bowen et al.	7,602,746 B2	10/2009	Calhoun et al.
7,403,763 B2	7/2008	Maes	7,606,918 B2	10/2009	Holzman et al.
7,409,447 B1	8/2008	Assadzadeh	7,607,041 B2	10/2009	Kraemer et al.
7,409,569 B2	8/2008	Illowsky et al.	7,609,650 B2	10/2009	Roskowski et al.
7,411,930 B2	8/2008	Montejo et al.	7,609,700 B1	10/2009	Ying et al.
7,418,253 B2	8/2008	Kavanah	7,610,047 B2	10/2009	Hicks, III et al.
7,418,257 B2	8/2008	Kim	7,610,057 B2	10/2009	Bahl et al.
7,421,004 B2	9/2008	Fehér	7,610,328 B2	10/2009	Haase et al.
7,423,971 B1	9/2008	Mohaban et al.	7,610,396 B2	10/2009	Taglienti et al.
7,428,750 B1	9/2008	Dunn et al.	7,614,051 B2	11/2009	Glaum et al.
7,433,362 B2	10/2008	Mallya et al.	7,616,962 B2	11/2009	Oswal et al.
7,436,816 B2	10/2008	Mehta et al.	7,617,516 B2	11/2009	Huslak et al.
7,440,433 B2	10/2008	Rink et al.	7,620,041 B2	11/2009	Dunn et al.
7,444,669 B1	10/2008	Bahl et al.	7,620,065 B2	11/2009	Falardeau
7,450,591 B2	11/2008	Korling et al.	7,620,162 B2	11/2009	Aaron et al.
7,450,927 B1	11/2008	Creswell et al.	7,620,383 B2	11/2009	Taglienti et al.
7,454,191 B2	11/2008	Dawson et al.	7,627,314 B2	12/2009	Carlson et al.
7,457,265 B2	11/2008	Julka et al.	7,627,600 B2	12/2009	Citron et al.
7,457,870 B1	11/2008	Lownsbrough et al.	7,627,767 B2	12/2009	Sherman et al.
7,460,837 B2	12/2008	Diener	7,627,872 B2	12/2009	Hebeler et al.
7,466,652 B2	12/2008	Lau et al.	7,633,438 B2	12/2009	Tysowski
7,467,160 B2	12/2008	McIntyre	7,633,909 B1	12/2009	Jones et al.
7,472,189 B2	12/2008	Mallya et al.	7,634,388 B2	12/2009	Archer et al.
7,478,420 B2	1/2009	Wright et al.	7,636,574 B2	12/2009	Poosala
7,486,185 B2	2/2009	Culpepper et al.	7,636,626 B2	12/2009	Oesterling et al.
7,486,658 B2	2/2009	Kumar	7,643,411 B2	1/2010	Andreasen et al.
7,493,659 B1	2/2009	Wu et al.	7,644,151 B2	1/2010	Jerrim et al.
7,496,652 B2	2/2009	Pezzutti	7,644,267 B2	1/2010	Ylikoski et al.
7,499,438 B2	3/2009	Hinman et al.	7,644,414 B2	1/2010	Smith et al.
7,499,537 B2	3/2009	Elsey et al.	7,647,047 B2	1/2010	Moghaddam et al.
7,502,672 B1	3/2009	Kolls	7,650,137 B2	1/2010	Jobs et al.
7,505,756 B2	3/2009	Bahl	7,653,394 B2	1/2010	McMillin
7,505,795 B1	3/2009	Lim et al.	7,656,271 B2	2/2010	Ehrman et al.
7,508,799 B2	3/2009	Sumner et al.	7,657,920 B2	2/2010	Arseneau et al.
7,512,128 B2	3/2009	DiMambro et al.	7,660,419 B1	2/2010	Ho
7,512,131 B2	3/2009	Svensson et al.	7,661,124 B2	2/2010	Ramanathan et al.
7,515,608 B2	4/2009	Yuan et al.	7,664,494 B2	2/2010	Jiang
7,515,926 B2	4/2009	Bu et al.	7,668,176 B2	2/2010	Chuah
7,516,219 B2	4/2009	Moghaddam et al.	7,668,612 B1	2/2010	Okkonen
7,522,549 B2	4/2009	Karaoguz et al.	7,668,903 B2	2/2010	Edwards et al.
7,522,576 B2	4/2009	Du et al.	7,668,966 B2	2/2010	Klinker et al.
7,526,541 B2	4/2009	Roese et al.	7,672,695 B1	3/2010	Rainnie et al.
7,529,204 B2	5/2009	Bourlas et al.	7,676,673 B2	3/2010	Weller et al.
7,535,880 B1	5/2009	Hinman et al.	7,680,086 B2	3/2010	Eglin
7,536,695 B2	5/2009	Alam et al.	7,681,226 B2	3/2010	Kraemer et al.
7,539,132 B2	5/2009	Werner et al.	7,684,370 B2	3/2010	Kezys
7,539,862 B2	5/2009	Edgett et al.	7,685,131 B2	3/2010	Batra et al.
7,540,408 B2	6/2009	Levine et al.	7,685,254 B2	3/2010	Pandya
7,545,782 B2	6/2009	Rayment et al.	7,685,530 B2	3/2010	Sherrard et al.
7,546,460 B2	6/2009	Maes	7,688,792 B2	3/2010	Babbar et al.
7,546,629 B2	6/2009	Albert et al.	7,693,107 B2	4/2010	De Froment
7,548,875 B2	6/2009	Mikkelsen et al.	7,693,720 B2	4/2010	Kennewick et al.
7,548,976 B2	6/2009	Bahl et al.	7,697,540 B2	4/2010	Haddad et al.
			7,710,932 B2	5/2010	Muthuswamy et al.
			7,711,848 B2	5/2010	Maes
			7,719,966 B2	5/2010	Luft et al.
			7,720,206 B2	5/2010	Devolites et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

7,720,464 B2	5/2010	Batta	7,881,199 B2	2/2011	Krstulich
7,720,505 B2	5/2010	Gopi et al.	7,881,267 B2	2/2011	Crosswy et al.
7,720,960 B2	5/2010	Pruss et al.	7,881,697 B2	2/2011	Baker et al.
7,721,296 B2	5/2010	Ricagni	7,882,029 B2	2/2011	White
7,724,716 B2	5/2010	Fadell	7,882,247 B2	2/2011	Sturniolo et al.
7,725,570 B1	5/2010	Lewis	7,882,560 B2	2/2011	Kraemer et al.
7,729,326 B2	6/2010	Sekhar	7,886,047 B1	2/2011	Potluri
7,730,123 B1	6/2010	Erickson et al.	7,889,384 B2	2/2011	Armentrout et al.
7,734,784 B1	6/2010	Araujo et al.	7,890,084 B1	2/2011	Dudziak et al.
7,742,406 B1	6/2010	Muppala	7,890,111 B2	2/2011	Bugenhagen
7,746,854 B2	6/2010	Ambe et al.	7,894,431 B2	2/2011	Goring et al.
7,747,240 B1	6/2010	Briscoe et al.	7,899,039 B2	3/2011	Andreasen et al.
7,747,699 B2	6/2010	Prueitt et al.	7,899,438 B2	3/2011	Baker et al.
7,747,730 B1	6/2010	Harlow	7,903,553 B2	3/2011	Liu
7,752,330 B2	7/2010	Olsen et al.	7,907,970 B2	3/2011	Park et al.
7,756,056 B2	7/2010	Kim et al.	7,908,358 B1	3/2011	Prasad et al.
7,756,534 B2	7/2010	Anupam et al.	7,911,975 B2	3/2011	Droz et al.
7,756,757 B1	7/2010	Oakes, III	7,912,025 B2	3/2011	Pattenden et al.
7,760,137 B2	7/2010	Martucci et al.	7,912,056 B1	3/2011	Brassem
7,760,711 B1	7/2010	Kung et al.	7,920,529 B1	4/2011	Mahler et al.
7,760,861 B1	7/2010	Croak et al.	7,921,463 B2	4/2011	Sood et al.
7,765,294 B2	7/2010	Edwards et al.	7,925,740 B2	4/2011	Nath et al.
7,769,397 B2	8/2010	Funato et al.	7,925,778 B1	4/2011	Wijnands et al.
7,770,785 B2	8/2010	Jha et al.	7,929,959 B2	4/2011	DeAtley et al.
7,774,323 B2	8/2010	Helfman	7,929,960 B2	4/2011	Martin et al.
7,774,412 B1	8/2010	Schnepel	7,929,973 B2	4/2011	Zavalkovsky et al.
7,774,456 B1	8/2010	Lownsbrough et al.	7,930,327 B2	4/2011	Craft et al.
7,778,176 B2	8/2010	Morford	7,930,446 B2	4/2011	Kesselman et al.
7,778,643 B2	8/2010	Laroia et al.	7,930,553 B2	4/2011	Satarasinghe et al.
7,792,257 B1	9/2010	Vanier et al.	7,933,274 B2	4/2011	Verma et al.
7,792,538 B2	9/2010	Kozisek	7,936,736 B2	5/2011	Proctor, Jr. et al.
7,792,708 B2	9/2010	Alva	7,937,069 B2	5/2011	Rassam
7,797,019 B2	9/2010	Friedmann	7,937,450 B2	5/2011	Janik
7,797,060 B2	9/2010	Grgic et al.	7,940,685 B1	5/2011	Breslau et al.
7,797,204 B2	9/2010	Balent	7,940,751 B2	5/2011	Hansen
7,797,401 B2	9/2010	Stewart et al.	7,941,184 B2	5/2011	Prendergast et al.
7,801,523 B1	9/2010	Kenderov	7,944,948 B2	5/2011	Chow et al.
7,801,783 B2	9/2010	Kende et al.	7,945,238 B2	5/2011	Baker et al.
7,801,985 B1	9/2010	Pitkow et al.	7,945,240 B1	5/2011	Klock et al.
7,802,724 B1	9/2010	Nohr	7,945,945 B2	5/2011	Graham et al.
7,805,140 B2	9/2010	Friday et al.	7,948,952 B2	5/2011	Hurta et al.
7,805,522 B2	9/2010	Schlüter et al.	7,948,953 B2	5/2011	Melkote et al.
7,805,606 B2	9/2010	Birger et al.	7,948,968 B2	5/2011	Voit et al.
7,809,351 B1	10/2010	Panda et al.	7,949,529 B2	5/2011	Weider et al.
7,809,372 B2	10/2010	Rajaniemi	7,953,808 B2	5/2011	Sharp et al.
7,813,746 B2	10/2010	Rajkotia	7,953,877 B2	5/2011	Vemula et al.
7,817,615 B1	10/2010	Breau et al.	7,957,020 B2	6/2011	Mine et al.
7,817,983 B2	10/2010	Cassett et al.	7,957,381 B2	6/2011	Clermidy et al.
7,822,837 B1	10/2010	Urban et al.	7,957,511 B2	6/2011	Drudis et al.
7,822,849 B2	10/2010	Titus	7,958,029 B1	6/2011	Bobich et al.
7,826,427 B2	11/2010	Sood et al.	7,962,622 B2	6/2011	Friend et al.
7,826,607 B1	11/2010	De Carvalho Resende et al.	7,962,798 B2	6/2011	Locasto et al.
7,835,275 B1	11/2010	Swan et al.	7,965,983 B1	6/2011	Swan et al.
7,843,831 B2	11/2010	Morrill et al.	7,966,405 B2	6/2011	Sundaresan et al.
7,843,843 B1	11/2010	Papp, III et al.	7,969,950 B2	6/2011	Iyer et al.
7,844,034 B1	11/2010	Oh et al.	7,970,350 B2	6/2011	Sheynman
7,844,728 B2	11/2010	Anderson et al.	7,970,426 B2	6/2011	Poe et al.
7,848,768 B2	12/2010	Omori et al.	7,974,624 B2	7/2011	Gallagher et al.
7,849,161 B2	12/2010	Koch et al.	7,975,184 B2	7/2011	Goff et al.
7,849,170 B1	12/2010	Hargens et al.	7,978,627 B2	7/2011	Taylor et al.
7,849,310 B2	12/2010	Watt et al.	7,978,686 B2	7/2011	Goyal et al.
7,849,477 B2	12/2010	Cristofalo et al.	7,979,069 B2	7/2011	Hupp et al.
7,853,255 B2	12/2010	Karaoguz et al.	7,979,889 B2	7/2011	Gladstone et al.
7,853,656 B2	12/2010	Yach et al.	7,979,896 B2	7/2011	McMurtry et al.
7,856,226 B2	12/2010	Wong et al.	7,984,130 B2	7/2011	Bogineni et al.
7,860,088 B2	12/2010	Lioy	7,984,511 B2	7/2011	Kocher et al.
7,865,182 B2	1/2011	Macaluso	7,986,935 B1	7/2011	D'Souza et al.
7,865,187 B2	1/2011	Ramer et al.	7,987,496 B2	7/2011	Bryce et al.
7,868,778 B2	1/2011	Kenwright	7,987,510 B2	7/2011	Kocher et al.
7,873,001 B2	1/2011	Silver	7,990,049 B2	8/2011	Shioya
7,873,344 B2	1/2011	Bowser et al.	8,000,276 B2	8/2011	Scherzer et al.
7,873,346 B2	1/2011	Petersson et al.	8,000,318 B2	8/2011	Wiley et al.
7,873,540 B2	1/2011	Arumugam	8,005,009 B2	8/2011	McKee et al.
7,873,705 B2	1/2011	Kalish	8,005,459 B2	8/2011	Balsillie
7,877,090 B2	1/2011	Maes	8,005,726 B1	8/2011	Bao
			8,005,913 B1	8/2011	Carlander
			8,005,988 B2	8/2011	Maes
			8,010,080 B1	8/2011	Thenthiruperai et al.
			8,010,081 B1	8/2011	Roskowski

(56)

References Cited

U.S. PATENT DOCUMENTS

8,010,082 B2	8/2011	Sutaria et al.	8,131,281 B1	3/2012	Hildner et al.
8,010,990 B2	8/2011	Ferguson et al.	8,131,840 B1	3/2012	Denker
8,015,133 B1	9/2011	Wu et al.	8,131,858 B2	3/2012	Agulnik et al.
8,015,234 B2	9/2011	Lum et al.	8,132,256 B2	3/2012	Bari
8,015,249 B2	9/2011	Nayak et al.	8,134,954 B2	3/2012	Godfrey et al.
8,019,687 B2	9/2011	Wang et al.	8,135,388 B1	3/2012	Gailloux et al.
8,019,820 B2	9/2011	Son et al.	8,135,392 B2	3/2012	Marcellino et al.
8,019,846 B2	9/2011	Roelens et al.	8,135,657 B2	3/2012	Kapoor et al.
8,019,868 B2	9/2011	Rao et al.	8,140,690 B2	3/2012	Ly et al.
8,019,886 B2	9/2011	Harrang et al.	8,144,591 B2	3/2012	Ghai et al.
8,023,425 B2	9/2011	Raleigh	8,145,194 B2	3/2012	Yoshikawa et al.
8,024,397 B1	9/2011	Erickson et al.	8,146,142 B2	3/2012	Lortz et al.
8,024,424 B2	9/2011	Freimuth et al.	8,149,748 B2	4/2012	Bata et al.
8,027,339 B2	9/2011	Short et al.	8,149,823 B2	4/2012	Turcan et al.
8,031,601 B2	10/2011	Feroz et al.	8,150,394 B2	4/2012	Bianconi et al.
8,032,168 B2	10/2011	Ikaheimo	8,150,431 B2	4/2012	Wolovitz et al.
8,032,409 B1	10/2011	Mikurak	8,151,205 B2	4/2012	Follmann et al.
8,032,899 B2	10/2011	Archer et al.	8,155,155 B1	4/2012	Chow et al.
8,036,387 B2	10/2011	Kudelski et al.	8,155,620 B2	4/2012	Wang et al.
8,036,600 B2	10/2011	Garrett et al.	8,155,666 B2	4/2012	Alizadeh-Shabdiz
8,044,792 B2	10/2011	Orr et al.	8,155,670 B2	4/2012	Fullam et al.
8,045,973 B2	10/2011	Chambers	8,156,206 B2	4/2012	Kiley et al.
8,046,449 B2	10/2011	Yoshiuchi	8,159,520 B1	4/2012	Dhanoa et al.
8,050,275 B1	11/2011	Iyer	8,160,015 B2	4/2012	Rashid et al.
8,050,690 B2	11/2011	Neeraj	8,160,056 B2	4/2012	Van der Merwe et al.
8,050,705 B2	11/2011	Sicher et al.	8,160,598 B2	4/2012	Savoor
8,059,530 B1	11/2011	Cole	8,165,576 B2	4/2012	Raju et al.
8,060,017 B2	11/2011	Schlicht et al.	8,166,040 B2	4/2012	Brindisi et al.
8,060,463 B1	11/2011	Spiegel	8,166,554 B2	4/2012	John
8,060,603 B2	11/2011	Caunter et al.	8,170,553 B2	5/2012	Bennett
8,064,417 B2	11/2011	Maki	8,174,378 B2	5/2012	Richman et al.
8,064,418 B2	11/2011	Maki	8,174,970 B2	5/2012	Adamczyk et al.
8,064,896 B2	11/2011	Bell et al.	8,175,574 B1	5/2012	Panda et al.
8,065,365 B2	11/2011	Saxena et al.	8,180,333 B1	5/2012	Wells et al.
8,068,824 B2	11/2011	Shan et al.	8,180,881 B2	5/2012	Seo et al.
8,068,829 B2	11/2011	Lemond et al.	8,180,886 B2	5/2012	Overcash et al.
8,073,427 B2	12/2011	Koch et al.	8,184,530 B1	5/2012	Swan et al.
8,073,721 B1	12/2011	Lewis	8,184,590 B2	5/2012	Rosenblatt
8,078,140 B2	12/2011	Baker et al.	8,185,088 B2	5/2012	Klein et al.
8,078,163 B2	12/2011	Lemond et al.	8,185,093 B2	5/2012	Jheng et al.
8,085,808 B2	12/2011	Brusca et al.	8,185,127 B1	5/2012	Cai et al.
8,086,398 B2	12/2011	Sanchez et al.	8,185,152 B1	5/2012	Goldner
8,086,497 B1	12/2011	Oakes, III	8,185,158 B2	5/2012	Tamura et al.
8,086,791 B2	12/2011	Caulkins	8,190,087 B2	5/2012	Fisher et al.
8,090,359 B2	1/2012	Proctor, Jr. et al.	8,190,122 B1	5/2012	Alexander et al.
8,090,361 B2	1/2012	Hagan	8,190,675 B2	5/2012	Tribbett
8,090,616 B2	1/2012	Proctor, Jr. et al.	8,191,106 B2	5/2012	Choyi et al.
8,091,087 B2	1/2012	Ali et al.	8,191,116 B1	5/2012	Gazzard
8,094,551 B2	1/2012	Huber et al.	8,191,124 B2	5/2012	Wynn et al.
8,095,112 B2	1/2012	Chow et al.	8,194,549 B2	6/2012	Huber et al.
8,095,124 B2	1/2012	Balia	8,194,553 B2	6/2012	Liang et al.
8,095,640 B2	1/2012	Guingo et al.	8,194,572 B2	6/2012	Horvath et al.
8,095,666 B2	1/2012	Schmidt et al.	8,194,581 B1	6/2012	Schroeder et al.
8,098,579 B2	1/2012	Ray et al.	8,195,093 B2	6/2012	Garrett et al.
8,099,077 B2	1/2012	Chowdhury et al.	8,195,153 B1	6/2012	Frencel et al.
8,099,517 B2	1/2012	Jia et al.	8,195,163 B2	6/2012	Gisby et al.
8,102,814 B2	1/2012	Rahman et al.	8,195,661 B2	6/2012	Kalavade
8,103,285 B2	1/2012	Kalhan	8,196,199 B2	6/2012	Hrastar et al.
8,104,080 B2	1/2012	Burns et al.	8,200,163 B2	6/2012	Hoffman
8,107,953 B2	1/2012	Zimmerman et al.	8,200,200 B1	6/2012	Belser et al.
8,108,520 B2	1/2012	Ruutu et al.	8,200,509 B2	6/2012	Kenedy et al.
8,108,680 B2	1/2012	Murray	8,200,775 B2	6/2012	Moore
8,112,435 B2	2/2012	Epstein et al.	8,200,818 B2	6/2012	Freund et al.
8,116,223 B2	2/2012	Tian et al.	8,204,190 B2	6/2012	Bang et al.
8,116,749 B2	2/2012	Proctor, Jr. et al.	8,204,505 B2	6/2012	Jin et al.
8,116,781 B2	2/2012	Chen et al.	8,208,788 B2	6/2012	Ando et al.
8,122,128 B2	2/2012	Burke, II et al.	8,208,919 B2	6/2012	Kotecha
8,122,249 B2	2/2012	Falk et al.	8,213,296 B2	7/2012	Shannon et al.
8,125,897 B2	2/2012	Ray et al.	8,213,363 B2	7/2012	Mng et al.
8,126,123 B2	2/2012	Cai et al.	8,214,536 B2	7/2012	Zhao
8,126,396 B2	2/2012	Bennett	8,214,890 B2	7/2012	Kirovski et al.
8,126,476 B2	2/2012	Vardi et al.	8,219,134 B2	7/2012	Maharajh et al.
8,126,722 B2	2/2012	Robb et al.	8,223,655 B2	7/2012	Heinz et al.
8,130,793 B2	3/2012	Edwards et al.	8,223,741 B1	7/2012	Bartlett et al.
8,131,256 B2	3/2012	Martti et al.	8,224,382 B2	7/2012	Bultman
			8,224,773 B2	7/2012	Spiegel
			8,228,818 B2	7/2012	Chase et al.
			8,229,394 B2	7/2012	Karlberg
			8,229,914 B2	7/2012	Ramer et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,233,433 B2	7/2012	Kalhan	8,355,570 B2	1/2013	Karsanbhai et al.
8,233,883 B2	7/2012	De Froment	8,355,696 B1	1/2013	Olding et al.
8,233,895 B2	7/2012	Tysowski	8,356,336 B2	1/2013	Johnston et al.
8,234,583 B2	7/2012	Sloo et al.	8,358,638 B2	1/2013	Scherzer et al.
8,238,287 B1	8/2012	Gopi et al.	8,358,975 B2	1/2013	Bahl et al.
8,238,913 B1	8/2012	Bhattacharyya et al.	8,363,658 B1	1/2013	Delker et al.
8,239,520 B2	8/2012	Grah	8,363,799 B2	1/2013	Gruchala et al.
8,242,959 B2	8/2012	Mia et al.	8,364,089 B2	1/2013	Phillips
8,244,241 B2	8/2012	Montemurro	8,364,806 B2	1/2013	Short et al.
8,249,601 B2	8/2012	Emberson et al.	8,369,274 B2	2/2013	Sawai
8,254,880 B2	8/2012	Aaltonen et al.	8,370,477 B2	2/2013	Short et al.
8,254,915 B2	8/2012	Kozisek	8,370,483 B2	2/2013	Choong et al.
8,255,515 B1	8/2012	Melman et al.	8,374,090 B2	2/2013	Morrill et al.
8,255,534 B2	8/2012	Assadzadeh	8,374,592 B2	2/2013	Proctor, Jr. et al.
8,255,669 B2	8/2012	Kim et al.	8,375,128 B2	2/2013	Tofighbakhsh et al.
8,259,692 B2	9/2012	Bajko	8,375,136 B2	2/2013	Roman et al.
8,260,252 B2	9/2012	Agarwal	8,380,247 B2	2/2013	Engstrom
8,264,965 B2	9/2012	Dolganow et al.	8,385,199 B1	2/2013	Coward et al.
8,265,004 B2	9/2012	Toutonghi	8,385,896 B2	2/2013	Proctor, Jr. et al.
8,266,249 B2	9/2012	Hu	8,385,964 B2	2/2013	Haney
8,266,681 B2	9/2012	Deshpande et al.	8,385,975 B2	2/2013	Forutanpour et al.
8,270,955 B2	9/2012	Ramer et al.	8,386,386 B1	2/2013	Zhu
8,270,972 B2	9/2012	Otting et al.	8,391,262 B2	3/2013	Maki et al.
8,271,025 B2	9/2012	Brisebois et al.	8,391,834 B2	3/2013	Raleigh
8,271,045 B2	9/2012	Parolkar et al.	8,392,982 B2	3/2013	Harris et al.
8,271,049 B2	9/2012	Silver et al.	8,396,458 B2	3/2013	Raleigh
8,271,992 B2	9/2012	Chatley et al.	8,396,929 B2	3/2013	Helfman et al.
8,275,415 B2	9/2012	Huslak	8,401,968 B1	3/2013	Schattauer et al.
8,275,830 B2	9/2012	Raleigh	8,402,165 B2	3/2013	Deu-Ngoc et al.
8,279,067 B2	10/2012	Berger et al.	8,402,540 B2	3/2013	Kapoor et al.
8,279,864 B2	10/2012	Wood	8,406,427 B2	3/2013	Chand et al.
8,280,354 B2	10/2012	Smith et al.	8,406,736 B2	3/2013	Das et al.
8,284,740 B2	10/2012	O'Connor	8,407,763 B2	3/2013	Weller et al.
8,285,249 B2	10/2012	Baker et al.	8,411,587 B2	4/2013	Curtis et al.
8,285,992 B2	10/2012	Mathur et al.	8,411,691 B2	4/2013	Aggarwal
8,290,820 B2	10/2012	Plastina et al.	8,412,798 B1	4/2013	Wang
8,291,238 B2	10/2012	Ginter et al.	8,413,245 B2	4/2013	Kraemer et al.
8,291,439 B2	10/2012	Jethi et al.	8,418,168 B2	4/2013	Tyhurst et al.
8,296,404 B2	10/2012	McDysan et al.	8,422,988 B1	4/2013	Keshav
8,300,575 B2	10/2012	Willars	8,423,016 B2	4/2013	Buckley et al.
8,306,518 B1	11/2012	Gailloux	8,429,403 B2	4/2013	Moret et al.
8,306,741 B2	11/2012	Tu	8,437,734 B2	5/2013	Ray et al.
8,307,067 B2	11/2012	Ryan	8,442,015 B2	5/2013	Behzad et al.
8,310,943 B2	11/2012	Mehta et al.	8,446,831 B2	5/2013	Kwan et al.
8,315,198 B2	11/2012	Corneille et al.	8,447,324 B2	5/2013	Shuman et al.
8,315,593 B2	11/2012	Gallant et al.	8,447,607 B2	5/2013	Weider et al.
8,315,594 B1	11/2012	Mausser et al.	8,447,980 B2	5/2013	Godfrey et al.
8,315,718 B2	11/2012	Caffrey et al.	8,448,015 B2	5/2013	Gerhart
8,315,999 B2	11/2012	Chatley et al.	8,452,858 B2	5/2013	Wu et al.
8,320,244 B2	11/2012	Muqattash et al.	8,457,603 B2	6/2013	El-Kadri et al.
8,320,949 B2	11/2012	Matta	8,461,958 B2	6/2013	Saenz et al.
8,325,638 B2	12/2012	Jin et al.	8,463,194 B2	6/2013	Erlenback et al.
8,325,906 B2	12/2012	Fullarton et al.	8,463,232 B2	6/2013	Tuli et al.
8,326,319 B2	12/2012	Davis	8,468,337 B2	6/2013	Gaur et al.
8,326,828 B2	12/2012	Zhou et al.	8,472,371 B1	6/2013	Bari et al.
8,331,223 B2	12/2012	Hill et al.	8,477,778 B2	7/2013	Lehmann, Jr. et al.
8,331,293 B2	12/2012	Sood	8,478,840 B2	7/2013	Skutela et al.
8,332,375 B2	12/2012	Chatley et al.	8,483,057 B2	7/2013	Cuervo
8,339,991 B2	12/2012	Biswas et al.	8,483,135 B2	7/2013	Cai et al.
8,340,625 B1	12/2012	Johnson et al.	8,483,694 B2	7/2013	Lewis et al.
8,340,628 B2	12/2012	Taylor et al.	8,484,327 B2	7/2013	Werner et al.
8,340,678 B1	12/2012	Pandey	8,488,597 B2	7/2013	Nie et al.
8,340,718 B2	12/2012	Colonna et al.	8,489,110 B2	7/2013	Frank et al.
8,346,210 B2	1/2013	Balsan et al.	8,489,720 B1	7/2013	Morford et al.
8,346,225 B2	1/2013	Raleigh	8,494,559 B1	7/2013	Malmi
8,346,923 B2	1/2013	Rowles et al.	8,495,181 B2	7/2013	Venkatraman et al.
8,347,104 B2	1/2013	Pathiyal	8,495,207 B2	7/2013	Lee
8,347,362 B2	1/2013	Cai et al.	8,495,227 B2	7/2013	Kaminsky et al.
8,347,378 B2	1/2013	Merkin et al.	8,495,360 B2	7/2013	Falk et al.
8,350,700 B2	1/2013	Fast et al.	8,495,700 B2	7/2013	Shahbazi
8,351,592 B2	1/2013	Freeny, Jr. et al.	8,495,743 B2	7/2013	Kraemer et al.
8,351,898 B2	1/2013	Raleigh	8,499,087 B2	7/2013	Hu
8,352,360 B2	1/2013	De Judicibus et al.	RE44,412 E	8/2013	Naqvi et al.
8,352,980 B2	1/2013	Howcroft	8,500,533 B2	8/2013	Lutnick et al.
8,353,001 B2	1/2013	Herrod	8,503,358 B2	8/2013	Hanson et al.
			8,503,455 B2	8/2013	Heikens
			8,504,032 B2	8/2013	Lott et al.
			8,504,574 B2	8/2013	Dvorak et al.
			8,504,687 B2	8/2013	Maffione et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,504,690 B2	8/2013	Shah et al.	8,694,772 B2	4/2014	Kao et al.
8,504,729 B2	8/2013	Pezzutti	8,700,729 B2	4/2014	Dua
8,505,073 B2	8/2013	Taglienti et al.	8,701,015 B2	4/2014	Bonnat
8,509,082 B2	8/2013	Heinz et al.	8,705,361 B2	4/2014	Venkataraman et al.
8,514,927 B2	8/2013	Sundararajan et al.	8,706,863 B2	4/2014	Fadell
8,516,552 B2	8/2013	Raleigh	8,713,535 B2	4/2014	Malhotra et al.
8,520,589 B2	8/2013	Bhatt et al.	8,713,641 B1	4/2014	Pagan et al.
8,520,595 B2	8/2013	Yadav et al.	8,719,397 B2	5/2014	Levi et al.
8,521,110 B2	8/2013	Rofougaran	8,719,423 B2	5/2014	Wylid
8,521,775 B1	8/2013	Poh et al.	8,724,486 B2	5/2014	Seto et al.
8,522,039 B2	8/2013	Hyndman et al.	8,725,899 B2	5/2014	Short et al.
8,522,249 B2	8/2013	Beaule	8,730,842 B2	5/2014	Collins et al.
8,522,337 B2	8/2013	Adusumilli et al.	8,731,519 B2	5/2014	Flynn et al.
8,523,547 B2	9/2013	Pekrul	8,732,808 B2	5/2014	Sewall et al.
8,526,329 B2	9/2013	Mahany et al.	8,739,035 B2	5/2014	Trethewey
8,526,350 B2	9/2013	Xue et al.	8,744,339 B2	6/2014	Halfmann et al.
8,527,013 B2	9/2013	Guba et al.	8,761,711 B2	6/2014	Grignani et al.
8,527,410 B2	9/2013	Markki et al.	8,780,857 B2	7/2014	Balasubramanian et al.
8,527,662 B2	9/2013	Biswas et al.	8,787,249 B2	7/2014	Giaretta et al.
8,528,068 B1	9/2013	Weglein et al.	8,792,857 B2	7/2014	Cai et al.
8,531,954 B2	9/2013	McNaughton et al.	8,793,304 B2	7/2014	Lu et al.
8,531,995 B2	9/2013	Khan et al.	8,798,610 B2	8/2014	Prakash et al.
8,532,610 B2	9/2013	Manning Cassett et al.	8,799,440 B2	8/2014	Zhou et al.
8,533,775 B2	9/2013	Alcorn et al.	8,800,015 B2 *	8/2014	Christopher H04W 4/50 455/411
8,535,160 B2	9/2013	Lutnick et al.	8,804,695 B2	8/2014	Branam
8,538,394 B2	9/2013	Zimmerman et al.	8,811,338 B2	8/2014	Jin et al.
8,538,421 B2	9/2013	Brisebois et al.	8,811,991 B2	8/2014	Jain et al.
8,538,458 B2	9/2013	Haney	8,818,394 B2	8/2014	Bienas et al.
8,539,544 B2	9/2013	Garimella et al.	8,819,253 B2	8/2014	Simeloff et al.
8,543,265 B2	9/2013	Ekhaguere et al.	8,825,109 B2	9/2014	Montemurro et al.
8,543,814 B2	9/2013	Laitinen et al.	8,826,411 B2	9/2014	Moen et al.
8,544,105 B2	9/2013	Mclean et al.	8,831,561 B2	9/2014	Sutaria et al.
8,548,427 B2	10/2013	Chow et al.	8,838,752 B2	9/2014	Lor et al.
8,549,173 B1	10/2013	Wu et al.	8,843,849 B2	9/2014	Neil et al.
8,554,876 B2	10/2013	Winsor	8,845,415 B2	9/2014	Lutnick et al.
8,559,369 B2	10/2013	Barkan	8,849,297 B2	9/2014	Balasubramanian
8,561,138 B2	10/2013	Rothman et al.	8,855,620 B2	10/2014	Sievers et al.
8,565,746 B2	10/2013	Hoffman	8,862,751 B2	10/2014	Faccin et al.
8,566,236 B2	10/2013	Busch	8,863,111 B2	10/2014	Selitser et al.
8,571,474 B2	10/2013	Chavez et al.	8,875,042 B2	10/2014	LeJeune et al.
8,571,501 B2	10/2013	Miller et al.	8,880,047 B2	11/2014	Konicek et al.
8,571,598 B2	10/2013	Valavi	8,891,483 B2	11/2014	Connelly et al.
8,571,993 B2	10/2013	Kocher et al.	8,898,748 B2	11/2014	Burks et al.
8,572,117 B2	10/2013	Rappaport	8,908,516 B2	12/2014	Tzamaloukas et al.
8,572,256 B2	10/2013	Babbar	8,929,374 B2	1/2015	Tönsing et al.
8,583,499 B2	11/2013	De Judicibus et al.	8,930,238 B2	1/2015	Coffman et al.
8,588,240 B2	11/2013	Ramankutty et al.	8,930,551 B2	1/2015	Pandya et al.
8,589,955 B2	11/2013	Roundtree et al.	8,943,551 B2	1/2015	Ganapathy et al.
8,594,665 B2	11/2013	Anschutz	8,948,726 B2	2/2015	Smith et al.
8,595,186 B1	11/2013	Mandyam et al.	8,949,382 B2	2/2015	Cornett et al.
8,600,895 B2	12/2013	Felsher	8,949,597 B1	2/2015	Reeves et al.
8,601,125 B2	12/2013	Huang et al.	8,955,038 B2	2/2015	Nicodemus et al.
8,605,691 B2	12/2013	Soomro et al.	8,966,018 B2	2/2015	Bugwadia et al.
8,615,507 B2	12/2013	Varadarajulu et al.	8,971,912 B2	3/2015	Chou et al.
8,619,735 B2	12/2013	Montemurro et al.	8,977,284 B2	3/2015	Reed
8,620,257 B2	12/2013	Qiu et al.	8,995,952 B1	3/2015	Baker et al.
8,630,630 B2	1/2014	Raleigh	9,002,342 B2	4/2015	Tenhunen et al.
8,630,925 B2	1/2014	Bystrom et al.	9,014,973 B2	4/2015	Ruckart
8,631,428 B2	1/2014	Scott et al.	9,015,331 B2	4/2015	Lai et al.
8,634,425 B2	1/2014	Gorti et al.	9,026,100 B2	5/2015	Castro et al.
8,635,164 B2	1/2014	Rosenhaft et al.	9,030,934 B2	5/2015	Shah et al.
8,639,215 B2	1/2014	McGregor et al.	9,032,192 B2	5/2015	Frank
8,644,702 B1	2/2014	Kalajan	9,049,010 B2	6/2015	Jueneman et al.
8,644,813 B1	2/2014	Gailloux et al.	9,064,275 B1	6/2015	Lu et al.
8,645,518 B2	2/2014	David	9,094,774 B2 *	7/2015	Smith H04L 41/0803
8,655,357 B1	2/2014	Gazzard et al.	9,105,031 B2	8/2015	Shen et al.
8,656,472 B2	2/2014	McMurtry et al.	9,111,088 B2	8/2015	Ghai et al.
8,660,853 B2	2/2014	Robb et al.	9,137,286 B1	9/2015	Yuan
8,666,395 B2	3/2014	Silver	9,172,553 B2	10/2015	Dawes et al.
8,667,542 B1	3/2014	Bertz et al.	9,177,455 B2	11/2015	Remer
8,670,334 B2	3/2014	Keohane et al.	9,204,282 B2	12/2015	Raleigh
8,675,852 B2	3/2014	Maes	9,282,460 B2	3/2016	Souissi
8,676,682 B2	3/2014	Kalliola	9,286,469 B2	3/2016	Kraemer et al.
8,676,925 B1	3/2014	Liu et al.	9,286,604 B2	3/2016	Aabye et al.
8,693,323 B1	4/2014	McDysan	9,313,708 B2	4/2016	Nam et al.
			9,325,737 B2	4/2016	Gutowski et al.
			9,326,173 B2	4/2016	Luft
			9,344,557 B2	5/2016	Gruchala et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,363,285 B2	6/2016	Kitamura	2004/0102182 A1	5/2004	Reith et al.
9,367,680 B2	6/2016	Mahaffey et al.	2004/0103193 A1	5/2004	Pandya et al.
9,413,546 B2	8/2016	Meier et al.	2004/0107360 A1	6/2004	Herrmann et al.
9,418,381 B2	8/2016	Ahuja et al.	2004/0116140 A1	6/2004	Babbar et al.
9,459,767 B2	10/2016	Cockcroft et al.	2004/0123153 A1	6/2004	Wright et al.
9,501,803 B2	11/2016	Bilac et al.	2004/0127200 A1	7/2004	Shaw et al.
9,544,397 B2	1/2017	Raleigh et al.	2004/0127208 A1	7/2004	Nair et al.
9,589,117 B2	3/2017	Ali et al.	2004/0127256 A1	7/2004	Goldthwaite et al.
9,609,459 B2	3/2017	Raleigh	2004/0132427 A1	7/2004	Lee et al.
9,712,476 B2	7/2017	Boynton et al.	2004/0133668 A1	7/2004	Nicholas, III
9,942,796 B2	4/2018	Raleigh	2004/0137890 A1	7/2004	Kalke
9,986,413 B2	5/2018	Raleigh	2004/0148237 A1	7/2004	Bittmann et al.
10,021,251 B2	7/2018	Aaron et al.	2004/0165596 A1	8/2004	Garcia et al.
10,285,025 B1	5/2019	Baker et al.	2004/0167958 A1	8/2004	Stewart et al.
10,326,800 B2	6/2019	Raleigh et al.	2004/0168052 A1	8/2004	Clisham et al.
10,492,102 B2	11/2019	Raleigh et al.	2004/0170191 A1	9/2004	Guo et al.
10,582,375 B2	3/2020	Raleigh	2004/0176104 A1	9/2004	Arcens
10,779,177 B2	9/2020	Raleigh	2004/0198331 A1	10/2004	Coward et al.
2001/0048738 A1	12/2001	Baniak et al.	2004/0203755 A1	10/2004	Brunet et al.
2001/0053694 A1	12/2001	Igarashi et al.	2004/0203833 A1	10/2004	Rathunde et al.
2002/0013844 A1	1/2002	Garrett et al.	2004/0225561 A1	11/2004	Hertzberg et al.
2002/0022472 A1	2/2002	Wattler et al.	2004/0225898 A1	11/2004	Frost et al.
2002/0022483 A1	2/2002	Thompson et al.	2004/0236547 A1	11/2004	Rappaport et al.
2002/0049074 A1	4/2002	Eisinger et al.	2004/0243680 A1	12/2004	Mayer
2002/0099848 A1	7/2002	Lee	2004/0243992 A1	12/2004	Gustafson et al.
2002/0116338 A1	8/2002	Gonthier et al.	2004/0249918 A1	12/2004	Sunshine
2002/0120370 A1	8/2002	Parupudi et al.	2004/0255145 A1	12/2004	Chow
2002/0120540 A1	8/2002	Kende et al.	2004/0259534 A1	12/2004	Chaudhari et al.
2002/0131404 A1	9/2002	Mehta et al.	2004/0260766 A1	12/2004	Barros et al.
2002/0138599 A1	9/2002	Dilman et al.	2004/0267872 A1	12/2004	Serdy et al.
2002/0138601 A1	9/2002	Piponijs et al.	2005/0007993 A1	1/2005	Chambers et al.
2002/0154751 A1	10/2002	Thompson et al.	2005/0009499 A1	1/2005	Koster
2002/0161601 A1	10/2002	Nauer et al.	2005/0021995 A1	1/2005	Lal et al.
2002/0164983 A1	11/2002	Raviv et al.	2005/0041617 A1	2/2005	Huotari et al.
2002/0176377 A1	11/2002	Hamilton	2005/0048950 A1	3/2005	Morper
2002/0188732 A1	12/2002	Buckman et al.	2005/0055291 A1	3/2005	Bevente et al.
2002/0191573 A1	12/2002	Whitehill et al.	2005/0055309 A1	3/2005	Williams et al.
2002/0199001 A1	12/2002	Wenocur et al.	2005/0055595 A1	3/2005	Frazer et al.
2003/0004937 A1	1/2003	Salmenkaita et al.	2005/0060266 A1	3/2005	Demello et al.
2003/0005112 A1	1/2003	Krautkremer	2005/0060525 A1	3/2005	Schwartz et al.
2003/0013434 A1	1/2003	Rosenberg et al.	2005/0075115 A1	4/2005	Corneille et al.
2003/0018524 A1	1/2003	Fishman et al.	2005/0079863 A1	4/2005	Macaluso
2003/0028623 A1	2/2003	Hennessey et al.	2005/0091505 A1	4/2005	Riley et al.
2003/0046396 A1	3/2003	Richter et al.	2005/0096024 A1	5/2005	Bicker et al.
2003/0050070 A1	3/2003	Mashinsky et al.	2005/0097516 A1	5/2005	Donnelly et al.
2003/0050837 A1	3/2003	Kim	2005/0101323 A1	5/2005	De Beer
2003/0084321 A1	5/2003	Tarquini et al.	2005/0107091 A1	5/2005	Vannithamby et al.
2003/0088671 A1	5/2003	Klinker et al.	2005/0108075 A1	5/2005	Douglis et al.
2003/0133408 A1	7/2003	Cheng et al.	2005/0108534 A1	5/2005	Bajikar et al.
2003/0134650 A1	7/2003	Sundar et al.	2005/0111463 A1	5/2005	Leung et al.
2003/0159030 A1	8/2003	Evans	2005/0128967 A1	6/2005	Scobbie
2003/0161265 A1	8/2003	Cao et al.	2005/0135264 A1	6/2005	Popoff et al.
2003/0171112 A1	9/2003	Lupper et al.	2005/0144505 A1	6/2005	Takeuchi et al.
2003/0182420 A1	9/2003	Jones et al.	2005/0163320 A1	7/2005	Brown et al.
2003/0182435 A1	9/2003	Redlich et al.	2005/0166043 A1	7/2005	Zhang et al.
2003/0184793 A1	10/2003	Pineau	2005/0183143 A1	8/2005	Anderholm et al.
2003/0188006 A1	10/2003	Bard	2005/0186948 A1	8/2005	Gallagher et al.
2003/0188117 A1	10/2003	Yoshino et al.	2005/0198377 A1	9/2005	Ferguson et al.
2003/0220984 A1	11/2003	Jones et al.	2005/0216421 A1	9/2005	Barry et al.
2003/0224781 A1	12/2003	Milford et al.	2005/0228985 A1	10/2005	Ylikoski et al.
2003/0229900 A1	12/2003	Reisman	2005/0238046 A1	10/2005	Hassan et al.
2003/0233332 A1	12/2003	Keeler et al.	2005/0239447 A1	10/2005	Holzman et al.
2003/0236745 A1	12/2003	Hartsell et al.	2005/0245241 A1	11/2005	Durand et al.
2004/0019539 A1	1/2004	Raman et al.	2005/0246282 A1	11/2005	Naslund et al.
2004/0019564 A1	1/2004	Goldthwaite et al.	2005/0250508 A1	11/2005	Guo et al.
2004/0021697 A1	2/2004	Beaton et al.	2005/0250536 A1	11/2005	Deng et al.
2004/0024756 A1	2/2004	Rickard	2005/0254435 A1	11/2005	Moakley et al.
2004/0030705 A1	2/2004	Bowman-Amuah	2005/0266825 A1	12/2005	Clayton
2004/0039792 A1	2/2004	Nakanishi	2005/0266880 A1	12/2005	Gupta
2004/0044623 A1	3/2004	Wake et al.	2006/0014519 A1	1/2006	Marsh et al.
2004/0047358 A1	3/2004	Chen et al.	2006/0015749 A1	1/2006	Mittal
2004/0054779 A1	3/2004	Takeshima et al.	2006/0019632 A1	1/2006	Cunningham et al.
2004/0073672 A1	4/2004	Fascenda	2006/0020781 A1	1/2006	Scarlata et al.
2004/0082346 A1	4/2004	Skytt et al.	2006/0020787 A1	1/2006	Choyi et al.
2004/0098715 A1	5/2004	Aghera et al.	2006/0026679 A1	2/2006	Zakas
			2006/0030306 A1	2/2006	Kuhn
			2006/0034256 A1	2/2006	Addagatla et al.
			2006/0035631 A1	2/2006	White et al.
			2006/0040642 A1	2/2006	Boris et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2006/0045245 A1	3/2006	Aaron et al.	2007/0117538 A1	5/2007	Weiser et al.
2006/0048223 A1	3/2006	Lee et al.	2007/0130283 A1	6/2007	Klein et al.
2006/0068796 A1	3/2006	Millen et al.	2007/0130315 A1	6/2007	Friend et al.
2006/0072451 A1	4/2006	Ross	2007/0140113 A1	6/2007	Gemelos
2006/0072550 A1	4/2006	Davis et al.	2007/0140145 A1	6/2007	Kumar et al.
2006/0072646 A1	4/2006	Feher	2007/0140275 A1	6/2007	Bowman et al.
2006/0075506 A1	4/2006	Sanda et al.	2007/0143824 A1	6/2007	Shahbazi
2006/0085543 A1	4/2006	Hrastar et al.	2007/0143827 A1	6/2007	Nicodemus et al.
2006/0093107 A1	5/2006	Chien	2007/0147317 A1	6/2007	Smith et al.
2006/0095517 A1	5/2006	O'Connor et al.	2007/0147324 A1	6/2007	McGary
2006/0098627 A1	5/2006	Karaoguz et al.	2007/0149252 A1	6/2007	Jobs et al.
2006/0099970 A1	5/2006	Morgan et al.	2007/0155365 A1	7/2007	Kim et al.
2006/0101507 A1	5/2006	Camenisch	2007/0165630 A1	7/2007	Rasanen et al.
2006/0112016 A1	5/2006	Ishibashi	2007/0168499 A1	7/2007	Chu
2006/0112427 A1	5/2006	Shahbazi	2007/0171856 A1	7/2007	Bruce et al.
2006/0114821 A1	6/2006	Willey et al.	2007/0174490 A1	7/2007	Choi et al.
2006/0114832 A1	6/2006	Hamilton et al.	2007/0178888 A1	8/2007	Alfano et al.
2006/0126562 A1	6/2006	Liu	2007/0191006 A1	8/2007	Carpenter
2006/0135144 A1	6/2006	Jothipragsam	2007/0192460 A1	8/2007	Choi et al.
2006/0136882 A1	6/2006	Noonan et al.	2007/0198656 A1	8/2007	Mazzafferri et al.
2006/0143066 A1	6/2006	Calabria	2007/0201502 A1	8/2007	Abramson
2006/0143098 A1	6/2006	Lazaridis	2007/0213054 A1	9/2007	Han
2006/0156398 A1	7/2006	Ross et al.	2007/0220251 A1	9/2007	Rosenberg et al.
2006/0160536 A1	7/2006	Chou	2007/0226225 A1	9/2007	Yiu et al.
2006/0165060 A1	7/2006	Dua	2007/0226775 A1	9/2007	Andreasen et al.
2006/0168128 A1	7/2006	Sistla et al.	2007/0234402 A1	10/2007	Khosravi et al.
2006/0173959 A1	8/2006	Mckelvie et al.	2007/0243862 A1	10/2007	Coskun et al.
2006/0174035 A1	8/2006	Tufail	2007/0248100 A1	10/2007	Zuberi et al.
2006/0178917 A1	8/2006	Merriam et al.	2007/0254646 A1	11/2007	Sokondar
2006/0178918 A1	8/2006	Mikurak	2007/0254675 A1	11/2007	Zorlu Ozer et al.
2006/0182137 A1	8/2006	Zhou et al.	2007/0255769 A1	11/2007	Agrawal et al.
2006/0183461 A1	8/2006	Pearce	2007/0255797 A1	11/2007	Dunn et al.
2006/0183462 A1	8/2006	Kolehmainen	2007/0255848 A1	11/2007	Sewall et al.
2006/0190314 A1	8/2006	Hernandez	2007/0257767 A1	11/2007	Beeson
2006/0190987 A1	8/2006	Ohta et al.	2007/0259656 A1	11/2007	Jeong
2006/0193280 A1	8/2006	Lee et al.	2007/0259673 A1	11/2007	Willars et al.
2006/0199608 A1	9/2006	Dunn et al.	2007/0263558 A1	11/2007	Salomone
2006/0200663 A1	9/2006	Thornton	2007/0266422 A1	11/2007	Germano et al.
2006/0206709 A1	9/2006	Labrou et al.	2007/0274327 A1	11/2007	Kaarela et al.
2006/0206904 A1	9/2006	Watkins et al.	2007/0280453 A1	12/2007	Kelley
2006/0218395 A1	9/2006	Maes	2007/0282896 A1	12/2007	Wydroug et al.
2006/0233108 A1	10/2006	Krishnan	2007/0288989 A1	12/2007	Aarnos et al.
2006/0233166 A1	10/2006	Bou-Diab et al.	2007/0293191 A1	12/2007	Mir et al.
2006/0236095 A1	10/2006	Smith et al.	2007/0294395 A1	12/2007	Strub et al.
2006/0242685 A1	10/2006	Heard et al.	2007/0294410 A1	12/2007	Pandya et al.
2006/0258341 A1	11/2006	Miller et al.	2007/0297378 A1	12/2007	Poyhonen et al.
2006/0274706 A1	12/2006	Chen et al.	2007/0298764 A1	12/2007	Clayton
2006/0277590 A1	12/2006	Limont et al.	2007/0299965 A1	12/2007	Nieh et al.
2006/0291419 A1	12/2006	McConnell et al.	2007/0300252 A1	12/2007	Acharya et al.
2006/0291477 A1	12/2006	Croak et al.	2008/0005285 A1	1/2008	Robinson et al.
2007/0005795 A1	1/2007	Gonzalez	2008/0005561 A1	1/2008	Brown et al.
2007/0006282 A1	1/2007	Durham et al.	2008/0010379 A1	1/2008	Zhao
2007/0006289 A1	1/2007	Limont et al.	2008/0010452 A1	1/2008	Holtzman et al.
2007/0019670 A1	1/2007	Falardeau	2008/0018494 A1	1/2008	Waite et al.
2007/0022289 A1	1/2007	Alt et al.	2008/0020738 A1	1/2008	Ho et al.
2007/0025301 A1	2/2007	Petersson et al.	2008/0022354 A1	1/2008	Grewal et al.
2007/0033194 A1	2/2007	Srinivas et al.	2008/0025230 A1	1/2008	Patel et al.
2007/0033197 A1	2/2007	Scherzer et al.	2008/0032715 A1	2/2008	Jia et al.
2007/0035390 A1	2/2007	Thomas et al.	2008/0034063 A1	2/2008	Yee
2007/0036312 A1	2/2007	Cai et al.	2008/0034419 A1	2/2008	Mullick et al.
2007/0055694 A1	3/2007	Ruge et al.	2008/0039102 A1	2/2008	Sewall et al.
2007/0060200 A1	3/2007	Boris et al.	2008/0049630 A1	2/2008	Kozisek et al.
2007/0061243 A1	3/2007	Ramer et al.	2008/0050715 A1	2/2008	Golczewski et al.
2007/0061535 A1	3/2007	Xu et al.	2008/0051076 A1	2/2008	O'Shaughnessy et al.
2007/0061800 A1	3/2007	Cheng et al.	2008/0052387 A1	2/2008	Heinz et al.
2007/0061878 A1	3/2007	Hagiu et al.	2008/0052395 A1	2/2008	Wright et al.
2007/0073899 A1	3/2007	Judge et al.	2008/0056273 A1	3/2008	Pelletier et al.
2007/0076616 A1	4/2007	Ngo et al.	2008/0059474 A1	3/2008	Lim
2007/0093243 A1	4/2007	Kapadekar et al.	2008/0059743 A1	3/2008	Bychkov et al.
2007/0100981 A1	5/2007	Adameczyk et al.	2008/0060066 A1	3/2008	Wynn et al.
2007/0101426 A1	5/2007	Lee et al.	2008/0062900 A1	3/2008	Rao
2007/0104126 A1	5/2007	Calhoun et al.	2008/0064367 A1	3/2008	Nath et al.
2007/0104169 A1	5/2007	Polson	2008/0066149 A1	3/2008	Lim
2007/0109983 A1	5/2007	Shankar et al.	2008/0066150 A1	3/2008	Lim
2007/0111740 A1	5/2007	Wandel	2008/0066181 A1	3/2008	Haveson et al.
			2008/0070550 A1	3/2008	Hose
			2008/0077705 A1	3/2008	Li et al.
			2008/0080457 A1	4/2008	Cole
			2008/0081606 A1	4/2008	Cole

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0082643	A1	4/2008	Storrie et al.	2009/0005005	A1	1/2009	Forstall et al.
2008/0083013	A1	4/2008	Soliman et al.	2009/0006116	A1	1/2009	Baker et al.
2008/0085707	A1	4/2008	Fadell	2009/0006200	A1	1/2009	Baker et al.
2008/0089295	A1	4/2008	Keeler et al.	2009/0006229	A1	1/2009	Sweeney et al.
2008/0089303	A1	4/2008	Wirtanen et al.	2009/0013157	A1	1/2009	Beaule
2008/0095339	A1	4/2008	Elliott et al.	2009/0016310	A1	1/2009	Rasal
2008/0096559	A1	4/2008	Phillips et al.	2009/0017809	A1	1/2009	Jethi et al.
2008/0098062	A1	4/2008	Balia	2009/0036111	A1	2/2009	Danford et al.
2008/0109679	A1	5/2008	Wright et al.	2009/0042536	A1	2/2009	Bernard et al.
2008/0117958	A1	5/2008	Pattenden et al.	2009/0044185	A1	2/2009	Krivopaltsev
2008/0120129	A1	5/2008	Seubert et al.	2009/0046707	A1	2/2009	Smires et al.
2008/0120174	A1	5/2008	Li	2009/0046723	A1	2/2009	Rahman et al.
2008/0120668	A1	5/2008	Yau	2009/0047989	A1	2/2009	Harmon et al.
2008/0120688	A1	5/2008	Qiu et al.	2009/0048913	A1	2/2009	Shenfield et al.
2008/0122796	A1	5/2008	Jobs et al.	2009/0049156	A1	2/2009	Aronsson et al.
2008/0125079	A1	5/2008	O'Neil et al.	2009/0049518	A1	2/2009	Roman et al.
2008/0126287	A1	5/2008	Cox et al.	2009/0054030	A1	2/2009	Golds
2008/0127304	A1	5/2008	Ginter et al.	2009/0065571	A1	3/2009	Jain
2008/0130534	A1	6/2008	Tomioka	2009/0066999	A1	3/2009	Ito
2008/0130656	A1	6/2008	Kim et al.	2009/0067372	A1	3/2009	Shah et al.
2008/0132201	A1	6/2008	Karlberg	2009/0068984	A1	3/2009	Burnett
2008/0132268	A1	6/2008	Chol-Grogan et al.	2009/0070379	A1	3/2009	Rappaport
2008/0134330	A1	6/2008	Kapoor et al.	2009/0077622	A1	3/2009	Baum et al.
2008/0139210	A1	6/2008	Gisby et al.	2009/0077643	A1	3/2009	Schmidt et al.
2008/0147454	A1	6/2008	Walker et al.	2009/0079699	A1	3/2009	Sun
2008/0160958	A1	7/2008	Abichandani et al.	2009/0113514	A1	4/2009	Hu
2008/0162637	A1	7/2008	Adameczyk et al.	2009/0125619	A1	5/2009	Antani
2008/0162704	A1	7/2008	Poplett et al.	2009/0132860	A1	5/2009	Liu et al.
2008/0164304	A1	7/2008	Narasimhan et al.	2009/0149154	A1	6/2009	Bhasin et al.
2008/0166993	A1	7/2008	Gautier et al.	2009/0149165	A1	6/2009	Minborg et al.
2008/0167027	A1	7/2008	Gautier et al.	2009/0157792	A1	6/2009	Fiatl
2008/0167033	A1	7/2008	Beckers	2009/0163173	A1	6/2009	Williams
2008/0168275	A1	7/2008	DeAtley et al.	2009/0172077	A1	7/2009	Roxburgh et al.
2008/0168523	A1	7/2008	Ansari et al.	2009/0180391	A1	7/2009	Petersen et al.
2008/0177998	A1	7/2008	Apsangi et al.	2009/0181662	A1	7/2009	Fleischman et al.
2008/0178300	A1	7/2008	Brown et al.	2009/0197585	A1	8/2009	Aaron
2008/0181117	A1	7/2008	Acke et al.	2009/0197612	A1	8/2009	Kiiskinen
2008/0183812	A1	7/2008	Paul et al.	2009/0203352	A1	8/2009	Fordon et al.
2008/0184127	A1	7/2008	Rafey et al.	2009/0217065	A1	8/2009	Araujo, Jr.
2008/0189760	A1	8/2008	Rosenberg et al.	2009/0217364	A1	8/2009	Salmela et al.
2008/0201266	A1	8/2008	Chua et al.	2009/0219170	A1	9/2009	Clark et al.
2008/0207167	A1	8/2008	Bugenhagen	2009/0248883	A1	10/2009	Suryanarayana et al.
2008/0212470	A1	9/2008	Castaneda et al.	2009/0254465	A1	10/2009	Oester
2008/0212751	A1	9/2008	Chung	2009/0254857	A1	10/2009	Romine et al.
2008/0219268	A1	9/2008	Dennison	2009/0257379	A1	10/2009	Robinson et al.
2008/0221951	A1	9/2008	Stanforth et al.	2009/0262715	A1	10/2009	Juang
2008/0222692	A1	9/2008	Andersson et al.	2009/0265754	A1	10/2009	Hinds
2008/0225748	A1	9/2008	Khemani et al.	2009/0271514	A1	10/2009	Thomas et al.
2008/0229385	A1	9/2008	Feder et al.	2009/0282127	A1	11/2009	Leblanc et al.
2008/0229388	A1	9/2008	Maes	2009/0286507	A1	11/2009	O'Neil et al.
2008/0235511	A1	9/2008	O'Brien et al.	2009/0287921	A1	11/2009	Zhu et al.
2008/0240373	A1	10/2008	Wilhelm	2009/0288140	A1	11/2009	Huber et al.
2008/0244018	A1	10/2008	Chen et al.	2009/0291665	A1	11/2009	Gaskarth et al.
2008/0250053	A1	10/2008	Aaltonen et al.	2009/0299857	A1	12/2009	Brubaker
2008/0256593	A1	10/2008	Vinberg et al.	2009/0300450	A1	12/2009	Tzannes
2008/0259924	A1	10/2008	Gooch et al.	2009/0307696	A1	12/2009	Vals et al.
2008/0262798	A1	10/2008	Kim et al.	2009/0307746	A1	12/2009	Di et al.
2008/0263348	A1	10/2008	Zaltsman et al.	2009/0315735	A1	12/2009	Bhavani et al.
2008/0268813	A1	10/2008	Maes	2009/0320110	A1	12/2009	Nicolson et al.
2008/0270212	A1	10/2008	Blight et al.	2010/0017506	A1	1/2010	Fadell
2008/0279216	A1	11/2008	Sharif-Ahmadi et al.	2010/0020822	A1	1/2010	Zerillo et al.
2008/0282319	A1	11/2008	Fontijn et al.	2010/0027469	A1	2/2010	Gurajala et al.
2008/0291872	A1	11/2008	Henriksson	2010/0027559	A1	2/2010	Lin et al.
2008/0293395	A1	11/2008	Mathews et al.	2010/0030890	A1	2/2010	Dutta et al.
2008/0298230	A1	12/2008	Luft et al.	2010/0041364	A1	2/2010	Lott et al.
2008/0305793	A1	12/2008	Gallagher et al.	2010/0041365	A1	2/2010	Lott et al.
2008/0311885	A1	12/2008	Dawson et al.	2010/0041391	A1	2/2010	Spivey et al.
2008/0313315	A1	12/2008	Karaoguz et al.	2010/0042675	A1	2/2010	Fujii
2008/0313730	A1	12/2008	Ifimie et al.	2010/0043068	A1	2/2010	Varadhan et al.
2008/0316923	A1	12/2008	Fedders et al.	2010/0069074	A1	3/2010	Kodialam et al.
2008/0318547	A1	12/2008	Ballou et al.	2010/0071053	A1	3/2010	Ansari et al.
2008/0318550	A1	12/2008	DeAtley	2010/0075666	A1	3/2010	Garner
2008/0319879	A1	12/2008	Carroll et al.	2010/0077035	A1	3/2010	Li et al.
2008/0320497	A1	12/2008	Tarkoma et al.	2010/0080202	A1	4/2010	Hanson
2009/0005000	A1	1/2009	Baker et al.	2010/0082431	A1	4/2010	Ramer et al.
				2010/0088387	A1	4/2010	Calamera
				2010/0103820	A1	4/2010	Fuller et al.
				2010/0113020	A1	5/2010	Subramanian et al.
				2010/0121744	A1	5/2010	Belz et al.

(56)	References Cited			CN	1867024	A	11/2006
				CN	1878160	A	12/2006
	U.S. PATENT DOCUMENTS			CN	1937511	A	3/2007
				CN	101123553	A	9/2007
2010/0131584	A1	5/2010	Johnson	CN	101080055	A	11/2007
2010/0142478	A1	6/2010	Forssell et al.	CN	101115248	A	1/2008
2010/0144310	A1	6/2010	Bedingfield	CN	101127988	A	2/2008
2010/0151866	A1	6/2010	Karpov et al.	CN	101183958	A	5/2008
2010/0153781	A1	6/2010	Hanna	CN	101335666	A	12/2008
2010/0167696	A1	7/2010	Smith et al.	CN	101341764	A	1/2009
2010/0183132	A1	7/2010	Satyavolu et al.	CN	101815275	A	8/2010
2010/0188975	A1	7/2010	Raleigh	EP	1098490	A2	5/2001
2010/0188990	A1	7/2010	Raleigh	EP	1247411	A2	10/2002
2010/0188992	A1	7/2010	Raleigh	EP	1289326	A1	3/2000
2010/0188994	A1	7/2010	Raleigh	EP	1463238		9/2004
2010/0190469	A1	7/2010	Vanderveen et al.	EP	1503548	A1	2/2005
2010/0191576	A1	7/2010	Raleigh	EP	1545114	A1	6/2005
2010/0191612	A1	7/2010	Raleigh	EP	1739518		1/2007
2010/0191846	A1	7/2010	Raleigh	EP	1772988		4/2007
2010/0192170	A1	7/2010	Raleigh	EP	1850575	A1	10/2007
2010/0192212	A1	7/2010	Raleigh	EP	1887732	A1	2/2008
2010/0195503	A1	8/2010	Raleigh	EP	1942698	A1	7/2008
2010/0197268	A1	8/2010	Raleigh	EP	1978772		10/2008
2010/0198698	A1	8/2010	Raleigh et al.	EP	2007065	A1	12/2008
2010/0198939	A1	8/2010	Raleigh	EP	2026514	A1	2/2009
2010/0235329	A1	9/2010	Koren et al.	JP	3148713	B2	3/2001
2010/0241544	A1	9/2010	Benson et al.	JP	2005339247	A	12/2005
2010/0248719	A1	9/2010	Scholaert	JP	2006041989		2/2006
2010/0284327	A1	11/2010	Miklos	JP	2006155263	A	6/2006
2010/0284388	A1	11/2010	Fantini et al.	JP	2006197137		7/2006
2010/0287599	A1	11/2010	He et al.	JP	2006344007	A	12/2006
2010/0311402	A1	12/2010	Srinivasan et al.	JP	2007318354	A	12/2007
2010/0325420	A1	12/2010	Kanekar	JP	2008301121	A	12/2008
2011/0004917	A1	1/2011	Salsa et al.	JP	2009111919		5/2009
2011/0013569	A1	1/2011	Scherzer et al.	JP	2009212707	A	9/2009
2011/0019574	A1	1/2011	Malomsooky et al.	JP	2009218773		9/2009
2011/0081881	A1	4/2011	Baker et al.	JP	2009232107	A	10/2009
2011/0082790	A1	4/2011	Baker et al.	KR	20040053858	A	6/2004
2011/0110309	A1	5/2011	Bennett	KR	100958566	B1	5/2010
2011/0126141	A1	5/2011	King et al.	WO	1998058505		12/1998
2011/0145920	A1	6/2011	Mahaffey et al.	WO	1999027723		6/1999
2011/0159818	A1	6/2011	Scherzer et al.	WO	1999065185	A3	5/2001
2011/0173678	A1	7/2011	Kaippallimalil et al.	WO	0208863		1/2002
2011/0177811	A1	7/2011	Heckman et al.	WO	2002045315	A2	6/2002
2011/0182220	A1	7/2011	Black et al.	WO	2002067616	A1	8/2002
2011/0185202	A1	7/2011	Black et al.	WO	2002093877	A1	11/2002
2011/0244837	A1	10/2011	Murata et al.	WO	03017065	A2	2/2003
2011/0249668	A1	10/2011	Milligan et al.	WO	2003014891		2/2003
2011/0264923	A1	10/2011	Kocher et al.	WO	2003017063	A2	2/2003
2011/0277019	A1	11/2011	Pritchard, Jr.	WO	2003017065	A2	2/2003
2012/0011017	A1	1/2012	Wolcott et al.	WO	2003058880	A1	7/2003
2012/0020296	A1	1/2012	Scherzer et al.	WO	2004028070	A1	4/2004
2012/0144025	A1	6/2012	Melander et al.	WO	2004064306	A2	7/2004
2012/0166364	A1	6/2012	Ahmad et al.	WO	2004095753	A3	1/2005
2012/0196644	A1	8/2012	Scherzer et al.	WO	2005008995		1/2005
2012/0238287	A1	9/2012	Scherzer	WO	2005053335	A1	6/2005
2013/0029653	A1	1/2013	Baker et al.	WO	2005083934	A1	9/2005
2013/0058274	A1	3/2013	Scherzer et al.	WO	2006004467	A1	1/2006
2013/0065555	A1	3/2013	Baker et al.	WO	2006004784	A1	1/2006
2013/0072177	A1	3/2013	Ross et al.	WO	2006012610	A2	2/2006
2013/0084835	A1	4/2013	Scherzer et al.	WO	2006050758	A1	5/2006
2013/0144789	A1	6/2013	Aaltonen et al.	WO	2006077481	A1	7/2006
2013/0225151	A1	8/2013	King et al.	WO	2006093961	A1	9/2006
2013/0326356	A9	12/2013	Zheng et al.	WO	2006120558		11/2006
2014/0073291	A1	3/2014	Hildner et al.	WO	2006130960	A1	12/2006
2014/0241342	A1	8/2014	Constantinof	WO	2007001833	A2	1/2007
2015/0181628	A1	6/2015	Haverinen et al.	WO	2007014630	A1	2/2007
				WO	2007018363	A1	2/2007
				WO	2007053848	A1	5/2007
				WO	2007068288		6/2007
				WO	2007097786	A	8/2007
CN	1345154	A	4/2002	WO	2007107701	A2	9/2007
CN	1508734	A	6/2004	WO	2007120310		10/2007
CN	1538730	A	10/2004	WO	2007124279		11/2007
CN	1567818	A	1/2005	WO	2007126352		11/2007
CN	101035308	A	3/2006	WO	2007129180	A2	11/2007
CN	1801829	A	7/2006	WO	2007133844	A	11/2007
CN	1802839	A	7/2006	WO	2004077797	A3	2/2008
CN	1889777	A	7/2006	WO	2008017837	A1	2/2008
CN	1011155343	B	9/2006				

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2008051379	A2	5/2008
WO	2008066419	A1	6/2008
WO	2008080139	A1	7/2008
WO	2008080430	A1	7/2008
WO	2008099802	A1	8/2008
WO	2009002949	A	12/2008
WO	2009008817	A1	1/2009
WO	2009002949	5	3/2009
WO	2006073837	A3	4/2009
WO	2007069245	A3	4/2009
WO	2009091295	A1	7/2009
WO	2010088413	A1	8/2010
WO	2010128391	A2	11/2010
WO	2010128391	A3	1/2011
WO	2011002450	A1	1/2011

OTHER PUBLICATIONS

Arm TrustZone Paper, TrustZone: Integrated Hardware and Software Security, dated Jul. 2004.

Limont Prosecution History Excerpt, U.S. Appl. No. 11/171,850, filed Jun. 30, 2005.

Plaintiff's Infringement Contention in Case No. 6:23-CV-00352-JRG-RSP, *Headwater Research LLC v. Celco Partnership, d/b/a Verizon Wireless, Verizon Corporate Services Group Inc.* (E.D. Tex, filed Jul. 28, 2023).

IPR2024-00809 Petition for Inter Partes Review of U.S. Pat. No. 9,198,042, filed Apr. 19, 2024.

IPR2024-00809 File History of Inter Partes Review of U.S. Pat. No. 9,198,042, filed Apr. 19, 2024.

"Ads and movies on the run," the Gold Coast Bulletin, Southport, Qld, Jan. 29, 2008.

"ASA/PIX: Allow Split Tunneling for VPN Clients on the ASA Configuration Example," Document ID 70917, Jan. 10, 2008.

"Communication Concepts for Mobile Agent Systems," by Joachim Baumann et al.; Inst. Of Parallel and Distributed High-Performance Systems, Univ. of Stuttgart, Germany, pp. 123-135, 1997.

"End to End QoS Solution for Real-time Multimedia Application;" Computer Engineering and Applications, 2007, 43 (4): 155-159, by Tan Zu-guo, Wang Wen-juan; Information and Science School, Zhanjian Normal College, Zhan jiang, Guangdong 524048, China. "Jentro Technologies launches Zenlet platform to accelerate location-based content delivery to mobile devices," The Mobile Internet, Boston, MA, Feb. 2008.

"The Construction of Intelligent Residential District in Use of Cable Television Network," Shandong Science, vol. 13, No. 2, Jun. 2000. 3rd Generation Partnership Project, "Technical Specification Group Services and System Aspects; General Packet Radio Service (GPRS) Enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Access," Release 8, Document No. 3GPP TS 23.401, V8.4.0, Dec. 2008.

3rd Generation Partnership Project, "Technical Specification Group Services and System Aspects; Policy and Charging Control Architecture," Release 8, Document No. 3GPP TS 23.203, V8.4.0, Dec. 2008.

Accuris Networks, "The Business Value of Mobile Data Offload—a White Paper", 2010.

Ahmed et al., "A Context-Aware Vertical Handover Decision Algorithm for Multimode Mobile Terminals and Its Performance," BenQ Mobile, Munich Germany; University of Klagenfurt, Klagenfurt, Austria; 2006.

Alonistioti et al., "Intelligent Architectures Enabling Flexible Service Provision and Adaptability," 2002.

Amazon Technologies, Inc., "Kindle™ User's Guide," 3rd Edition, Copyright 2004-2009.

Android Cupcake excerpts, The Android Open Source Project, Feb. 10, 2009.

Anton, B. et al., "Best Current Practices for Wireless Internet Service Provider (WISP) Roaming"; Release Date Feb. 2003, Version 1.0; Wi-Fi Alliance—Wireless ISP Roaming (WISPr).

Blackberry Mobile Data System, version 4.1, Technical Overview, 2006.

Byrd, "Open Secure Wireless," May 5, 2010.

Chandrasekhar et al., "Femtocell Networks: A Survey," Jun. 28, 2008.

Chaouchi et al., "Policy Based Networking in the Integration Effort of 4G Networks and Services," 2004 IEEE.

Cisco Systems, Inc., "Cisco Mobile Exchange (CMX) Solution Guide: Chapter 2—Overview of GSM, GPRS, and UMTS," Nov. 4, 2008.

Client Guide for Symantec Endpoint Protection and Symantec Network Access Control, 2007.

Dikaiaikos et al., "A Distributed Middleware Infrastructure for Personalized Services," Nov. 24, 2003.

Dixon et al., Triple Play Digital Services: Comcast and Verizon (Digital Phone, Television, and Internet), Aug. 2007.

Droid Wall 1.3.7 description 20100428 obtained from <https://www.freewarelovers.com/android/apps/droid-wall>.

Ehnert, "Small application to monitor IP traffic on a Blackberry—1.01.03", Mar. 27, 2008; <http://www.ehnert.net/MiniMoni/>.

European Commission, "Data Roaming Tariffs—Transparency Measures," obtained from Europa—Europe's Information Society Thematic Portal website, Jun. 24, 2011: "http://ec.europa.eu/information_society/activities/roaming/data/measures/index_en.htm."

Farooq et al., "An IEEE 802.16 WiMax Module for the NS-3 Simulator," Mar. 2-6, 2009.

Fujitsu, "Server Push Technology Survey and Bidirectional Communication in HTTP Browser," Jan. 9, 2008 (JP).

Han et al., "Information Collection Services for Qos-Aware Mobile Applications," 2005.

Hartmann et al., "Agent-Based Banking Transactions & Information Retrieval—What About Performance Issues?" 1999.

Hewlett-Packard Development Company, LP, "IP Multimedia Services Charging," white paper, Jan. 2006.

Hossain et al., "Gain-Based Selection of Ambient Media Services in Pervasive Environments," Mobile Networks and Applications. Oct. 3, 2008.

Jing et al., "Client-Server Computing in Mobile Environments," GTE Labs. Inc., Purdue University, ACM Computing Surveys, vol. 31, No. 2, Jun. 1999.

Kasper et al., "Subscriber Authentication in mobile cellular Networks with virtual software SIM Credentials using Trusted Computing," Fraunhofer—Institute for Secure Information Technology SIT, Darmstadt, Germany; ICACT 2008.

Kassar et al., "An overview of vertical handover decision strategies in heterogeneous wireless networks," ScienceDirect, University Pierre & Marie Curie, Paris, France, Jun. 5, 2007.

Kim, "Free wireless a high-wire act; MetroFi needs to draw enough ads to make service add profits," San Francisco Chronicle, Aug. 21, 2006.

Knight et al., "Layer 2 and 3 Virtual Private Networks: Taxonomy, Technology, and Standardization Efforts," IEEE Communications Magazine, Jun. 2004.

Koutsopoulou et al., "Charging, Accounting and Billing Management Schemes In Mobile Telecommunication Networks and the Internet," IEEE Communications Surveys & Tutorials, First Quarter 2004, vol. 6, No. 1.

Koutsopoulou et al., "Middleware Platform for the Support of Charging Reconfiguration Actions," 2005.

Kuntze et al., "Trustworthy content push," Fraunhofer—Institute for Secure Information Technology SIT; Germany; WCNC 2007 proceedings, IEEE.

Kyriakakos et al., "Ubiquitous Service Provision in Next Generation Mobile Networks," Proceedings of the 13th IST Mobile and Wireless Communications Summit, Lyon, France, Jun. 2004.

Li, Yu, "Dedicated E-Reading Device: The State of the Art and The Challenges," Scroll, vol. 1, No. 1, 2008.

Loopt User Guide, metroPCS, Jul. 17, 2008.

Muntermann et al., "Potentiale und Sicherheitsanforderungen mobiler Finanzinformationsdienste und deren Systeminfrastrukturen," Chair of Mobile Commerce & Multilateral Security, Goethe Univ. Frankfurt, 2004.

(56)

References Cited

OTHER PUBLICATIONS

- NetLimiter Lite 4.0.19.0; <http://www.heise.de/download/netlimiter-lite-3617703.html> from vol. 14/2007.
- Nilsson et al., "A Novel MAC Scheme for Solving the QoS Parameter Adjustment Problem in IEEE802.11e EDCA," Feb. 2006.
- Nuzman et al., "A compound model for TCP connection arrivals for LAN and WAN applications," Oct. 22, 2002.
- Open Mobile Alliance (OMA), Push Architecture, Candidate Version 2.2; Oct. 2, 2007; OMA-AD-Push-V2_2-20071002-C.
- Oppliger, Rolf, "Internet Security: Firewalls and Beyond," Communications of the ACM, May 1997, vol. 40, No. 5.
- Rao et al., "Evolution of Mobile Location-Based Services," Communication of the ACM, Dec. 2003.
- Richtel, "Cellphone consumerism; If even a debit card is too slow, now you have a new way to act on impulse: [National Edition]," National Post, Canada, Oct. 2, 2007.
- Rivadeneira et al., "A communication architecture to access data services through GSM," San Sebastian, Spain, 1998.
- Ruckus Wireless—White Paper; "Smarter Wi-Fi for Mobile Operator Infrastructures" 2010.
- Sabat, "The evolving mobile wireless value chain and market structure," Nov. 2002.
- Sadeh et al., "Understanding and Capturing People's Privacy Policies in a Mobile Social Networking Application," ISR School of Computer Science, Carnegie Mellon University, 2007.
- Schiller et al., "Location-Based Services," The Morgan Kaufmann Series in Data Management Systems, 2004.
- Sharkey, "Coding for Life—Battery Life, That Is," May 27, 2009.
- Steglich, Stephan, "I-Centric User Interaction," Nov. 21, 2003.
- Sun et al., "Towards Connectivity Management Adaptability: Context Awareness in Policy Representation and End-to-end Evaluation Algorithm," Dept. of Electrical and Information Engineering, Univ. of Oulu, Finland, 2004.
- Van Eijk, et al., "GigaMobile, Agent Technology for Designing Personalized Mobile Service Brokerage," Jul. 1, 2002.
- VerizonWireless.com news, "Verizon Wireless Adds to Portfolio of Consumer-Friendly Tools With Introduction of Usage Controls, Usage Controls and Chaperone 2.0 Offer Parents Full Family Security Solution," Aug. 18, 2008.
- Windows7 Power Management, published Apr. 2009.
- Wireless Broadband Alliance, "WISPr 2.0, Apr. 8, 2010"; Doc. Ref. No. WBA/RM/WISPr, Version 01.00.
- Zhu et al., "A Survey of Quality of Service in IEEE 802.11 Networks," IEEE Wireless Communications, Aug. 2004.
- Complaint, *Headwater Research LLC v. Samsung Elec-tronics Co., Ltd. et al.*, 2-24-cv-00228, E.D. Tex., filed Apr. 3, 2024.
- (Excerpts) Smith, et al., 2005. "Virtual Machines: Versatile Platforms for Systems and Processes," Elsevier, Inc, 2005, ISBN 1-55860-910-5.
- (Excerpts) Telecom Dictionary, Athos Publishing, 2007.
- (Excerpts) Eberspächer, Jörg (2001). GSM Switching, Services and Protocols, Second Edition. John Wiley & Sons Ltd. ISBN: 978-0-470-85394-8.
- 3rd Generation Partnership Project; Technical Specification Group Terminals; "Characteristics of the USIM application" (Release 7), 3GPP TS 31.102 V7.0.0 ("3GPP USIM").
- Kasper, et al., 2008, February. "Subscriber authentication in cellular networks with trusted virtual sims." In 2008 10th International Conference on Advanced Communication Technology (vol. 2, pp. 903-908). IEEE.
- TCG Mobile Reference Architecture, version 1.0, Revision 1, Jun. 12, 2007. ("TCG Mobile Reference Architecture").
- TCG Mobile Trusted Module Specification, version 1.0, Revision 6, Jun. 26, 2008. ("TCG Mobile Trusted Module Specification").
- Stone, G.N., Lundy, B. and Xie, G.G., 2001. Network policy languages: a survey and a new approach. IEEE network, 15(1), pp. 10-21.
- David K. Gifford. 1982. Cryptographic sealing for information secrecy and authentication. Commun. ACM 25, 4 (Apr. 1982), 274-286. <https://doi.org/10.1145/358468.358493>.
- Jansen, Wayne A. and Richard P. Ayers. "Forensic Tools for Mobile Phone Subscriber Identity Modules." J. Digit. Forensics Secur. Law 1 (2006): 75-94.
- National Institute of Standards and Technology. 2001. Security Requirements for Cryptographic Modules, downloaded from the Internet at <https://nvl-pubs.nist.gov/nistpubs/FIPS/NIST.FIPS.140-2.pdf> on Dec. 5, 2024.
- Verma, et al., (2002). Policy-based management of content distribution networks. IEEE network, 16(2), 34-39.
- Lobo, et al., (1999). A policy description language. AAAI/IAAI, 1999, 291-298.
- Westerinen, et al., IETF RFC 3198, Terminology for Policy-Based Management, Nov. 2001, downloaded from the Internet on May 27, 2024.
- (Excerpts) Keith Mayes and Konstantinos Markantonakis. 2008. Smart Cards, Tokens, Security and Applications (1st. ed.).
- (Excerpts) Gasser, Morrie. Building a Secure Computer System. New York, NY: Van Nostrand Reinhold, 1988. ("Gasser").
- (Excerpts) Malhotra, Ravi. 2002. IP Routing: Help for Network Administrators. O'Reilly Media. ISBN: 978-0-596-00275-0 ("Malhotra").
- Jude, Michael. "Policy-Based Management: Beyond the Hype." Business Communications Review 31.3 (2001): 52-56. ("Jude").
- Merkle, Ralph C. 1978. Secure communications over insecure channels. Commun. ACM 21, 4 (Apr. 1978), 294-299. <https://doi.org/10.1145/359460.359473> ("Merkle").
- ARM. 2004. PrimeCell Infrastructure AMBA 3 TrustZone Protection Controller (BP147) Revision: r0p0 Technical Overview, downloaded from the Internet at https://documentation-ser-vice.arm.com/static/5e9565afc8052b1608762aae%3Fto-ken%3D&ved=2ahUKEwiZq56_3pGKAxVUCnkGHcR9OGIQFnoECAwQAQ&usq=AOv-Vaw2PG8jUG9TU8fpiRumGKNyM on Dec. 5, 2024.
- Network Associates, Inc. 1999. PGP, Version 6.5.1. An Introduction to Cryptography.
- Stuart E. Madnick and John J. Donovan. 1973. Application and analysis of the virtual machine approach to information system security and isolation. In Proceedings of the workshop on virtual computer systems. Association for Computing Machinery, New York, NY, USA, 210-224. <https://doi.org/10.1145/800122.803961>.
- European Telecommunications Standards Institute. 1998. Terrestrial Trunked Radio (TETRA); Security Aspects; Subscriber Identity Module to Mobile Equipment (SIM—ME) interface. ETSI ETS 300 812 ed.1 (Nov. 1998), downloaded from the Internet at https://www.etsi.org/de-liver/etsi_i_ets/300800_300899/300812/01_20_9826/ets_300812e01c.pdf on Dec. 12, 2024.
- IETF RFC 1122, Requirements for Internet Hosts—Communication Layers, Oct. 1989, downloaded from the internet at <https://datacenter.ietf.org/doc/html/rfc1122> on Dec. 12, 2024.
- IETF RFC 793, Transmission Control Protocol, Sep. 1981, downloaded from the internet at <https://www.ietf.org/rfc/rfc793.txt> on Dec. 11, 2024.
- Smith, et al., 2005. "The architecture of virtual machines. Computer," 38(5).
- ISO/IEC 7498-1, "Information technology—Open Systems Interconnection—Basic Reference Model: The Basic Model," downloaded from the internet at <https://www.ecma-international.org/wp-content/uploads/s020269e.pdf> on Jan. 10, 2025.
- Gonçalves, et al., 2009, October. A graphical user interface for policy composition in CIM-SPL. In 2009 International Conference on Ultra Modern Telecommunications & Workshops (pp. 1-7). IEEE.
- Agrawal, et al., 2007, May. Issues in designing a policy language for distributed management of IT infrastructures. In 2007 10th IFIP/IEEE International Symposium on Integrated Network Management (pp. 30-39). IEEE.
- File History of IPR2025-00482, filed Jan. 28, 2025.
- Petition for Inter Partes Review in IPR2025-00482, filed Jan. 28, 2025.
- Declaration and Curriculum Vitae of Dr. Patrick McDaniel.
- Complaint, *Headwater Research LLC v. Samsung Electronics Co., Ltd. et al.*, 2-24-cv-00627, E.D. Tex., filed Aug. 2, 2024.

(56)

References Cited**OTHER PUBLICATIONS**

Excerpts from Smith, et al., 2005. "Virtual Machines: Versatile Platforms for Systems and Processes," Elsevier, Inc, 2005, ISBN 1-55860-910-5.

Declaration of June Munford.

Mallick, M., *Mobile and Wireless Design Essentials* (2003).

Anderson, R., *Security Engineering: A Guide to Building Dependable Distributed Systems*. (2001).

Jaeger, T., *Operating System Security (Synthesis Lectures on Information Security, Privacy, and Trust)* (2008).

Parziale, L., *TCP/IP Tutorial and Technical Overview* (IBM Redbooks) (2006).

McCanne, S. et al., *The BSD Packet Filter: A New Architecture for User-level Packet Capture* (1992).

Horak, R., *Webster's New World Telecom Dictionary*, (2008).

Ganz, A., et al., *Multimedia Wireless Networks: Technologies, Standards, and QoS* (2003).

Kwok, Y., et al., *Wireless Internet and Mobile Computing: Interoperability and Performance* (2007).

Brownlee et al., *Traffic Flow Measurement: Architecture* (1997).

Saltzer, J. H., *The Protection of Information in Computer Systems*, *Proceedings of the IEEE* (1975).

Tan, et al., *An Empirical Study on the Capacity and Performance of 3G Networks* (2008).

Wang, et al., *Shield: Vulnerability-Driven Network Filters for Preventing Known Vulnerability Exploits*, *ACM SIGCOMM Computer Communication Review* (2004).

Silberschatz et al., *Operating System Concepts* (7th ed. 2005).

Cheswick et al., *Firewalls and Internet Security* (2nd ed. 2003).

Koren et al., *Fault Tolerant Systems* (2007).

Loseri et al., *Voice and Video Telephony Services in Smartphone*, *EURASIP Journal on Wireless Communications and Networking* (2006).

Oxford University Press, *A Dictionary of Computing*, (6th ed. 2008) (defining "Agent").

Downing et al., *Dictionary of Computer and Internet Terms* (9th ed. 2006) (defining "Agent").

Kunie et al., "Low power architecture and design techniques for mobile handset LSI Medity M2," 748-753, 2008 Asia and South Pacific Design Automation Conference (2008).

M. Naruse et al., "A 65nm Single-Chip Application and Dual-Mode Baseband Processor with Partial Clock Activation and IP-MMU," 260-612, 2008 IEEE International Solid-State Circuits Conference—Digest of Technical Papers (2008).

Gertz et al., *Handbook of Database Security: Applications and Trends*, Springer (2007).

Oney, *Programming The Microsoft Windows Driver Model* (2nd ed. 2003).

Exhibit A, *Headwater Research LLC v. Samsung Electronics Co., Ltd. et al.*, 2-24-cv-00627, E.D. Tex., filed Aug. 2, 2024.

IEEE 100, *The Authoritative Dictionary of IEEE Standards Terms* (7th ed. 2000).

Tanenbaum et al., *Modern Operating Systems* (2001).

The Secure Sockets Layer ("SSL") Protocol, V. 3.0, available at <https://web.archive.org/web/19970614041044/http://home.netscape.com/eng/ssl3/ssl-toc.html> and <https://web.archive.org/web/19970617034012/http://home.netscape.com/eng/ssl3/3-SPEC.HTM#1>.

IPR2025-00963 Petition For Inter Partes Review Of U.S. Pat. No. 10,028,144, filed May 5, 2025.

File History of IPR2025-00963 for Inter Partes Review Of U.S. Pat. No. 10,028,144.

* cited by examiner

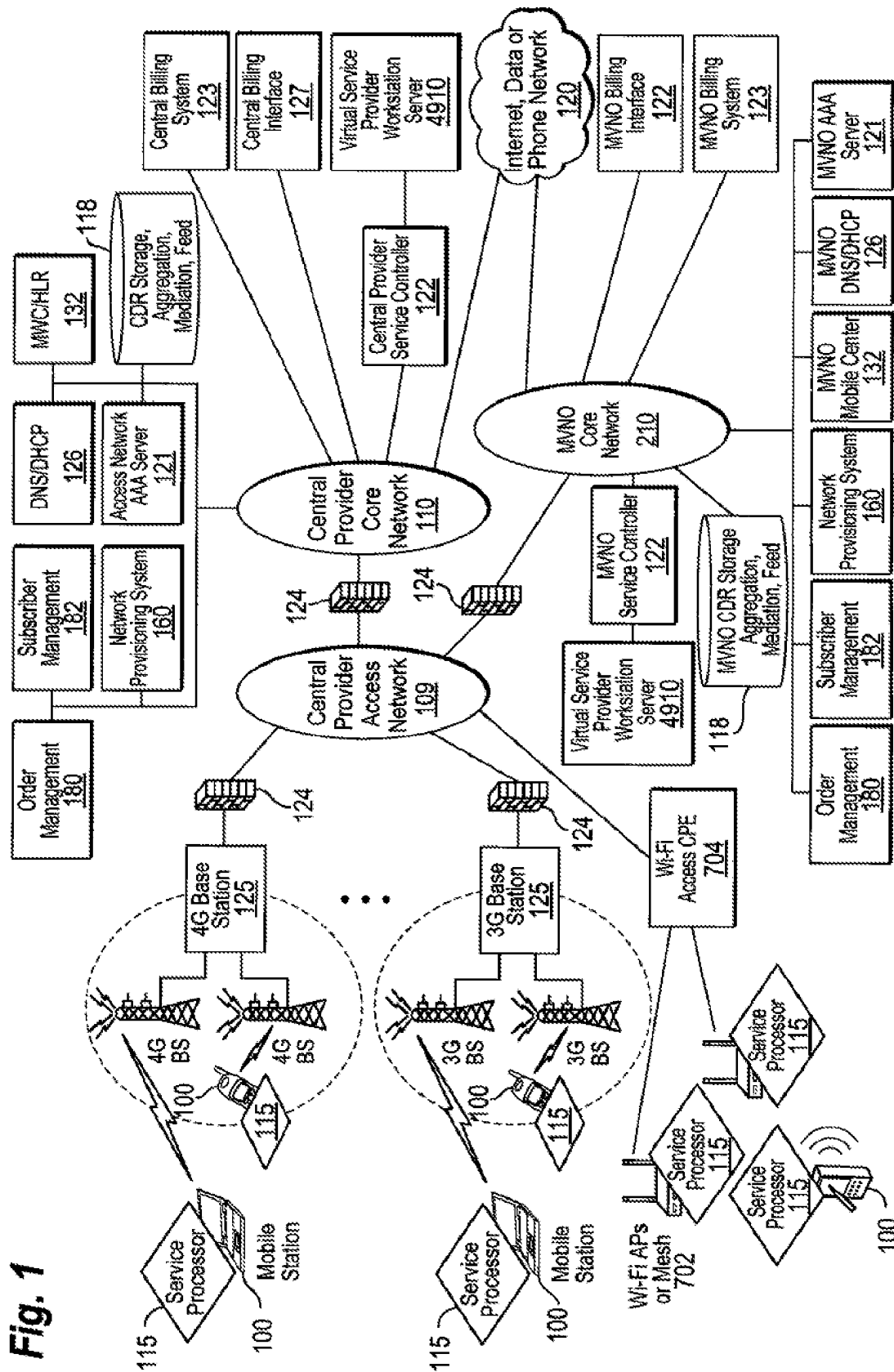
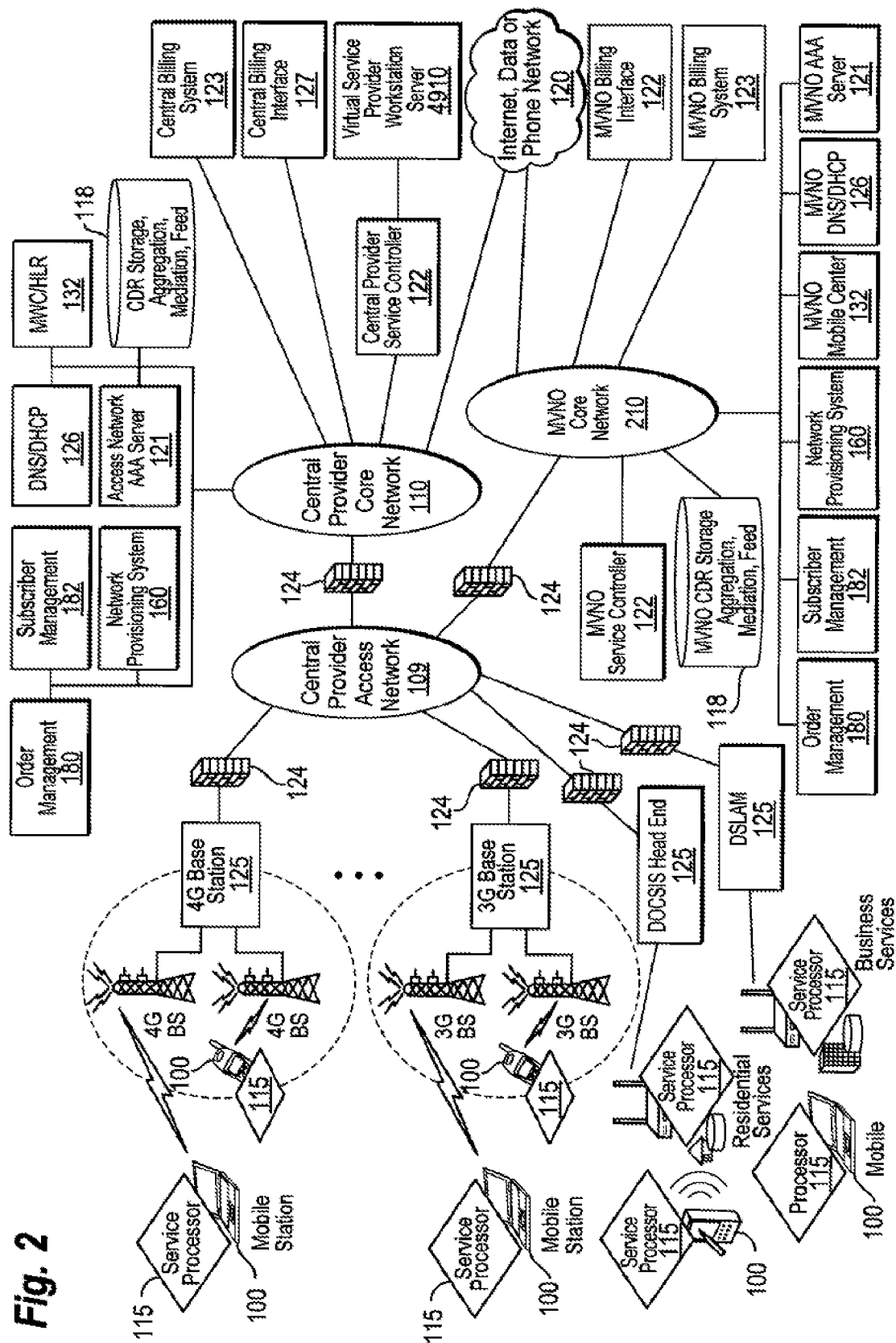


Fig. 2



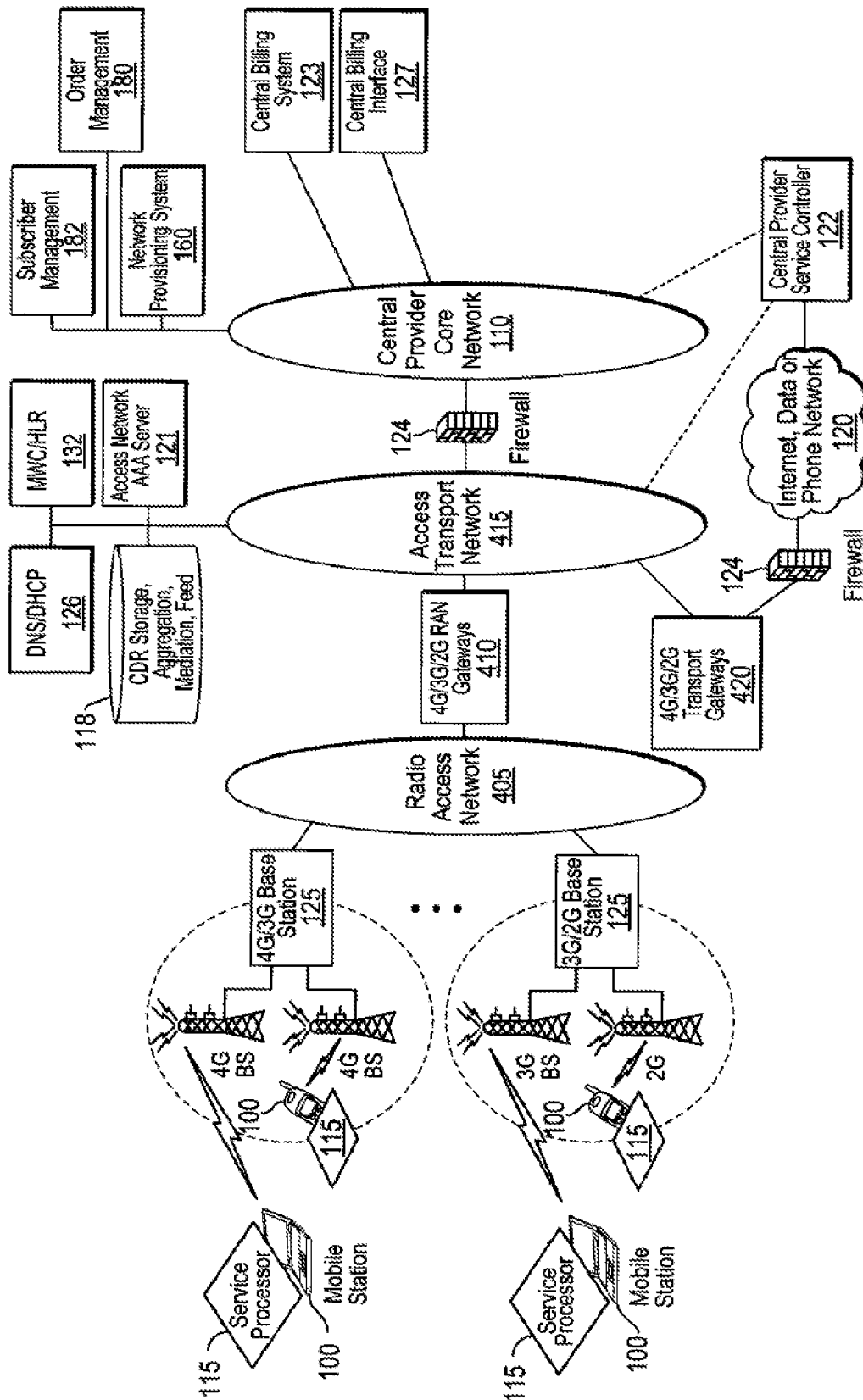
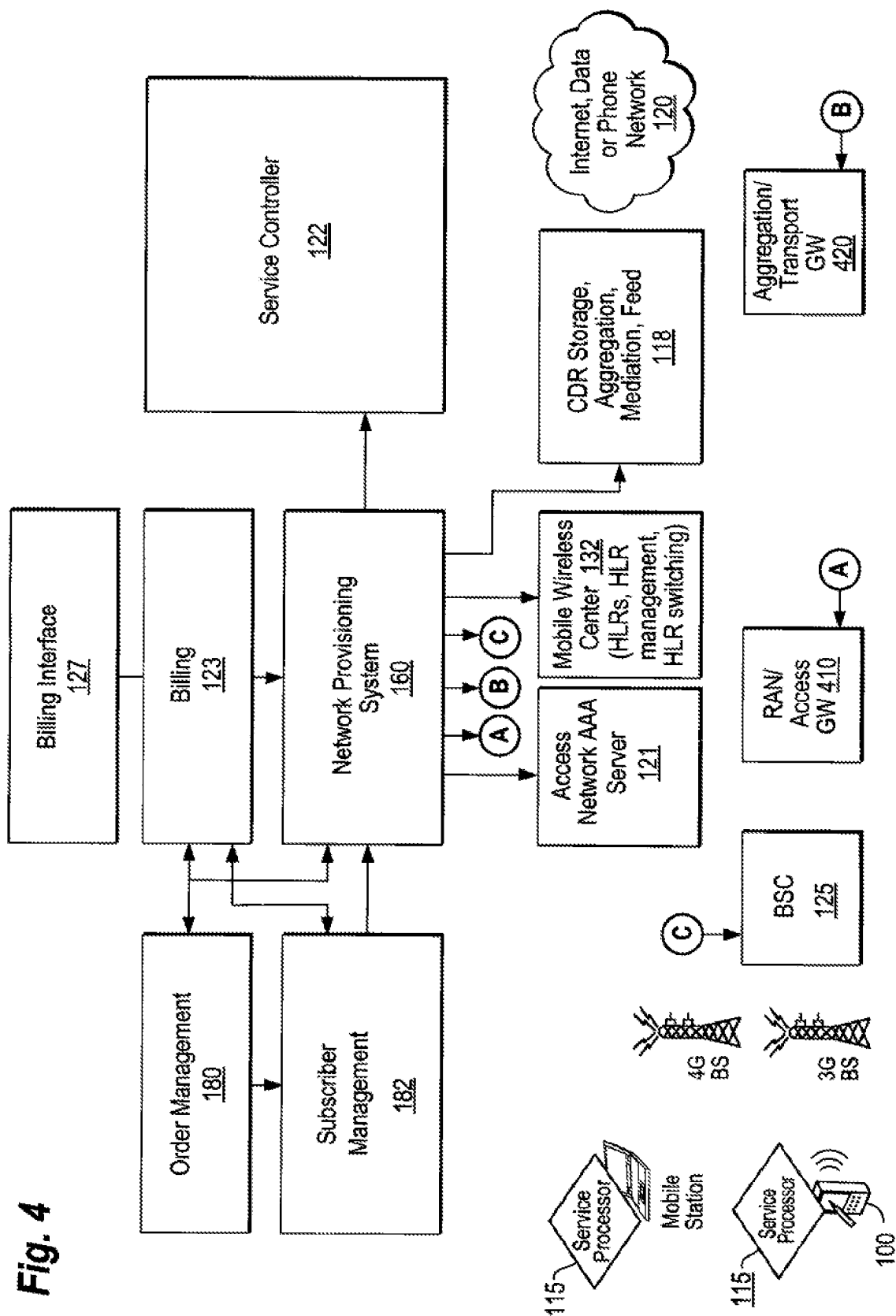


Fig. 3

Fig. 4



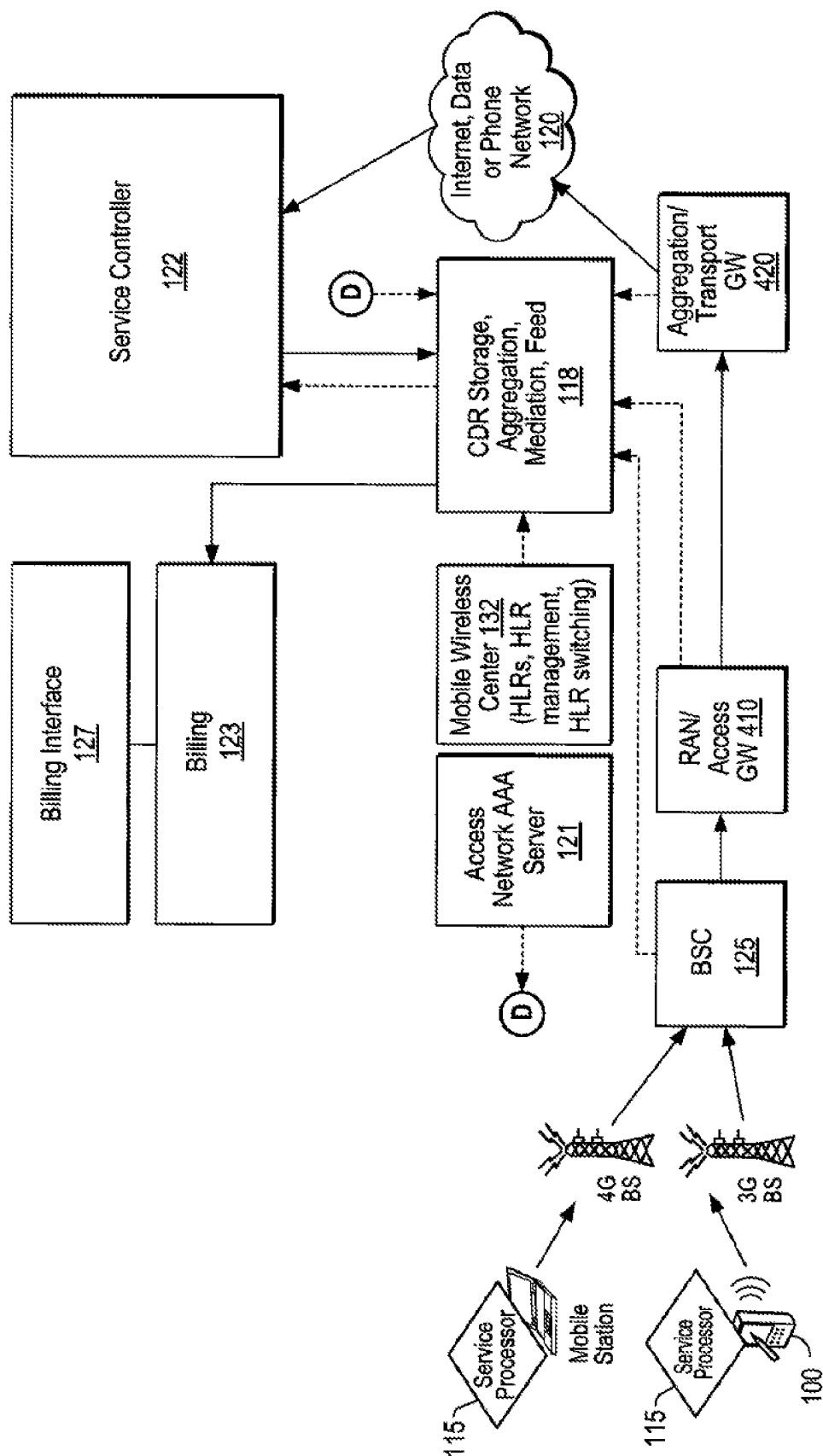


Fig. 5

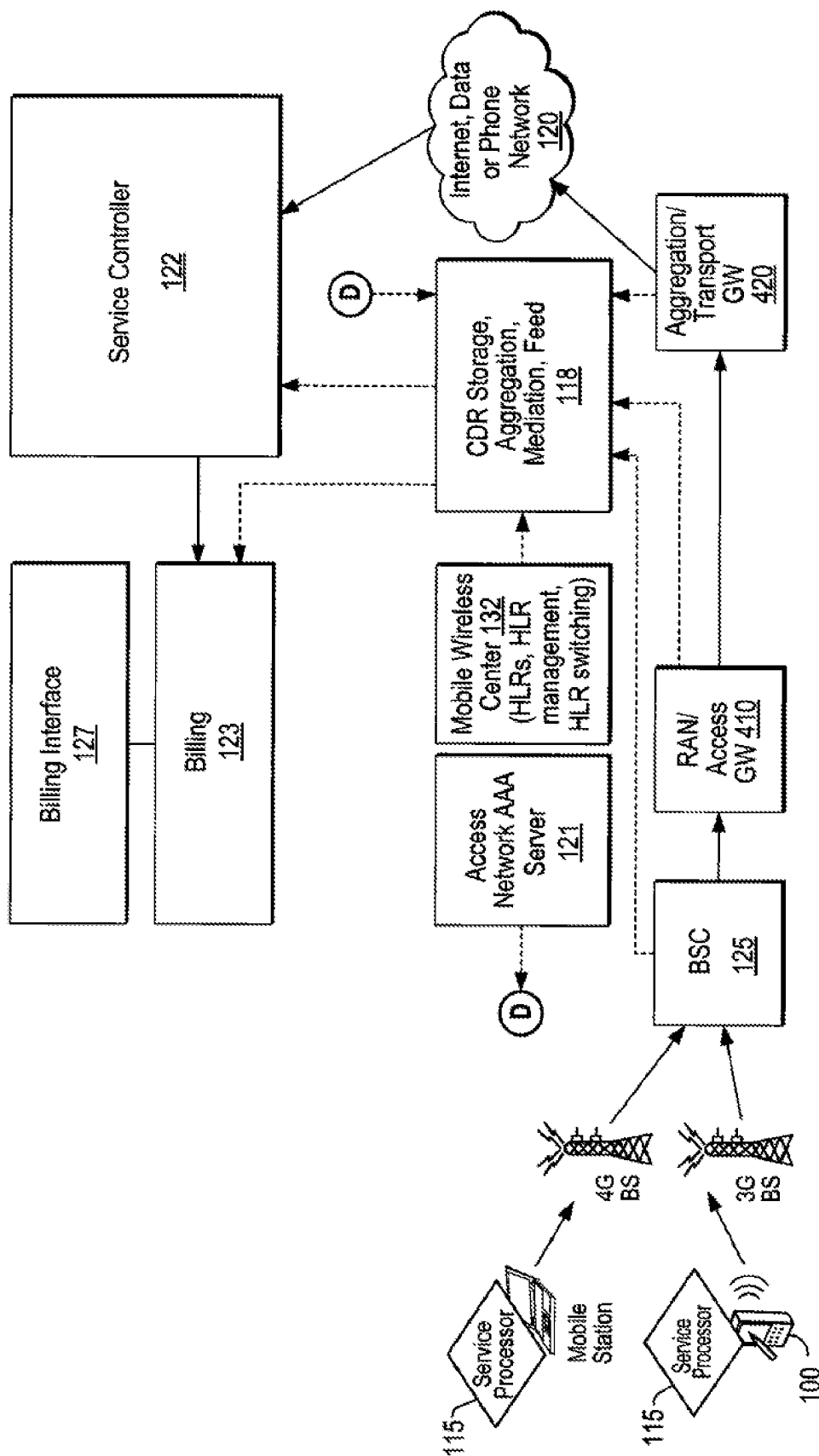


Fig. 6

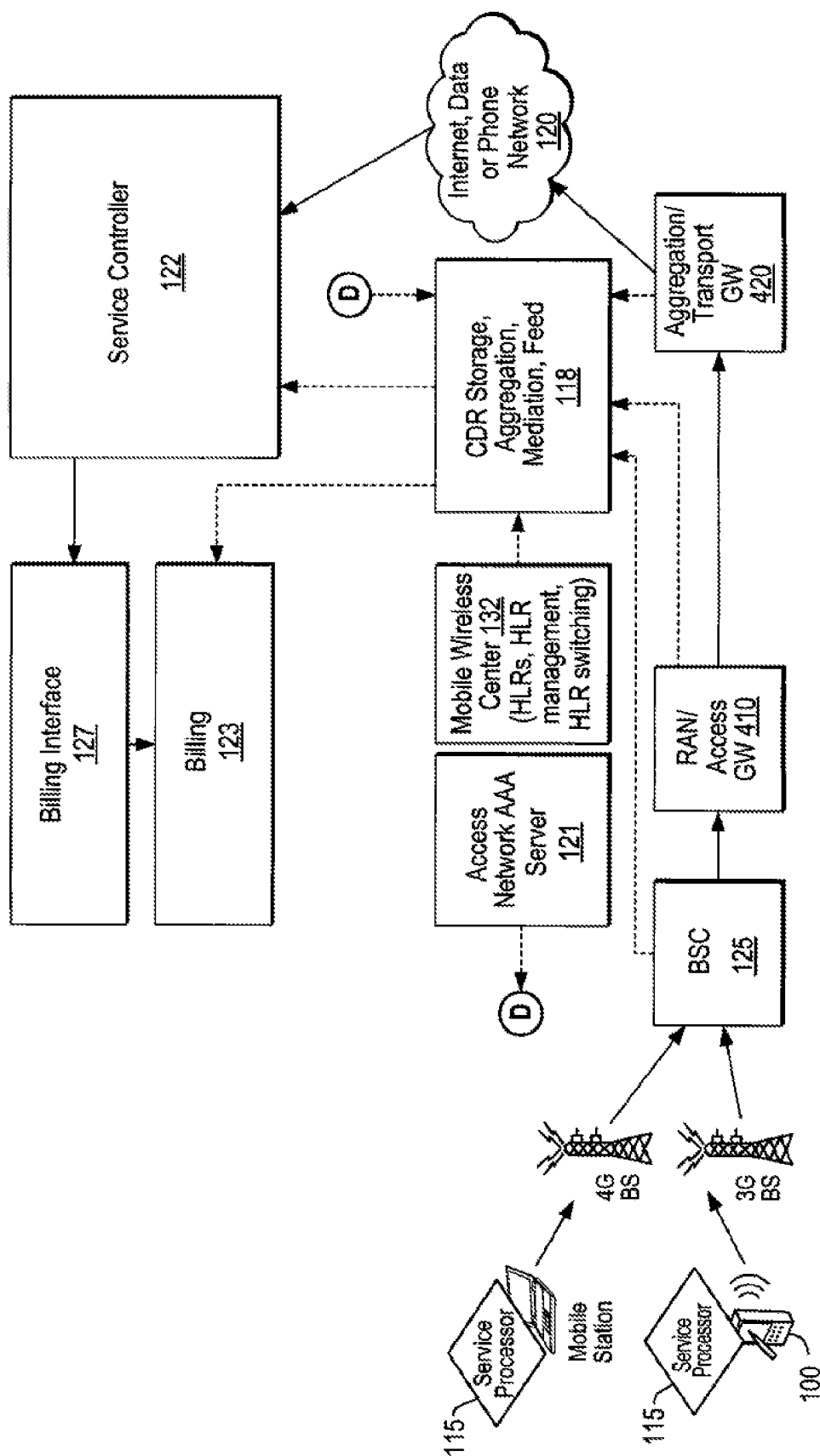


Fig. 7

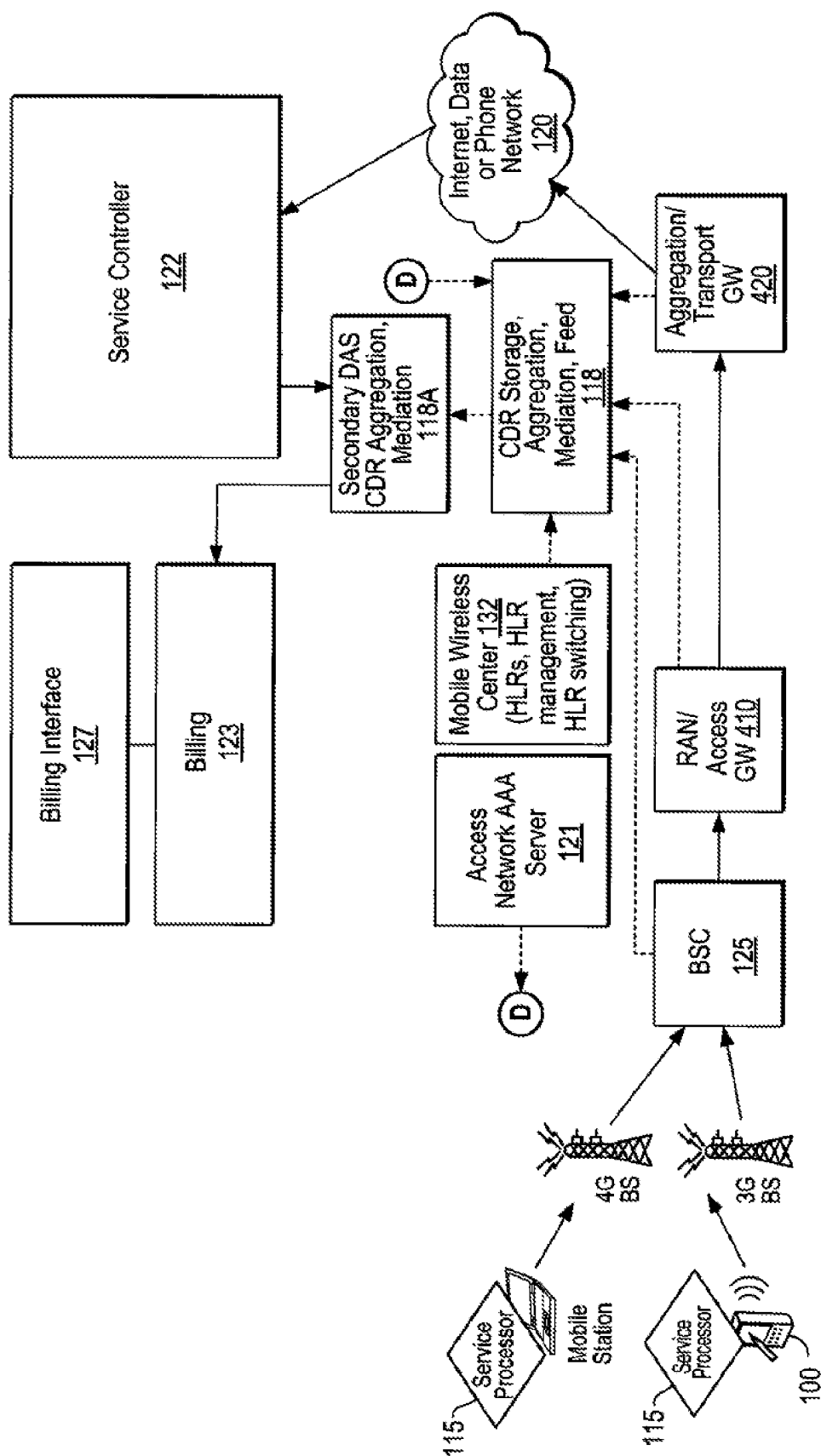


Fig. 8

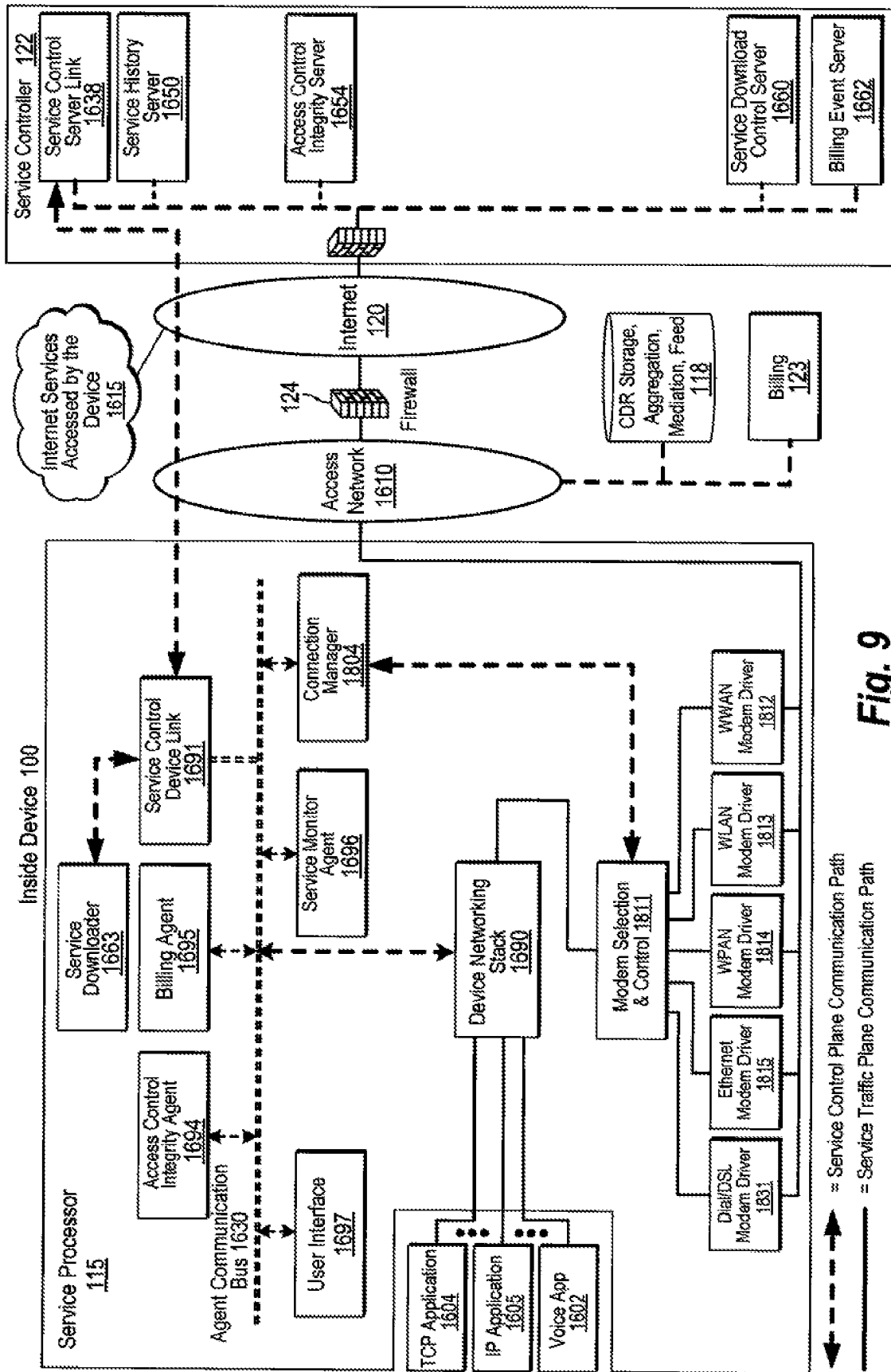


Fig. 9

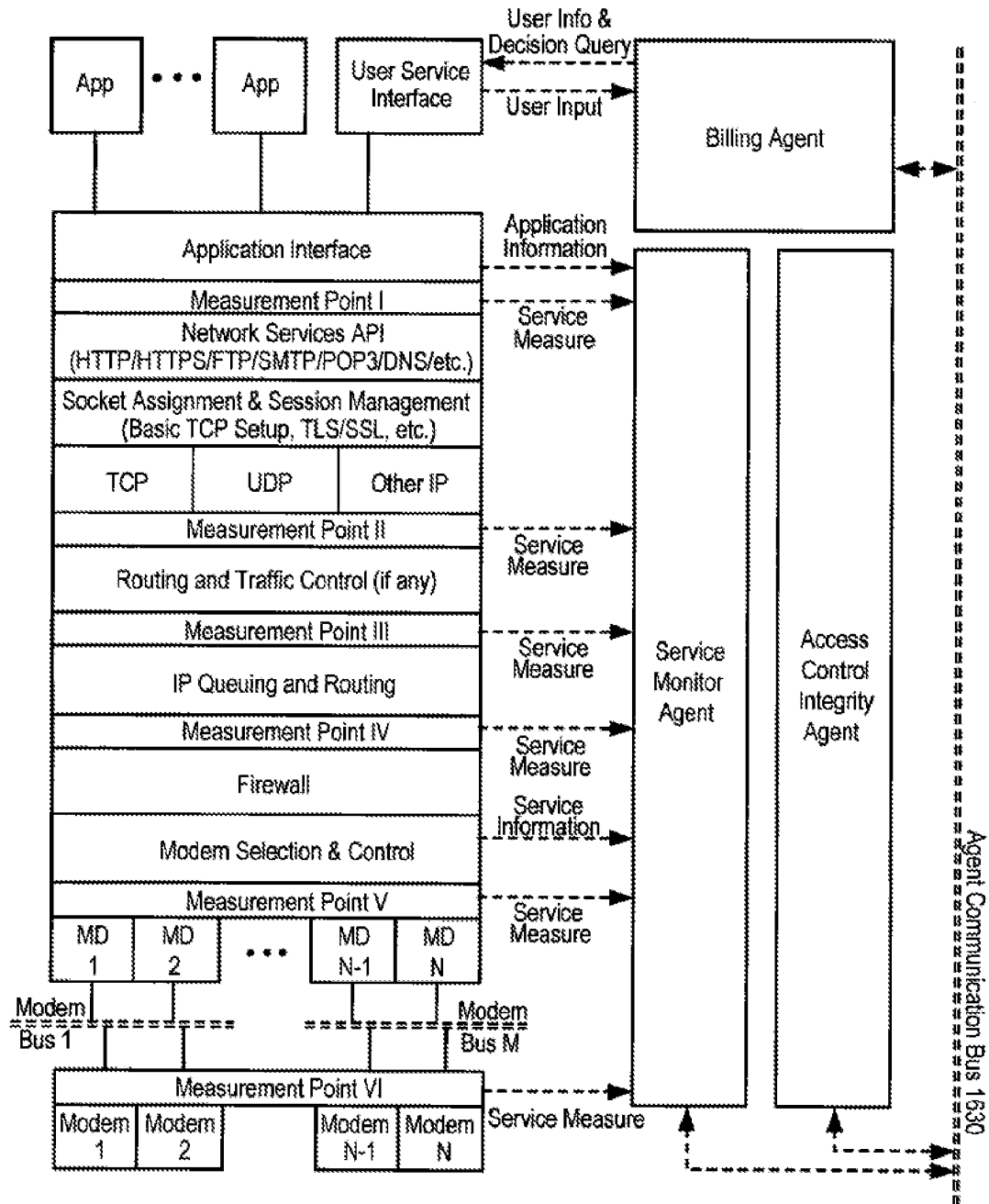
--- = Service Control Plane Communication Path
 — = Service Traffic Plane Communication Path

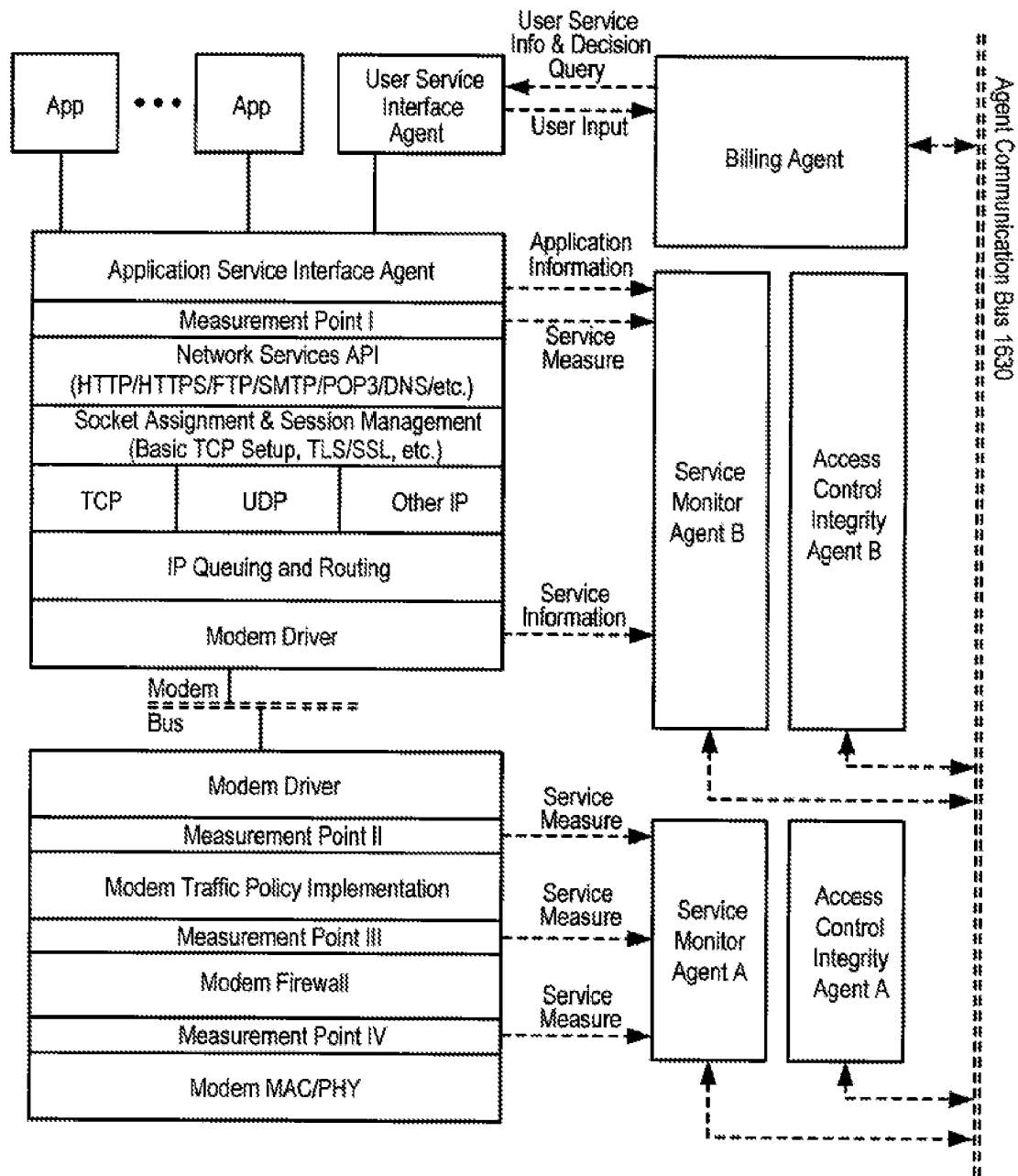
Service Processor 115 Embodiment	Partial Summary of Embodiment Functions
Service Control Device Link 1691	Device side control plane link for connecting Service Processor 415 to Service Controller. In some embodiments, also serves as the link for the agent heartbeat function.
Access Control Integrity Agent 1694	Collects device information on service policy, service usage, agent configuration and agent behavior. Cross checks this information to identify integrity breaches in the service policy implementation and control system. Initiates action when a service policy violation or a system integrity breach is suspected. In some embodiments, verifies configuration of other agents or performs challenge-response sequence testing. In some embodiments, monitors software loading activity, protected memory access or communication with Service Processor agents to detect unauthorized changes to Service Processor software or configuration.
Service Monitor Agent 1696	Records and reports device service usage. In some embodiments, assists in communicating application tagging of traffic flows through the networking stack policy implementation. In some embodiments, maintains a history and provides reports or summary reports of which networks in addition to the networks controlled by the Service Controller that the device has connected to. In some embodiments, this network activity summary may include a summary of the networks accessed, activity vs. time per connection, traffic vs. time per connection.
Billing Agent 1695	Detects and reports billing events. In some embodiments interacts with the User Interface Agent 497 to provide the user with service plan options, accept service plan selections, provide notification on service usage levels, provide options on service usage control policy, accept choices on service usage policy, provide transaction options or accept transaction choices. In some embodiments, interacts with Transaction Servers 134 to conduct ecommerce transactions with central billing.
User Interface 1697	Provide service interface to users.
Service Downloader 1663	Provides a download function to install or update service software elements on the device.
Connection Manager 1804	Provides a control and supervision function for one or more modem drivers or modems that connect to an access network.
Modem Selection and Control 1811	Selects the access network connection.
Modem Drivers 1831, 1815, 1814, 1813, 1812	Converts data traffic into modem bus traffic for one or more modems.
Modems 2141, 2125, 2124, 2123, 2122	Connects the device to one or more networks.

Fig. 10

Service Controller 122 Element	Partial Summary of Embodiment Functions
Service Control Server Link 1638	Network side control plane link for connecting Service Controller 422 Service Processor 415 device agents. In some embodiments, also serves as the link for the agent heartbeat function.
Access Control Integrity Server 1654	Collects device information on service policy, service usage, agent configuration and agent behavior. Cross checks this information to identify integrity breaches in the service policy implementation and control system. Initiates action when a service policy violation or a system integrity breach is suspected.
Service History Server 118	Collects and records service usage reports from the Access Network AAA Server 421 and the Service Monitor Agent 496. In some embodiments, maintains a history of which networks in addition to the networks controlled by the Service Controller that the device has connected to. In some embodiments, this network activity summary may include a summary of the networks accessed, activity vs. time per connection, traffic vs. time per connection. In some embodiments, this activity summary is further analyzed or reported to estimate the type of service plan associated with the traffic activity for the purpose of bill sharing reconciliation.
Central Provider Billing System 123	Provides mediation function for central provider billing events. Accepts service plan changes. In some embodiments, provides updates on device service usage, service plan limits or service policies.
Billing Event Server 1662	In some embodiments, collects billing events, provides service plan information to the Service Processor 415, provides service usage updates to the Service Processor 415, serves as interface between device and central Provider Billing System 123, or provides trusted third party function for certain ecommerce billing transactions.
Service Download Control Server 1660	Provides a download function to install or update service software elements on the device.

Fig. 11

**Fig. 12**

**Fig. 13**

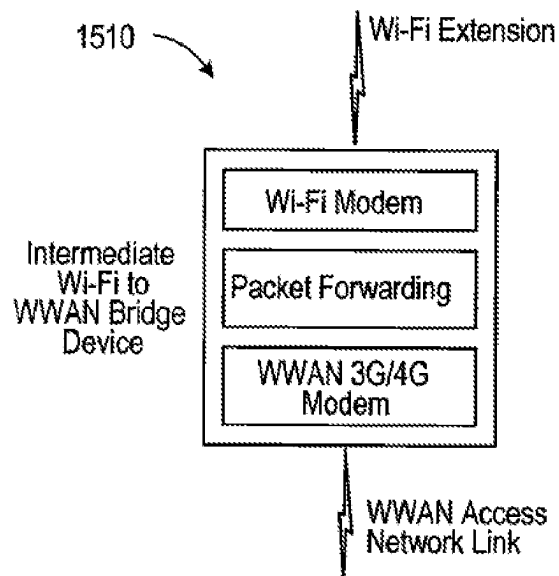


Fig. 14A

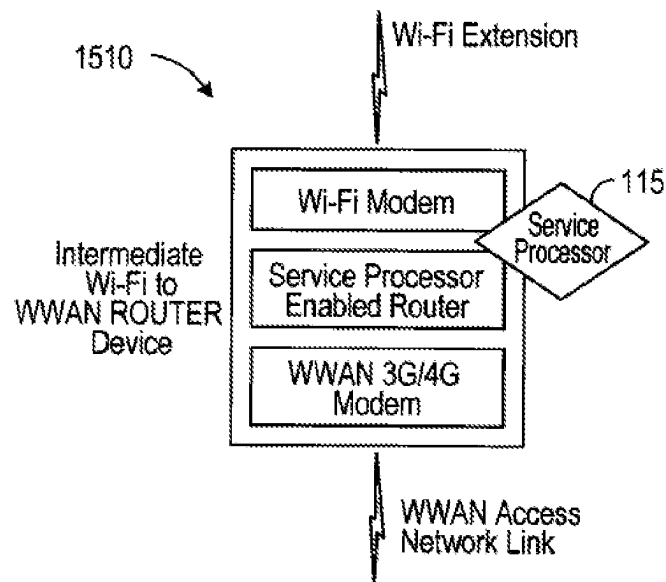


Fig. 14B

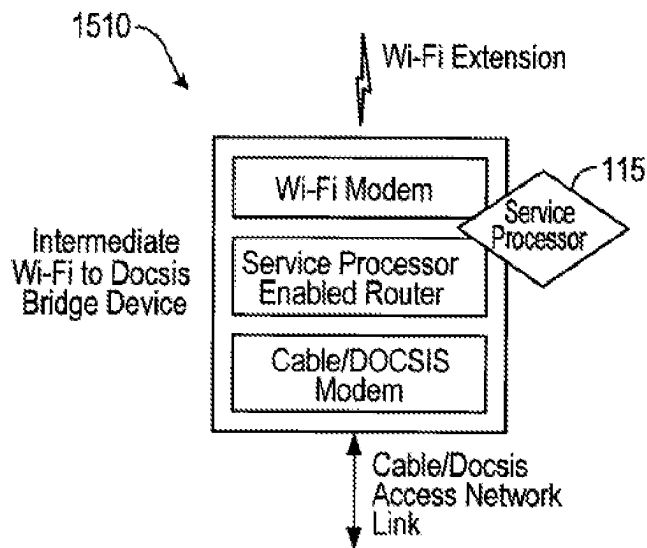


Fig. 14C

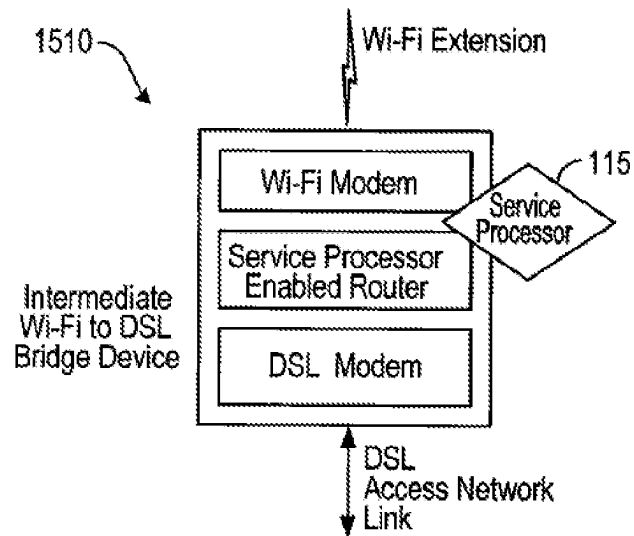


Fig. 14D

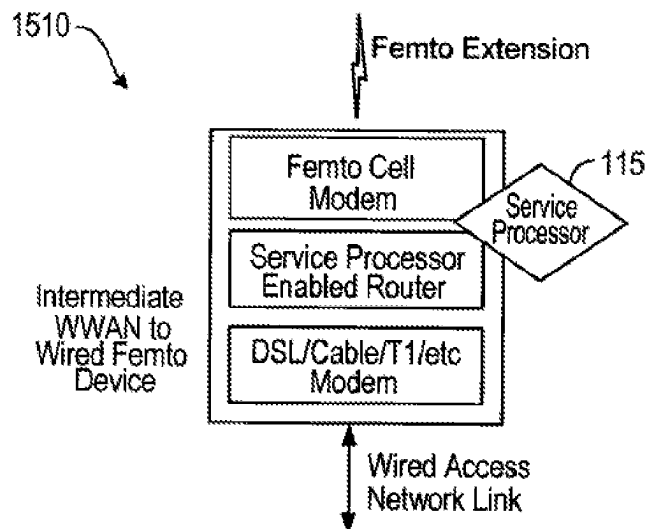


Fig. 14E

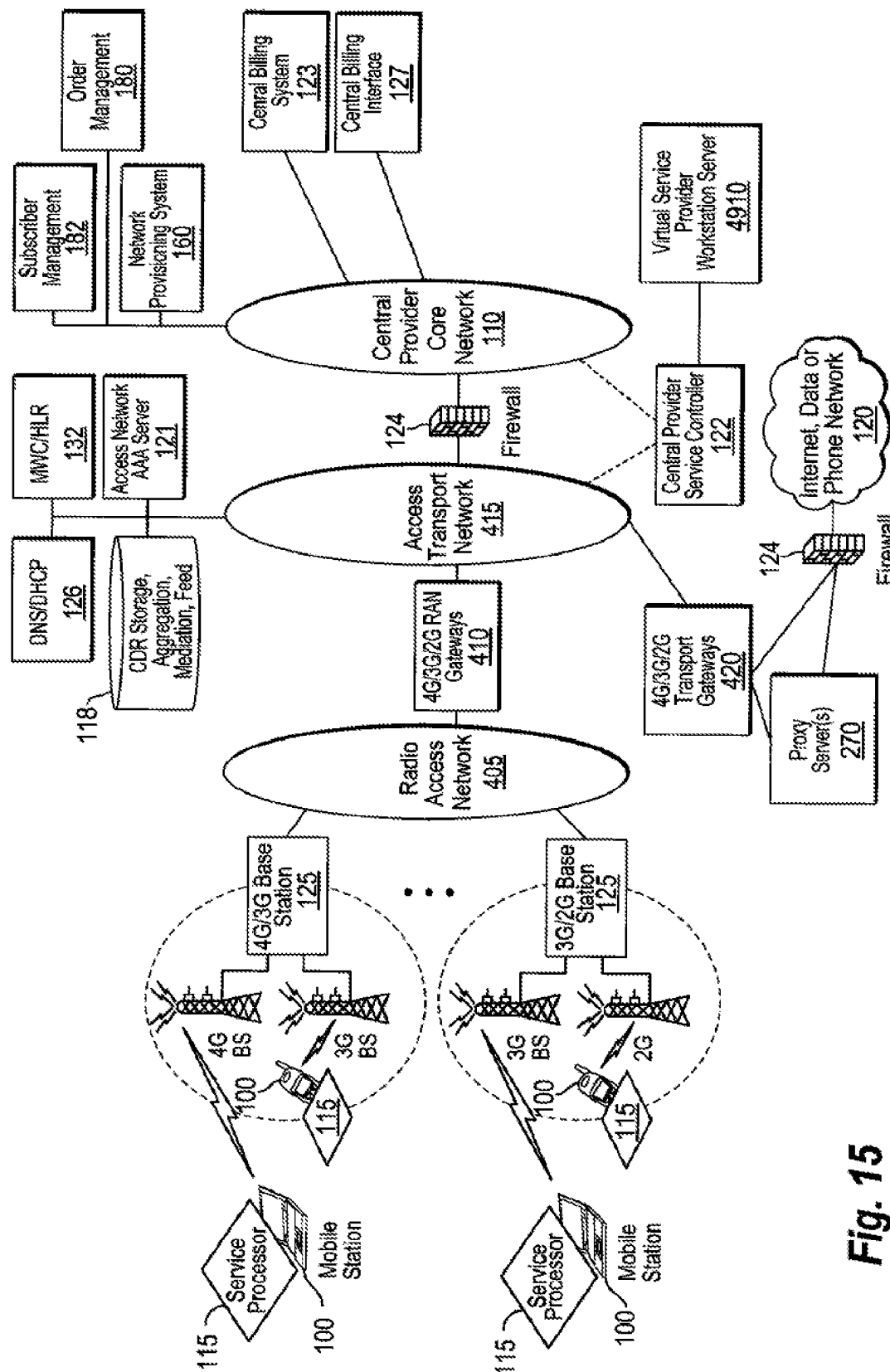
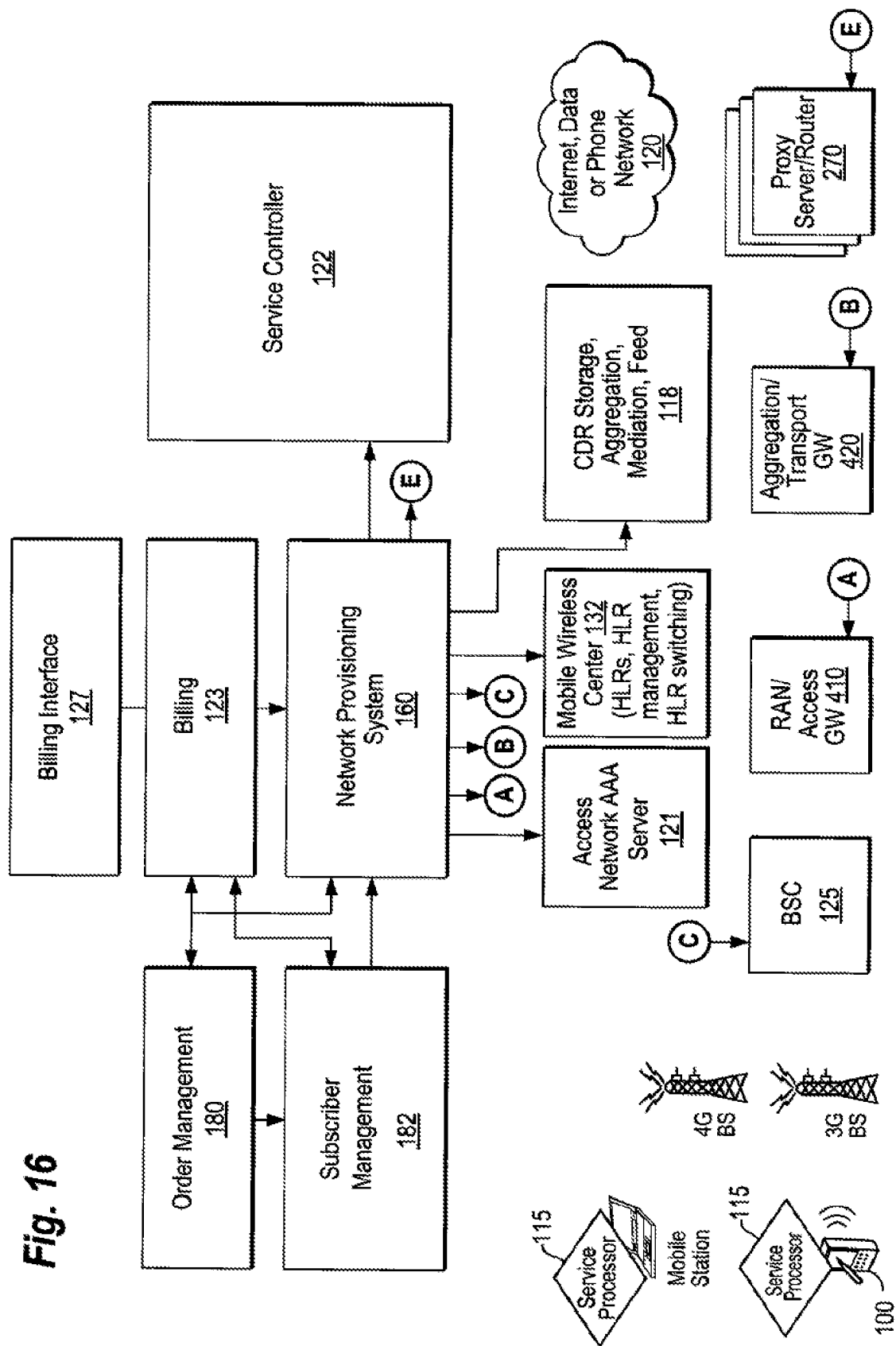


Fig. 15

Fig. 16



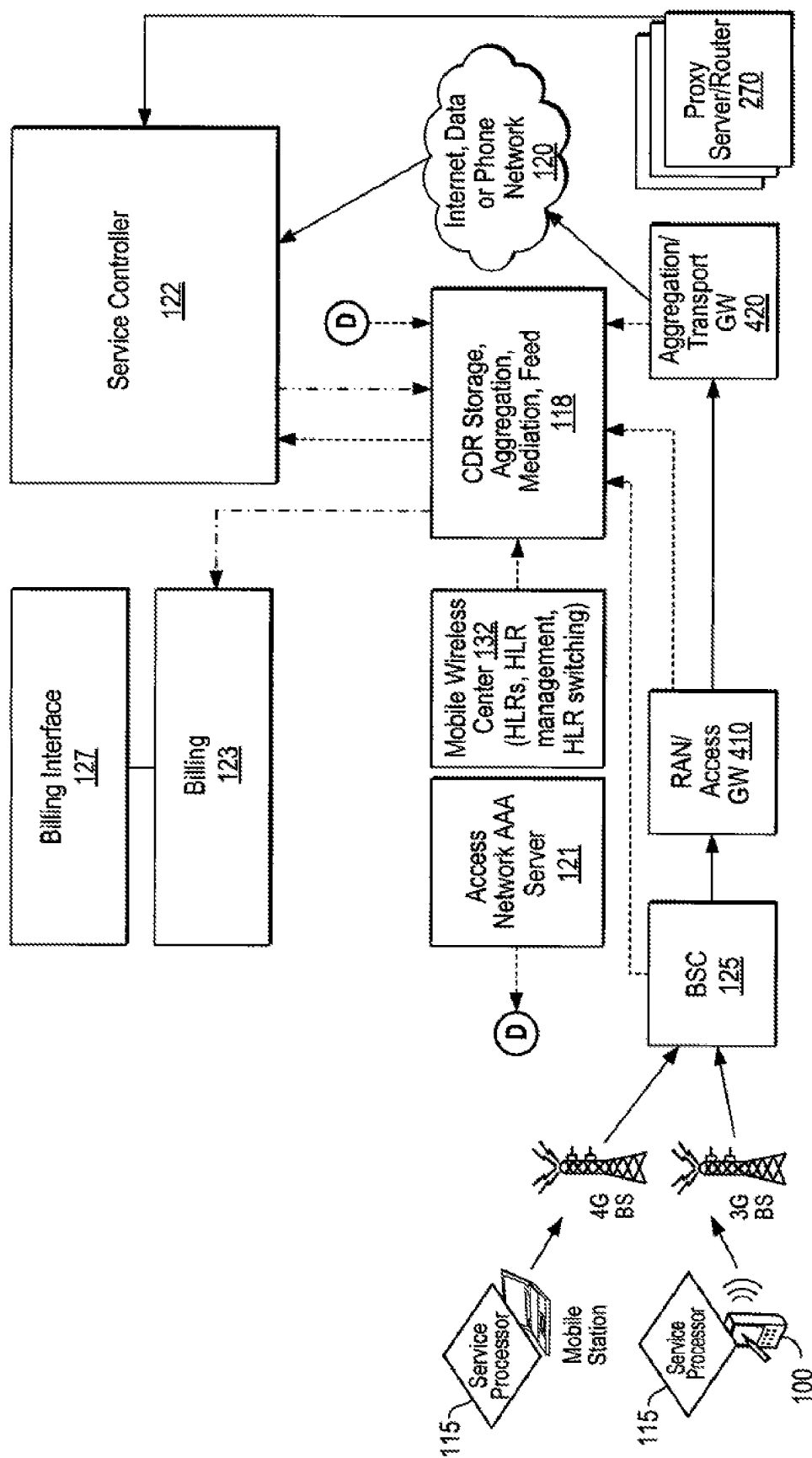


Fig. 17

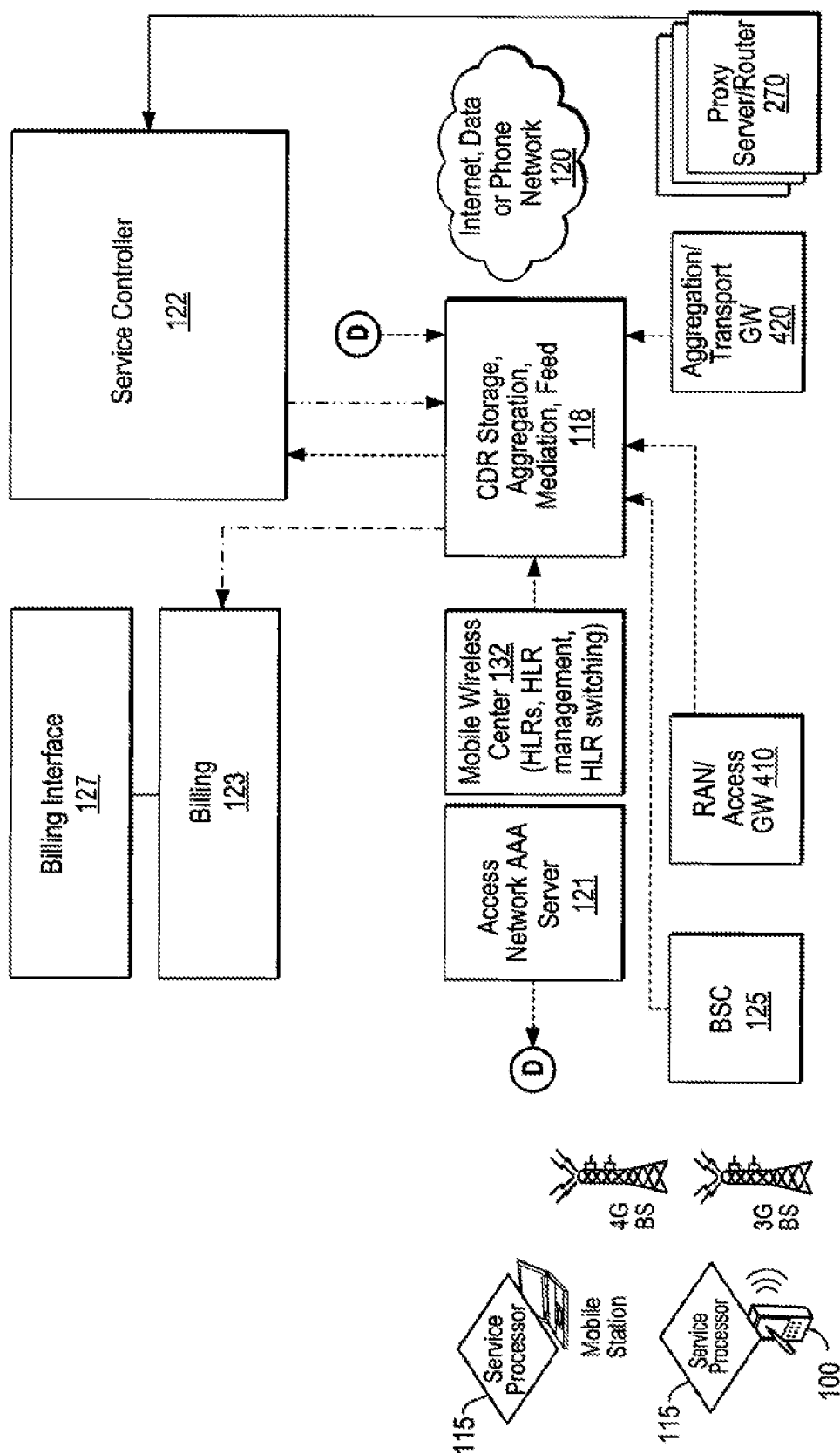
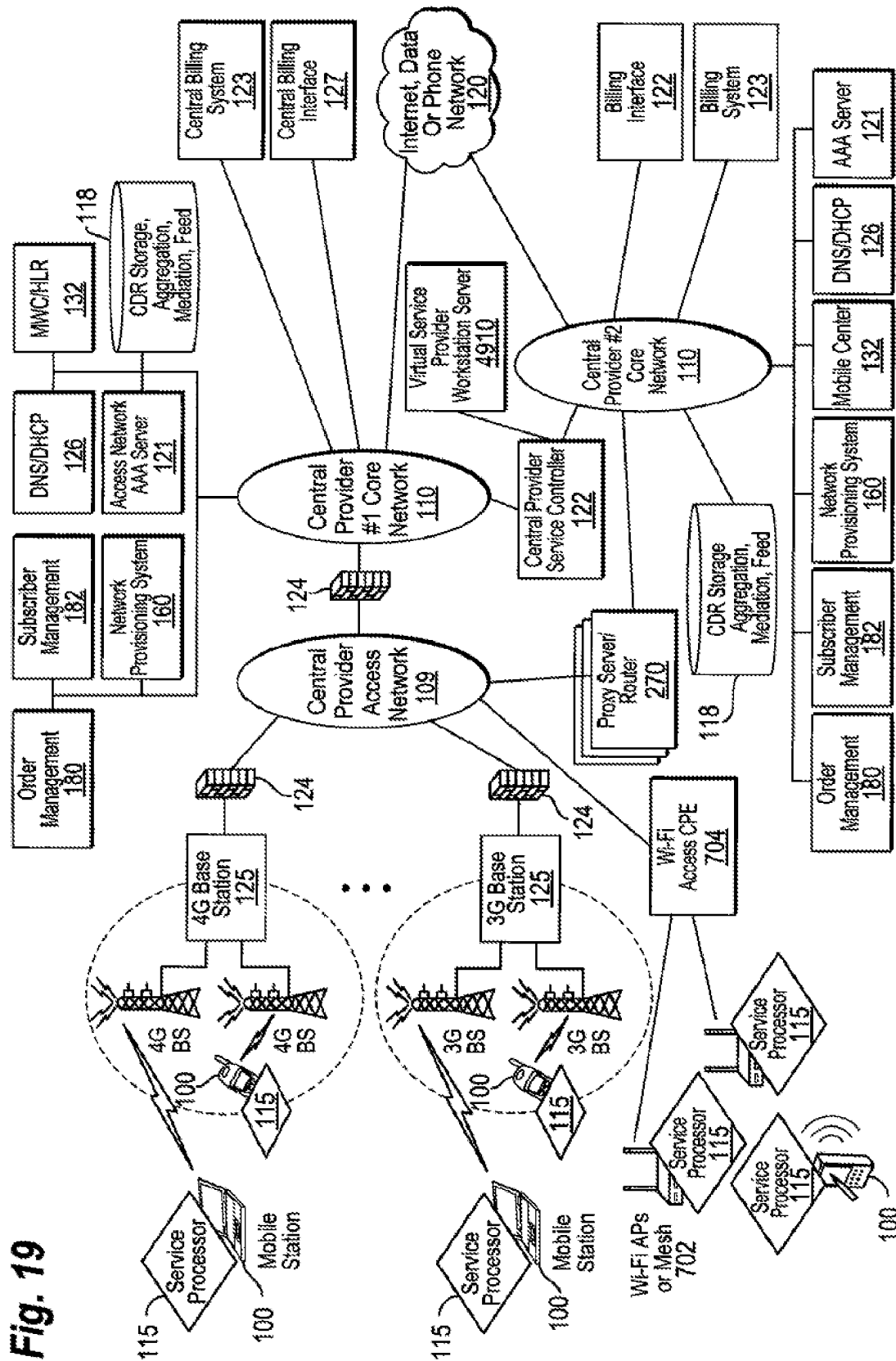
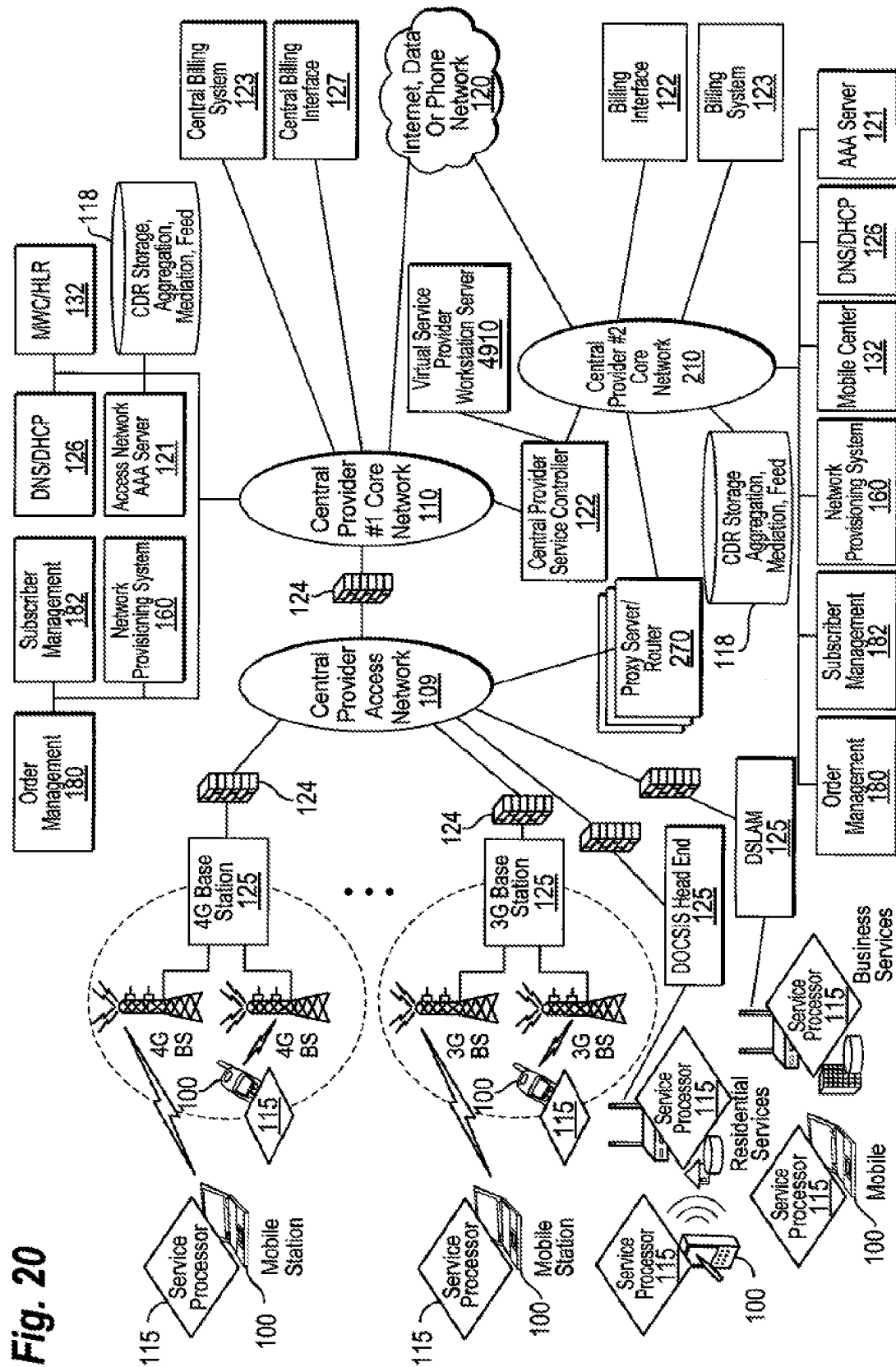


Fig. 18





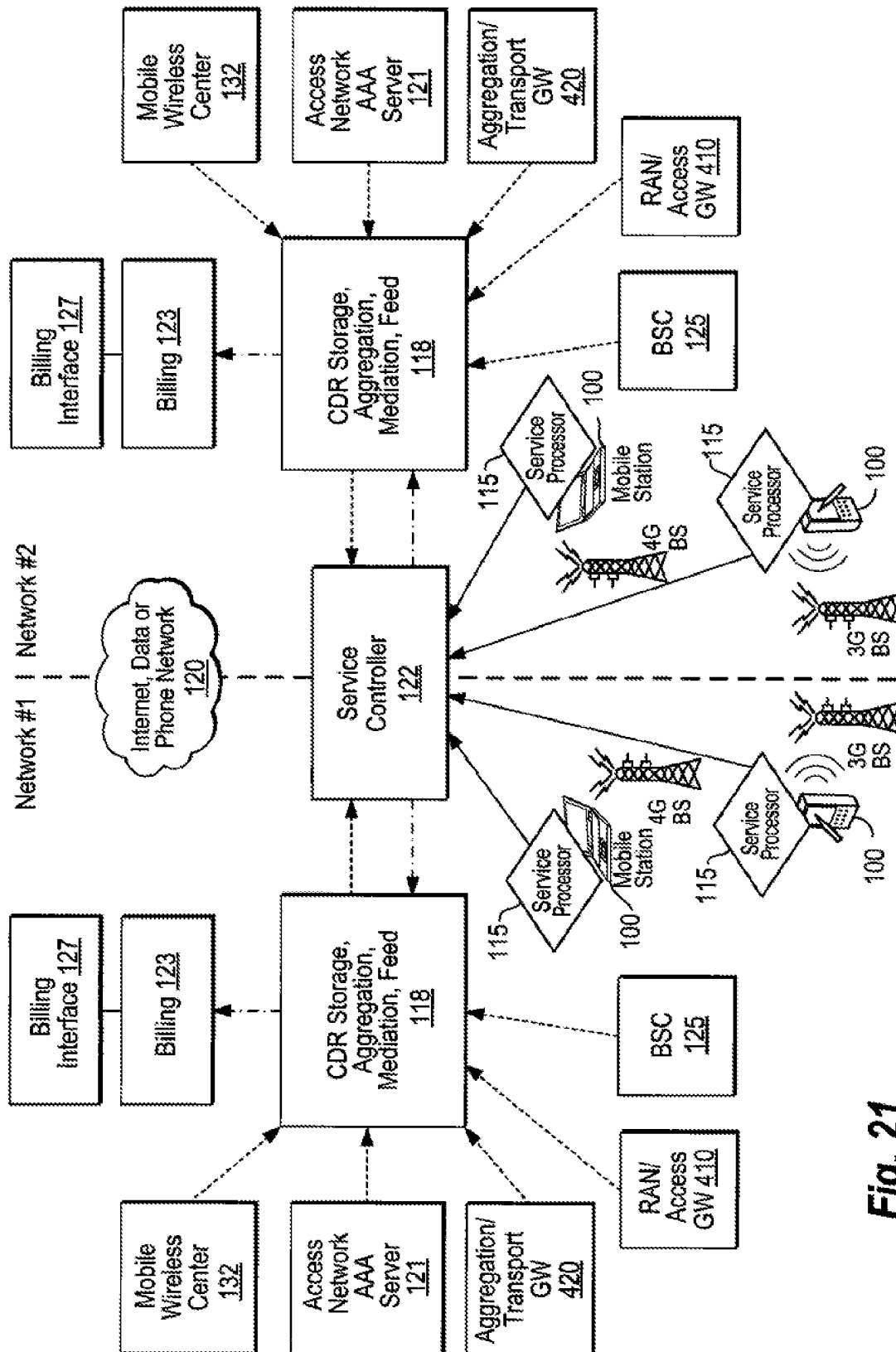


Fig. 21

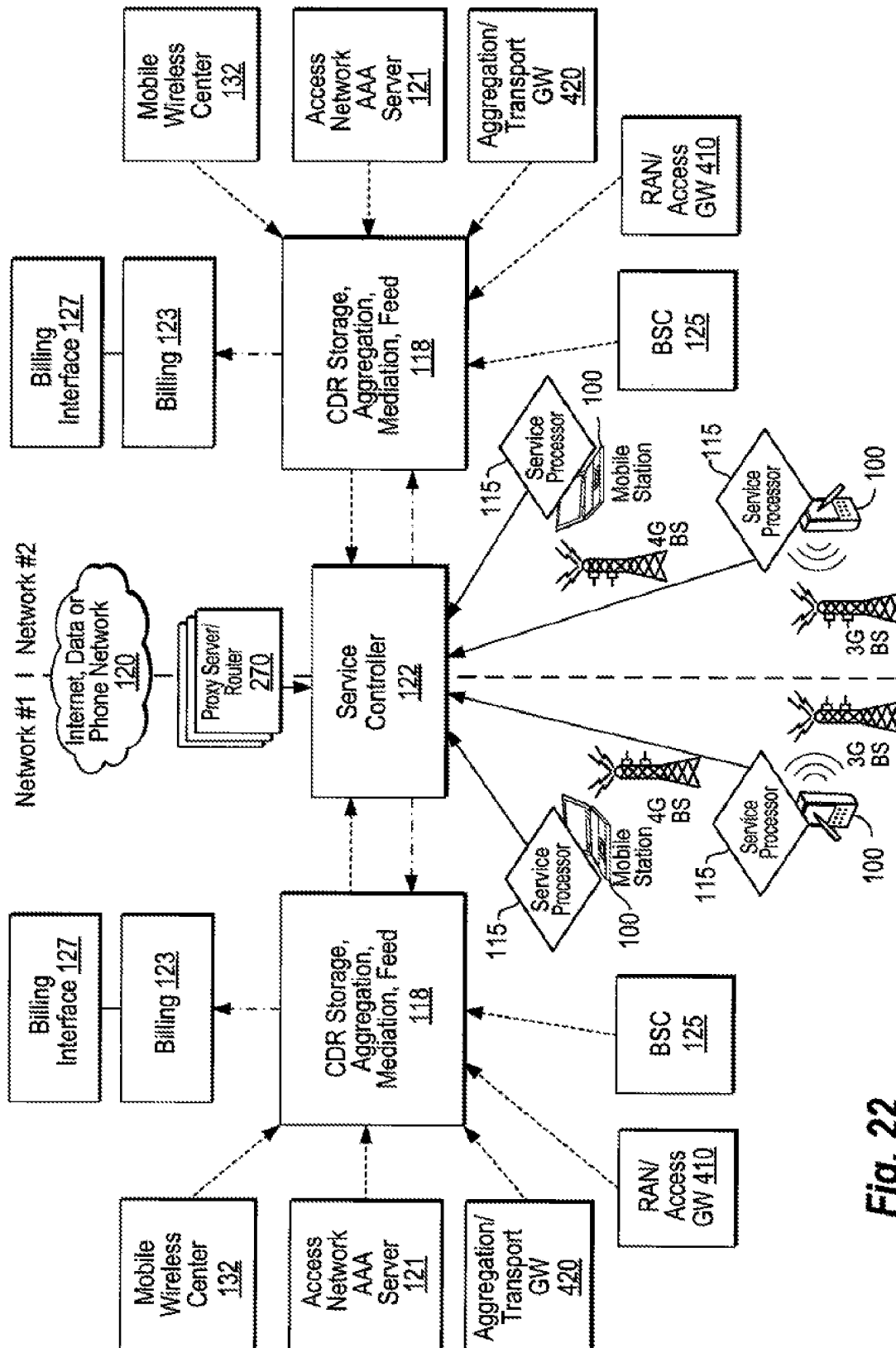


Fig. 22

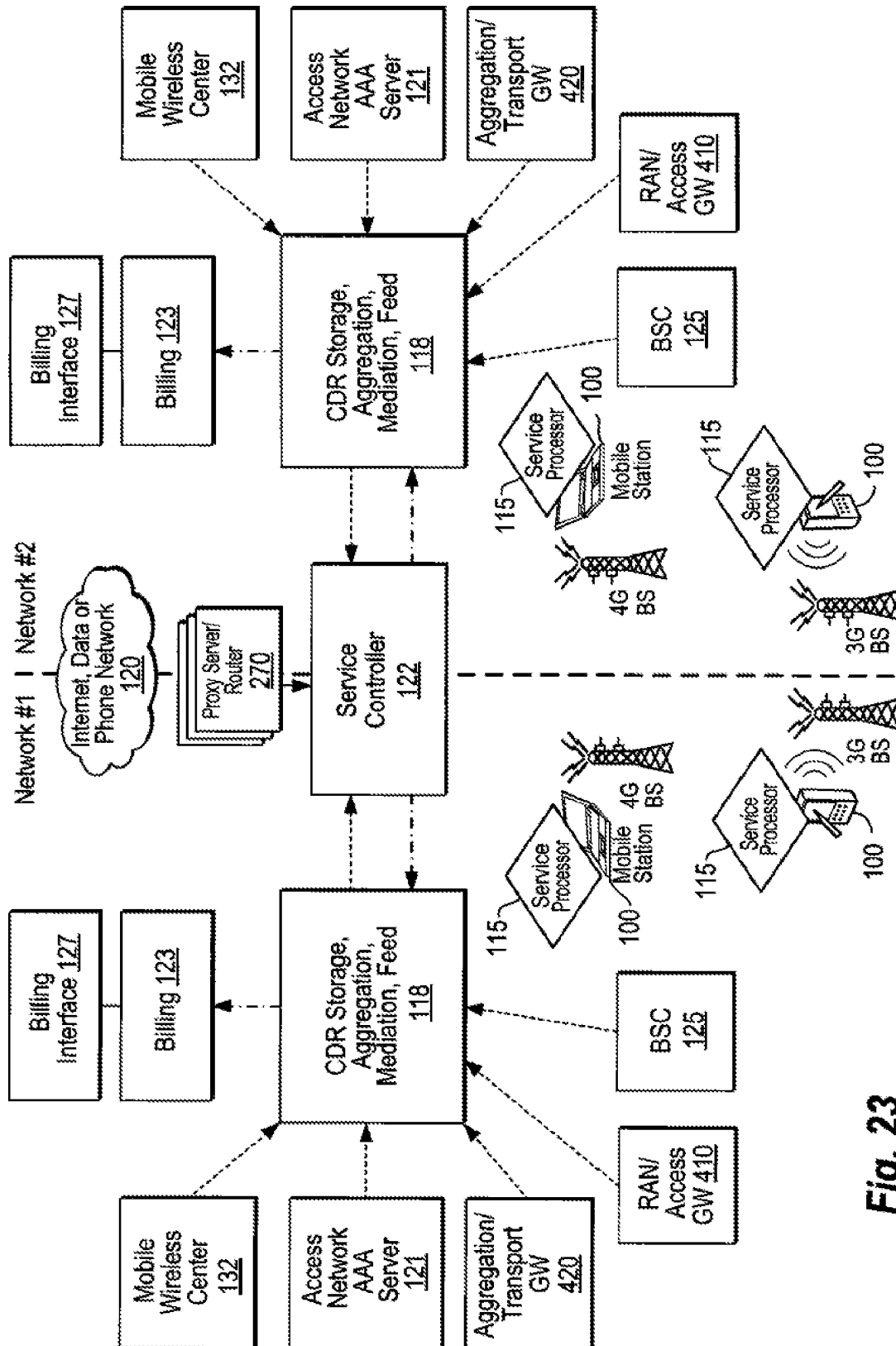


Fig. 23

1

ENHANCED ROAMING SERVICES AND CONVERGED CARRIER NETWORKS WITH DEVICE ASSISTED SERVICES AND A PROXY

BACKGROUND

With the advent of mass market digital communications and content distribution, many access networks such as wireless networks, cable networks and DSL (Digital Subscriber Line) networks are pressed for user capacity, with, for example, EVDO (Evolution-Data Optimized), HSPA (High Speed Packet Access), LTE (Long Term Evolution), WiMax (Worldwide Interoperability for Microwave Access), and Wi-Fi (Wireless Fidelity) wireless networks increasingly becoming user capacity constrained. Although wireless network capacity will increase with new higher capacity wireless radio access technologies, such as MIMO (Multiple-Input Multiple-Output), and with more frequency spectrum being deployed in the future, these capacity gains are likely to be less than what is required to meet growing digital networking demand.

Similarly, although wire line access networks, such as cable and DSL, can have higher average capacity per user, wire line user service consumption habits are trending toward very high bandwidth applications that can quickly consume the available capacity and degrade overall network service experience. Because some components of service provider costs go up with increasing bandwidth, this trend will also negatively impact service provider profits.

BRIEF DESCRIPTION OF THE DRAWINGS

Various embodiments are disclosed in the following detailed description and the accompanying drawings.

FIG. 1 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing in accordance with some embodiments.

FIG. 2 illustrates another wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing in accordance with some embodiments.

FIG. 3 illustrates another wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing in accordance with some embodiments.

FIG. 4 illustrates provisioning of a wireless network for providing device assisted CDR creation, aggregation, mediation and billing in accordance with some embodiments.

FIG. 5 illustrates a network architecture for providing device assisted CDRs in accordance with some embodiments.

FIG. 6 illustrates another network architecture for providing device assisted CDRs in accordance with some embodiments.

FIG. 7 illustrates another network architecture for providing device assisted CDRs in accordance with some embodiments.

FIG. 8 illustrates another network architecture for providing device assisted CDRs in accordance with some embodiments.

FIG. 9 is a functional diagram illustrating a device based service processor and a service controller in accordance with some embodiments.

2

FIG. 10 provides a table summarizing various service processor functional elements in accordance with some embodiments.

FIG. 11 provides a table summarizing various service controller functional elements in accordance with some embodiments.

FIG. 12 illustrates a device stack providing various service usage measurement from various points in the networking stack for a service monitor agent, a billing agent, and an access control integrity agent to assist in verifying the service usage measures and billing reports in accordance with some embodiments.

FIG. 13 illustrates an embodiment similar to FIG. 12 in which some of the service processor is implemented on the modem and some of the service processor is implemented on the device application processor in accordance with some embodiments.

FIGS. 14A through 14E illustrate various embodiments of intermediate networking devices that include a service processor for the purpose of verifiable service usage measurement, reporting, and billing reports in accordance with some embodiments.

FIG. 15 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a proxy server in accordance with some embodiments.

FIG. 16 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a service provider network in accordance with some embodiments, involving one or more of service controllers, service processors, and/or proxy server/routers.

FIG. 17 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a service provider network in accordance with some embodiments, involving one or more of service controllers, service processors, and/or proxy server/routers.

FIG. 18 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a service provider network in accordance with some embodiments, involving one or more of service controllers and/or proxy server/routers.

FIG. 19 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments.

FIG. 20 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments, involving one or more of service controllers and/or service processors.

FIG. 21 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments.

FIG. 22 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments, involving one or more of service controllers, service processors, and/or proxy server/routers.

FIG. 23 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments, involving one or more of service controllers and/or proxy server/routers.

The invention can be implemented in numerous ways, including as a process; an apparatus; a system; a composition of matter; a computer program product embodied on a computer readable storage medium; and/or a processor, such as a processor configured to execute instructions stored on and/or provided by a memory coupled to the processor. In this specification, these implementations, or any other form that the invention may take, may be referred to as techniques. In general, the order of the steps of disclosed processes may be altered within the scope of the invention. Unless stated otherwise, a component such as a processor or a memory described as being configured to perform a task may be implemented as a general component that is temporarily configured to perform the task at a given time or a specific component that is manufactured to perform the task. As used herein, the term “processor” refers to one or more devices, circuits, and/or processing cores configured to process data, such as computer program instructions.

A detailed description of one or more embodiments of the invention is provided below along with accompanying figures that illustrate the principles of the invention. The invention is described in connection with such embodiments, but the invention is not limited to any embodiment. The scope of the invention is limited only by the claims and the invention encompasses numerous alternatives, modifications and equivalents. Numerous specific details are set forth in the following description in order to provide a thorough understanding of the invention. These details are provided for the purpose of example and the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured.

There are many new types of digital devices where it is becoming desirable, for example, to connect these devices to wireless networks including wireless wide area networks (WWAN, such as 3G and 4G) and/or wireless local area (WLAN) networks. These devices include, for example, consumer electronics devices, business user devices, and machine to machine devices that benefit from flexible wide area data connections and the Internet. Example devices include netbooks, notebooks, mobile Internet devices, personal navigation (e.g., GPS enabled) devices, music and multimedia players, eReaders, industrial telemetry, automotive emergency response and diagnostics, 2-way home and industrial power metering and control, vending machines, parking meters, and many other devices. For example, it is highly advantageous to offer service usage and service billing plans for such devices that are more optimal for each type of device and each type of desired user experience. To accomplish this, more sophisticated service usage measuring and service usage billing systems are needed as compared to the conventional network based techniques in existence today. By providing more flexibility in service measurement and billing, more advantageous and cost effective service plans can be created for, for example, the new WWAN connected devices cited above for all three markets (e.g., consumer, business and machine to machine) that still maintain the necessary profit margins for the WWAN carriers to be successful with these various service businesses.

Accordingly, various embodiments disclosed herein provide for a new and flexible augmentation or replacement for

existing carrier network service usage measurement, service usage accounting, and service usage billing systems and techniques.

A charging data record (CDR) is a term that as used herein defines a formatted measure of device service usage information, typically generated by one or more network functions or device functions that supervise, monitor, control, account for, charge for, report, synchronize usage accounting or records and/or bill for network access or other service or purchase transactions for the device. To simplify the discussion, the term charging data record or CDR is often used interchangeably herein to refer to several different types of service usage records. For example, the term CDR is used herein interchangeably with one or more of other industry terms or record types including, for example, CDRs, flow data records (FDRs), start/stop records, interim accounting records, IP data records (IPDRs), call data records, micro-CDRs, service charging bucket accounting records, record types that include usage information identifying QoS level, QoS class, and/or other QoS classifier, records that include usage information identifying traffic type or class or other traffic identifier, records that identify traffic protocol, records that identify usage or purchase transactions, records that report a purchase transaction, etc. One of ordinary skill in the art will appreciate that in many of the embodiments the term CDR and/or many other terms can be interchanged for a given application or purpose. Any of these terms can refer in various embodiments to the lowest level of usage accounting records or can refer to various levels of charging record aggregation that may occur in the various network or device elements described herein.

CDRs typically form the basis for recording device network service usage, and often form the basis for billing for such usage. Various embodiments are provided herein for device assisted CDR creation, mediation, and billing. There are many limitations to the capabilities of service usage recording, aggregation and/or billing when CDRs are generated exclusively by network based functions or equipment. Accordingly, by either augmenting network based service usage measures with device based service usage measures, or by replacing network based service usage measures with device based service usage measures, it is possible to create a CDR generation, aggregation, mediation and/or billing solution that has superior or more desirable capabilities/features. While in theory, many of the service usage measures that can be evaluated on a device can also be evaluated in the network data path using various network equipment technologies including, for example, deep packet inspection (DPI), there are many examples where measuring service usage at the device is either more desirable or more practical, or in some cases it is the only way to obtain the desired measure. Such examples include but are not limited to the following:

- Application layer service usage measures (e.g., traffic usage categorized by application or by combinations of application, destination, and/or content type);

- Usage measures that do not involve user traffic but instead involve network overhead traffic (e.g., basic connection maintenance traffic, signaling traffic, network logon/AAA/authentication/monitoring traffic, service software update traffic);

- Usage that is associated with services that are charged to another entity other than the end user (e.g., basic network connection service offer traffic, traffic associated with providing network access to or downloading service marketing information, traffic associated with

5

advertiser sponsored services, traffic associated with content provider sponsored services, 911 service traffic);

Usage measures involving encrypted traffic (e.g., traffic that is run over encrypted networking protocols or between secure end points);

Implementing service usage measure collection and/or service usage billing across multiple networks that may have different and in some cases incompatible, inaccessible (to the CDR system of record) or incomplete service usage measurement capabilities;

Service usage measurement and/or service usage billing capabilities that are not supported by the present network gateways, routers, MWC/HLRs, AAA, CDR aggregation, CDR mediation, billing and/or provisioning systems;

New service usage measures and/or new service usage billing capabilities that are desirable to implement in a manner that does not require major changes or upgrades to the existing network gateways, routers, MWC/HLRs, AAA, CDR aggregation, CDR mediation, billing and/or provisioning systems;

New service usage measures and/or new service usage billing capabilities that are desirable to implement in a manner that allows for rapid definition and implementation of new service measures and/or billing plans;

New service usage measures and/or new service usage billing capabilities that are desirable to implement in a manner that may be implemented in a manner that enables multiple device group definitions in which each device group gets a customized programmable definition for service usage collection, accounting and/or billing;

Multi device billing;

Multi user billing;

Intermediate device billing with single user and multi user with and without multi device;

Content downloads from a specific source to a specific application with the content being of a specific type or even identified down to a particular content ID; and/or

Various other single event transactions used for billing purposes.

For these and other reasons, it is desirable to provide a system/process that utilizes device assisted service usage measures that provides either an enhancement of existing network based service usage CDR system capabilities and techniques and/or a replacement for network based CDR system capabilities and techniques.

In some embodiments, techniques, such as a system and/or process, that utilize device assisted service usage measures include one or more of the following: (1) receiving a service usage measure from a device in communication with a wireless network, (2) verifying or protecting the validity of the service usage measure, (3) generating a CDR based on the service usage measure (e.g., device assisted CDR), (4) aggregating CDRs, and (5) mediating the CDR with network CDRs. In some embodiments, the techniques also include providing a design and provisioning of devices/network equipment to recognize the CDRs. In some embodiments, the techniques also include provisioning to recognize that the device belongs to a Device Assisted Services (DAS) device group and that corresponding CDRs should be accepted and mediated. In some embodiments, the device assisted CDRs are also generated using formats, network communications protocols, network device authentication and/or provisioning to allow device assisted CDRs into the network CDR system, encryption, and/or signatures as

6

required by the network (e.g., to comply with network generated CDR requirements or based on any other network and/or service provider requirements and/or standards).

In some embodiments, mediation rules include multi device, multi user, single user devices, and/or intermediate networking devices that can be single user or multi user, as described herein.

In some embodiments, a device assisted CDR generator collects device based service usage measures that are used as the basis for, or as an enhancement (e.g., as a supplement or in addition) to, one or more (e.g., network generated) CDRs that provide one or more networking functions with properly formatted service usage reports that the network function(s) accepts as being transmitted from an authorized source, read, and utilized for helping to determine the service usage of a device or group of devices. In some embodiments, the network functions that the device assisted CDR generator shares CDRs with typically include one or more of the following: service usage/CDR aggregation and/or mediation servers, gateways, routers, communication nodes, Mobile Wireless Centers (MWCs, including HLRs), databases, AAA systems, billing interfaces, and billing systems. For example, the process of CDR creation in the CDR generator typically includes either using one or more device based measures of service usage, or one or more device based measures of service usage in combination with one or more network based measures of service usage, possibly processing one or more of such service usage measures according to a set of CDR creation, CDR aggregation, and/or CDR mediation rules to arrive at a final device usage measure that is, for example, then formatted with the proper syntax, framed, possibly encrypted and/or signed, and encapsulated in a communication protocol or packet suitable for sharing with network functions. In some embodiments, the CDR generator resides in the device. In some embodiments, the CDR generator resides in a network server function that receives the device assisted service usage measures, along with possibly network based usage measures, and then creates a CDR (e.g., in the service controller 122).

In some embodiments, the device assisted CDR generator can reside in the service processor (e.g., service processor 115), for example, in the service usage history or billing server functions. In some embodiments, the device assisted CDR generator resides in the device itself, for example, within the service processor functions, such as the billing agent or the service monitor agent.

There are several factors that are considered in the various embodiments in order to create a useful, reliable, and secure device assisted CDR system, including, for example, but not limited to:

- Identification of each device based service usage measure with one or more usage transaction codes;
- Verification of the device based usage measure(s);
- Secure communication of the device based usage measures to the network;
- Efficient (e.g., low bandwidth) communication of the device based service usage measure;
- Coordination/comparison/aggregation of the device based service usage measure with network based service usage measure(s);
- Formatting the device based service usage measure into a CDR that can be properly communicated to the network functions and/or equipment that process service usage information;
- Causing the network based functions and/or equipment used for CDR collection, aggregation, mediation and/or billing to recognize, authorize, and accept communi-

cations and CDRs from the device assisted CDR generator, reading and properly implementing the correct network session context for the CDR so that the CDR is properly associated with the correct device/user/session;

Implementing the CDR aggregation rules that determine how to collect and aggregate the device assisted CDRs as they are reported through the network CDR system hierarchy;

Implementing the mediation rules that determine how the various device based service usage transaction code measures are combined and mediated with the other device based service usage transaction code measures to result in consistent service usage information for each of the transaction code categories maintained in the network;

Implementing the mediation rules that determine how the device assisted CDRs are combined and mediated with network based CDRs to result in consistent service usage information for each of the transaction code categories maintained in the network;

Implementing mediation rules to reconcile the variances between network based CDR usage measures and device assisted CDR usage measures;

Classification of one or more device groups, with each group having the capability to uniquely define the service usage collection, accounting, and/or billing rules;

Collecting CDRs generated on networks other than the home network so that service usage may be measured, accounted for, and/or billed for across multiple networks;

Multi device billing;

Multi user billing; and/or

Intermediate device billing with single user and multi user with and without multi device.

In some embodiments, verification of the relative accuracy of the device assisted service usage measure is provided. Given that, for example, the service usage measure is often being generated on an end user device or a device that is readily physically accessed by the general public or other non-secure personnel from a network management viewpoint, in some embodiments, the device agents used in one or more of the service processor **115** agents are protected from hacking, spoofing, and/or other misuse. Various techniques are provided herein for protecting the integrity of the agents used for generating the device assisted service usage measures.

In some embodiments, the service usage measures are verified by network based cross checks using various techniques. For example, network based cross checks can provide valuable verification techniques, because, for example, it is generally not possible or at least very difficult to defeat well designed network based cross checks using various techniques, such as those described herein, even if, for example, the measures used to protect the device agents are defeated or if no device protection measures are employed. In some embodiments, network based cross checks used to verify the device assisted service usage measures include comparing network based service usage measures (e.g., CDRs generated by service usage measurement apparatus in the network equipment, such as the BTS/BSCs **125**, RAN Gateways **410**, Transport Gateways **420**, Mobile Wireless Center/HLRs **132**, AAA **121**, Service Usage History/CDR Aggregation, Mediation, Feed **118**, or other network equipment), sending secure query/response command sequences to the service processor **115** agent(s) involved in device

assisted CDR service usage measurement or CDR creation, sending test service usage event sequences to the device and verifying that the device properly reported the service usage, and using various other techniques, such as those described herein with respect to various embodiments.

In some embodiments, one or more of the following actions are taken if the device based service usage measure is found to be in error or inaccurate: bill the user for usage overage or an out of policy device, suspend the device, quarantine the device, SPAN the device, and/or report the device to a network administration function or person.

In some embodiments, the CDR syntax used to format the device assisted service usage information into a CDR and/or network communication protocols for transmitting CDRs are determined by industry standards (e.g., various versions of 3GPP TS 32.215 format and 3GPP2 TSG-X X.S0011 or TIA-835 format). In some embodiments, for a given network implementation the network designers will specify modifications of the standard syntax, formats and/or network communication/transmission protocols. In some embodiments, for a given network implementation the network designers will specify syntax, formats, and/or network communication/transmission protocols that are entirely different than the standards.

In some embodiments, within the syntax and formatting for the CDR the device assisted service usage is typically categorized by a transaction code. For example, the transaction code can be similar or identical to the codes in use by network equipment used to generate CDRs, or given that the device is capable of generating a much richer set of service usage measures, the transaction codes can be a superset of the codes used by network equipment used to generate CDRs (e.g., examples of the usage activities that can be labeled as transaction codes that are more readily supported by device assisted CDR systems as compared to purely network based CDR systems are provided herein).

In some embodiments, the device sends an identifier for a usage activity tag, an intermediate server determines how to aggregate into CDR transaction codes and which CDR transaction code to use.

In some embodiments, the device service processor **115** compartmentalizes usage by pre-assigned device activity transaction codes (e.g., these can be sub-transactions within the main account, transactions within a given bill-by-account transaction or sub-transactions within a bill-by-account transaction). The device implements bill-by-account rules to send different usage reports for each bill-by-account function. In some embodiments, the service controller **122** programs the device to instruct it on how to compartmentalize these bill-by-account service usage activities so that they can be mapped to a transaction code.

In some embodiments, the device reports less compartmentalized service usage information and the service controller **122** does the mapping of service usage activities to CDR transaction codes, including in some cases bill-by-account codes.

In some embodiments, the CDR sent to **118** or other network equipment, for example, can include various types of transaction codes including, for example, a raw device usage CDR, a bill-by-account (e.g., a sub-activity transaction code) CDR, a billing offset CDR, and/or a billing credit CDR. For example, the decision logic (also referred to as business rules or CDR aggregation and mediation rules) that determines how these various types of CDR transaction codes are to be aggregated and mediated by the core network and the billing system can be located in the network equip-

ment (e.g., a network element, such as service usage **118**), in the service controller **122**, and/or in the billing system **123**.

In some embodiments, the device assisted service usage measures to generate a CDR that includes service usage information, service usage transaction code(s), and, in some embodiments, network information context. In some embodiments, the service usage information, transaction code, and/or network information context is formatted into communication framing, syntax, encryption/signature, security and/or networking protocols that are compatible with the formatting used by conventional networking equipment to generate CDRs. For example, this allows networking equipment used for CDR collection, recording, aggregation, mediation, and/or conversion to billing records to properly accept, read, and interpret the CDRs that are generated with the assistance of device based service usage measurement. In some embodiments, the device assisted service measures are provided to an intermediate network server referred to as a service controller (e.g., service controller **122**). In some embodiments, the service controller uses a CDR feed aggregator for a wireless network to collect device generated usage information for one or more devices on the wireless network; and provides the device generated usage information in a syntax (e.g., charging data record (CDR)), and a communication protocol (e.g., 3GPP or 3GPP2, or other communication protocol(s)) that can be used by the wireless network to augment or replace network generated usage information for the one or more devices on the wireless network.

In some embodiments, mediation rules include multi device, multi user, single user devices, intermediate networking devices that can be single user or multi user. For example, the device assisted CDRs can be formatted by the device assisted CDR generator to include a transaction code for one user account, even though the CDRs originate from multiple devices that all belong to the same user. This is an example for a multi-user device assisted CDR billing solution. In another example for a multi-user device assisted CDR billing solution, device assisted CDRs from multiple devices and multiple users can all be billed to the same account (e.g., a family plan or a corporate account), but the bill-by-account CDR transaction records can be maintained through the billing system so that sub-account visibility is provided so that the person or entity responsible for the main account can obtain visibility about which users and/or devices are creating most of the service usage billing. For example, this type of multi-user, multi-device device assisted CDR billing solution can also be used to track types of service usage and/or bill for types of service usage that are either impossible or at least very difficult to account and/or bill for with purely network based CDR systems. In some embodiments, bill-by-account CDR transaction records can be used to provide sponsored transaction services, account for network chatter, provide service selection interfaces, and other services for multi-user or multi-device service plans.

In addition to conventional single user devices (e.g., cell phones, smart phones, netbooks/notebooks, mobile internet devices, personal navigation devices, music players, electronic eReaders, and other single user devices) device assisted service usage measurement and CDRs are also useful for other types of network capable devices and/or networking devices, such as intermediate networking devices (e.g., 3G/4G WWAN to WLAN bridges/routers/gateways, femto cells, DOCSIS modems, DSL modems, remote access/backup routers, and other intermediate network devices). For example, in such devices, particularly with a secure manner to verify that the device assisted

service usage measures are relatively accurate and/or the device service processor **115** software is not compromised or hacked, many new service provider service delivery and billing models can be supported and implemented using the techniques described herein. For example, in a WiFi to WWAN bridge or router device multiple user devices can be supported with the same intermediate networking device in a manner that is consistent and compatible with the central provider's CDR aggregation and/or billing system by sending device assisted CDRs as described herein that have a service usage and/or billing code referenced to the end user and/or the particular intermediate device.

In some embodiments, the device assisted CDRs generated for the intermediate networking device are associated with a particular end user in which there can be several or many end users using the intermediate networking device for networking access, and in some embodiments, with each end user being required to enter a unique log-in to the intermediate networking device. For example, in this way, all devices that connect using WiFi to the intermediate networking device to get WWAN access generate CDRs can either get billed to a particular end user who is responsible for the master account for that device, or the CDRs can get billed in a secure manner, with verified relative usage measurement accuracy to multiple end users from the same intermediate networking device. In another example, an end user can have one account that allows access to a number of intermediate networking devices, and each intermediate networking device can generate consistent device assisted CDRs with transaction codes for that end user regardless of which intermediate networking device the end user logs in on.

In some embodiments, some of the services provided by the intermediate networking device are billed to a specific end user device assisted CDR transaction code, while other bill-by-account services are billed to other transaction code accounts, such as sponsored partner transaction service accounts, network chatter accounts, sponsored advertiser accounts, and/or service sign up accounts. For example, in this manner, various embodiments are provided in which intermediate networking devices (e.g., a WWAN to WiFi router/bridge) can sold to one user but can service and be used to bill other users (e.g., and this can be covered in the first purchasing user's service terms perhaps in exchange for a discount), or such intermediate networking devices can be located wherever access is desired without concern that the device will be hacked into so that services can be acquired without charge.

In some embodiments, various types of service usage transactions are billed for on the intermediate networking device, to any of one or more users, in which the information required to bill for such services is not available to the central provider or MVNO network equipment, just as is the case with, for example, conventional single user devices. In view of the various embodiments and techniques described herein, those skilled in the art will appreciate that similar service models are equally applicable not just to WWAN to WiFi intermediate networking devices, but also to the Femto Cell, remote access router, DOCSIS, DSL and other intermediate WWAN to WiFi networking devices.

FIG. 1 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing in accordance with some embodiments. As shown, FIG. 1 includes a 4G/3G/2G wireless network operated by, for example, a central provider. As shown, various wireless devices **100** are in communication with base stations **125** for wireless network communication

11

with the wireless network, and other devices **100** are in communication with Wi-Fi Access Points (APs) or Mesh **702** for wireless communication to Wi-Fi Access CPE **704** in communication with central provider access network **109**. In some embodiments, each of the wireless devices **100** includes a service processor **115** (as shown), and each service processor connects through a secure control plane link to a service controller **122**. In some embodiments, the network based service usage information (e.g., CDRs) is obtained from one or more network elements. As shown, an MVNO core network **210** also includes a CDR storage, aggregation, mediation, feed **118**, a MVNO billing interface **122**, and a MVNO billing system **123** (and other network elements as shown in FIG. 1). A Virtual Service Provider Work Station **4910** (also referred to herein as a service design interface) provides a user interface to central provider service designers, MVNO service designers or other service designers for the purpose of simplifying and organizing the process of service design as described herein.

As shown in FIG. 1, a CDR storage, aggregation, mediation, feed **118** (e.g., service usage **118**, including a billing aggregation data store and rules engine) is a functional descriptor for, in some embodiments, a device/network level service usage information collection, aggregation, mediation, and reporting function located in one or more of the networking equipment components attached to one or more of the sub-networks shown in FIG. 1 (e.g., central provider access network **109** and/or central provider core network **110**), which is in communication with the service controller **122**, and a central billing interface **127**. As shown in FIG. 1, service usage **118** is shown as a function in communication with the central provider core network **110**. In some embodiments, the CDR storage, aggregation, mediation, feed **118** function is located elsewhere in the network or partially located in elsewhere or integrated with as part of other network elements. In some embodiments, CDR storage, aggregation, mediation, feed **118** functionality is located or partially located in the AAA server **121** and/or the mobile wireless center/Home Location Register (HLR) **132** (as shown, in communication with a DNS/DHCP server **126**). In some embodiments, service usage **118** functionality is located or partially located in the base station, base station controller and/or base station aggregator, collectively referred to as base station **125** in FIG. 1. In some embodiments, CDR storage, aggregation, mediation, feed **118** functionality is located or partially located in a networking component in the central provider access network **109**, a networking component in the core network **110**, the central billing system **123**, the central billing interface **127**, and/or in another network component or function. This discussion on the possible locations for the network based and device based service usage information collection, aggregation, mediation, and reporting function (e.g., CDR storage, aggregation, mediation, feed **118**) can be easily generalized as described herein and as shown in the other figures described herein by one of ordinary skill in the art. Also as shown in FIG. 1, the service controller **122** is in communication with the central billing interface **123** (also sometimes referred to as the external billing management interface or billing communication interface) **127**, which is in communication with the central billing system **123**. As shown, an order management **180** and subscriber management **182** are also in communication with the central provider core network **110** for facilitating order and subscriber management of services for the devices **100** in accordance with some embodiments.

In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or

12

combinations of network elements) provides a device/network level service usage information collection, aggregation, mediation, and reporting function. In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) collects device generated usage information for one or more devices on the wireless network (e.g., devices **100**); and provides the device generated usage information in a syntax and a communication protocol that can be used by the wireless network to augment or replace network generated usage information for the one or more devices on the wireless network. In some embodiments, the syntax is a charging data record (CDR), and the communication protocol is selected from one or more of the following: 3GPP, 3GPP2, or other communication protocols. In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) includes a service usage data store (e.g., a billing aggregator) and a rules engine for aggregating the collected device generated usage information. In some embodiments, the syntax is a charging data record (CDR), and the network device is a CDR feed aggregator, and the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) also aggregates CDRs for the one or more devices on the wireless network; applies a set of rules to the aggregated CDRs using a rules engine (e.g., bill by account, transactional billing, and/or any other billing or other rules for service usage information collection, aggregation, mediation, and reporting), and communicates a new set of CDRs for the one or more devices on the wireless network to a billing interface or a billing system (e.g., providing a CDR with a billing offset by account/service). In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) communicates a new set of CDRs for the one or more devices on the wireless network to a billing interface or a billing system. In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) communicates with a service controller to collect the device generated usage information for the one or more devices on the wireless network. In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) communicates with a service controller, in which the service controller is in communication with a billing interface or a billing system. In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) communicates the device generated usage information to a billing interface or a billing system. In some embodiments, the CDR storage, aggregation, mediation, feed (and/or other network elements or combinations of network elements) communicates with a transport gateway and/or a Radio Access Network (RAN) gateway to collect the network generated usage information for the one or more devices on the wireless network. In some embodiments, the service controller **122** communicates the device generated service usage information to the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements).

In some embodiments, the CDR storage, aggregation, mediation, feed **118** (and/or other network elements or combinations of network elements) performs rules for performing a bill by account aggregation and mediation function. In some embodiments, the service controller **122** in communication with the CDR storage, aggregation, media-

13

tion, feed 118 (and/or other network elements or combinations of network elements) performs a rules engine for aggregating and mediating the device generated usage information. In some embodiments, a rules engine device in communication with the CDR storage, aggregation, mediation, feed 118 (and/or other network elements or combinations of network elements) performs a rules engine for aggregating and mediating the device generated usage information.

In some embodiments, the rules engine is included in (e.g., integrated with/part of) the CDR storage, aggregation, mediation, feed 118. In some embodiments, the rules engine and associated functions, as described herein, is a separate function/device. In some embodiments, the service controller 122 performs some or all of these rules engine based functions, as described herein, and communicates with the central billing interface 127. In some embodiments, the service controller 122 performs some or all of these rules engine based functions, as described herein, and communicates with the central billing system 123.

In some embodiments, duplicate CDRs are sent from the network equipment to the billing system 123 that is used for generating service billing. In some embodiments, duplicate CDRs are filtered to send only those CDRs/records for devices controlled by the service controller and/or service processor (e.g., the managed devices). For example, this approach can provide for the same level of reporting, lower level of reporting, and/or higher level of reporting as compared to the reporting required by the central billing system 123.

In some embodiments, a bill-by-account billing offset is provided. For example, bill-by-account billing offset information can be informed to the central billing system 123 by providing a CDR aggregator feed that aggregates the device based service usage data feed to provide a new set of CDRs for the managed devices to the central billing interface 127 and/or the central billing system 123. In some embodiments, transaction billing is provided using similar techniques. For example, transaction billing log information can be provided to the central billing interface 127 and/or the central billing system 123.

In some embodiments, the rules engine (e.g., performed by the service usage 118 or another network element, as described herein) provides a bill-by-account billing offset. For example, device generated usage information (e.g., charging data records (CDRs)) includes a transaction type field (e.g., indicating a type of service for the associated service usage information). The rules engine can apply a rule or a set of rules based on the identified service associated with the device generated usage information to determine a bill-by-account billing offset (e.g., a new CDR can be generated to provide the determined bill-by-account billing offset). In some examples, the determined bill-by-account billing offset can be provided as a credit to the user's service usage account (e.g., a new CDR can be generated with a negative offset for the user's service usage account, such as for network chatter service usage, or transactional service usage, or for any other purposes based on one or more rules performed by the rules engine).

As another example, for a transactional service, a first new CDR can be generated with a negative offset for the user's service usage account for that transactional service related usage, and a second new CDR can be generated with a positive service usage value to charge that same service usage to the transactional service provider (e.g., Amazon, eBay, or another transactional service provider). In some embodiments, the service controller 122 generates these two

14

new CDRs, and the service usage 118 stores, aggregates, and communicates these two new CDRs to the central billing interface 127. In some embodiments, the service controller 122 generates these two new CDRs, and the service usage 118 stores, aggregates, and communicates these two new CDRs to the central billing interface 127, in which the central billing interface 127 applies rules (e.g., performs the rules engine for determining the bill-by-account billing offset).

In some embodiments, the service controller 122 sends the device generated CDRs to the rules engine (e.g., service usage 118), and the rules engine applies one or more rules, such as those described herein and/or any other billing/service usage related rules as would be apparent to one of ordinary skill in the art. In some embodiments, the service controller 122 generates CDRs similar to other network elements, and the rules (e.g., bill-by-account) are performed in the central billing interface 127. For example, for the service controller 122 to generate CDRs similar to other network elements, in some embodiments, the service controller 122 is provisioned on the wireless network and behaves substantially similar to other CDR generators on the network) as would be apparent to one of ordinary skill in the art.

In some embodiments, the service controller 122 is provisioned as a new type of networking function that is recognized as a valid and secure source for CDRs by the other necessary elements in the network (e.g., the Service Usage History/CDR Aggregation and Mediation Server 118). In some embodiments, in which the network apparatus typically only recognize CDRs from certain types of networking equipment (e.g., RAN Gateway 410 or Transport Gateway 420 (as shown in FIG. 3)), then the Service Controller 122 can provide authentication credentials to the other networking equipment that indicate it is one of the approved types of equipment (e.g., for purposes of generating/providing CDRs). In some embodiments, the link between the Service Controller 122 and the necessary CDR aggregation and mediation equipment is secured, authenticated, encrypted and/or signed.

In some embodiments, the CDR storage, aggregation, mediation, feed 118 discards the network based service usage information (e.g., network based CDRs) received from one or more network elements. In these embodiments, the service controller 122 can provide the device based service usage information (e.g., device based CDRs) to the CDR storage, aggregation, mediation, feed 118 (e.g., the CDR storage, aggregation, mediation, feed 118 can just provide a store, aggregate, and communication function(s)), and the device based service usage information is provided to the central billing interface 127 or the central billing system 123.

In some embodiments, the device based CDRs and/or new CDRs generated based on execution of a rules engine as described herein is provided only for devices that are managed and/or based on device group, service plan, or any other criteria, categorization, and/or grouping.

FIG. 2 illustrates another wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing in accordance with some embodiments. As shown in FIG. 2, some devices 100 are in communication with DOCSIS Head End 125 and some devices 100 are in communication with DSLAM 125, which are in communication with the central provider access network 109.

FIG. 3 illustrates another wireless network architecture for providing device assisted CDR creation, aggregation,

15

mediation and billing in accordance with some embodiments. Referring now to the 4G/3G/2G access network as shown in FIG. 3, the 4G/3G and 3G/2G base stations/nodes 125 are in communication with a 4G/3G/2G Radio Access Network (RAN) gateway 410 via a radio access network 405, which are in communication with a 4G/3G/2G transport gateway 420 via an access transport network 415. The central provider core network 110 is in network communication with the access transport network 415 (e.g., via a dedicated/leased line, and as shown, via a firewall 124). The Internet, data, and/or phone network 120 is available via a firewall 124 and the transport gateway(s) 420, as shown. Also, as shown, a network apparatus provisioning system 160, order management 180, and subscriber management 182 are in communication with the central provider core network 110. As shown, a AAA server 121, a mobile wireless center/Home Location Register (HLR) 132, a DNS/DHCP 126, and CDR storage, aggregation, mediation, feed 118 are also in communication with the access transport network 415. The central billing system 123 and the central billing interface 127 are shown in communication with the central provider core network 110.

As shown, FIG. 3 includes a 4G/3G/2G wireless network operated by, for example, a central provider. In some embodiments, each of the wireless devices 100 includes a service processor 115 (as shown), and each service processor connects through a secure control plane link to a service controller 122. In some embodiments, the network based service usage information (e.g., network generated CDRs) is obtained from Radio Access Network (RAN) gateway(s) 410 and/or transport gateway(s) 420. In some embodiments, device based service usage information (e.g., device assisted CDRs) are generated by the service processor 115 and/or service controller 122 for some or all of the wireless devices 100 using similar techniques as described herein, and in some embodiments, such device based service usage information (e.g., device assisted CDRs) is sent to the CDR storage, aggregation, mediation, feed 118 (e.g., the CDR storage, aggregation, mediation, feed 118 can just provide a store, aggregate, and communication function(s)), and/or to the central billing interface 127 or the central billing system 123, as similarly described herein with respect to various embodiments.

FIG. 4 illustrates provisioning of a wireless network for providing device assisted CDR creation, aggregation, mediation and billing in accordance with some embodiments. As shown in FIG. 4, the provisioning of various network equipment is provided as shown to recognize each other as an authorized source of CDRs (e.g., this can be done manually or in an automated manner). For example, order management 180, subscriber management, billing interface 127, billing system 123, network provisioning system 160, service controller 122, access network AAA server 121, mobile wireless center 132, and CDR storage, aggregation, mediation feed 118 communicate with each other for such provisioning, which can be implemented using various techniques. In some embodiments, the various network elements are provisioned to recognize device assisted CDRs being generated by the service controller 122, which, for example, can be provided to the billing interface 127 and/or the billing system 123. In some embodiments, network generated CDRs are provided by RAN/Access gateway 410, aggregation/transport gateway 425, and/or base station controller 125. In some embodiments, other network elements generate/receive/store device assisted CDRs.

In some embodiments, provisioning of various network equipment is provided to recognize a given device as

16

belonging to a device group that supports a service usage and/or billing plan that relies upon and/or utilizes device assisted CDRs.

In some embodiments, the CDR formats, transaction codes, and CDR transmission destinations are programmed for each device that generates CDRs, including the service controller 122 (e.g., in some embodiments, the service controller 122 is the intermediary for CDRs) and/or service processor 115 (e.g., in some embodiments, the device sends CDRs to network CDR aggregation or billing interface 127/billing system 123 with no intermediate server function).

While FIGS. 4 through 8 provide various embodiments in the context of wireless 3G/4G networks, it will be apparent to one of ordinary skill in the art that these techniques can similarly be applied to other types of access networks including, for example, WiFi, cable, DSL, fiber, satellite, and/or other networks.

FIG. 5 illustrates a network architecture for providing device assisted CDRs in accordance with some embodiments. As shown, network generated CDRs are sent from various network elements to the CDR storage, aggregation, mediation, feed 118 and the service controller 122, as shown in dashed lines with arrows in FIG. 5. In some embodiments, the network generated CDRs are used for verification of device assisted service (DAS) usage and/or billing information. In some embodiments, the network generated CDRs are provided to the service controller 122, and the service controller 122 implements aggregation and/or mediation rules to examine and, in some cases, aggregate and/or mediate network generated/based CDRs with device assisted/based CDRs.

In some embodiments, device assisted CDRs are sent from the service controller 122 to CDR storage, aggregation, mediation, feed 118 and communicated to the billing system 123, as shown in solid lines with arrows in FIG. 5. In some embodiments, CDR storage, aggregation, mediation, feed 118 uses DAS service usage CDRs to augment network generated/based CDRs with bill-by-account transaction codes (e.g., as similarly described herein). In some embodiments, CDR storage, aggregation, mediation, feed 118 implements aggregation and/or mediation rules to account for DAS CDR usage amount in a new bill-by-account transaction code and removes the same service usage amount from the bulk device account transaction code. In some embodiments, a first DAS CDR is sent for the new bill by account transaction code, and a second DAS CDR is sent to be used as a correction (credit) to the main device usage account transaction code, and CDR storage, aggregation, mediation, feed 118 implements the rules to perform this mediation. In some embodiments, a first DAS CDR is used for a given bill-by-account transaction code, and a second DAS CDR is used as the main device account transaction code, in which the service controller 122 (or device) has already implemented the mediation rules so that CDR storage, aggregation, mediation, feed 118 simply passes such DAS CDRs to billing after aggregating them.

FIG. 6 illustrates another network architecture for providing device assisted CDRs in accordance with some embodiments. FIG. 6 also shows the communication of device assisted CDRs and network generated CDRs using solid and dashed lines with arrows, respectively. As shown, in some embodiments, CDR storage, aggregation, mediation, feed 118 sends network based CDRs to service controller 122 for various purposes, such as those previously described herein.

17

In some embodiments, service controller **122** sends DAS CDRs to billing for various uses by the billing system **123**. In some embodiments, the billing system **123** uses DAS service usage CDRs to augment network based CDRs with bill-by-account transaction codes. In some embodiments, the billing system **123** implements aggregation and/or mediation rules to account for DAS CDR usage amount in a new bill-by-account transaction code and removes the same service usage amount from the bulk device account transaction code. In some embodiments, a first DAS CDR is sent for the new bill by account transaction code, and a second DAS CDR is sent to be used as a correction (credit) to the main device usage account transaction code, and the billing system **123** implements the rules to perform this mediation. In some embodiments, a first DAS CDR is used for a given bill-by-account transaction code, and a second is used as the main device account transaction code, in which the service controller **122** (or device) has already implemented the mediation rules so that the billing system **123** simply passes such DAS CDRs after aggregating them.

FIG. **7** illustrates another network architecture for providing device assisted CDRs in accordance with some embodiments. FIG. **7** also shows the communication of device assisted CDRs and network generated CDRs using solid and dashed lines with arrows, respectively. FIG. **7** is similar to FIG. **6**, except as shown in FIG. **7**, service usage information is passed through the billing interface **127** instead of the billing CDR aggregation interface. For example, the service usage detailed bill-by-account information and offset (credit) information can be formatted as a CDR or can be formatted in a higher level syntax as required by the billing interface **127**.

FIG. **8** illustrates another network architecture for providing device assisted CDRs in accordance with some embodiments. FIG. **8** also shows the communication of device assisted CDRs and network generated CDRs using solid and dashed lines with arrows, respectively. In some embodiments, as shown in FIG. **8**, the central provider need not modify the existing CDR storage, aggregation, mediation, feed **118**, so the additional aggregation and mediation rules discussed above with respect to FIG. **5** are implemented as a new layer of rules in a new network function, shown as secondary DAS CDR aggregation mediation **118A**, that is located between the billing system and the CDR storage, aggregation, mediation, feed **118**. For example, this new network function (e.g., secondary DAS CDR aggregation mediation **118A**) can reside in the network (as shown) or in the service processor **115**, in the service controller **122**, or elsewhere in the network or on the device. In some embodiments, the CDRs described herein include QoS tagging information specifying the QoS level for the service usage. In these embodiments, a service controller (e.g., or other network element providing similar functions for this purpose) is used to provide a centralized function to collect and organize QoS tagged charging records and aggregate, format and/or mediate those records into a format that is compatible with and can be processed by a billing system. In some embodiments, a service controller is used for collection, mediation, and/or transmission to billing of device assisted charging records that include charging level.

In some embodiments, charging records that include QoS tags are used to verify QoS policy adherence in the device. In some embodiments, charging records that include QoS tags are used to verify QoS policy adherence in other network elements (e.g., proxy router or server, other network elements that reserve, establish, provision or control QoS channels). In some embodiments, the QoS policies being

18

verified are one or more of QoS level authorization policy, QoS request policy, QoS reservation policy, QoS provisioning policy, QoS control policy, QoS charging or accounting policy, QoS billing policy, QoS roaming policy, and/or QoS notification policy.

In some embodiments, a service controller (e.g., or other network element providing similar functions for this purpose) is used for collection, aggregation and/or transmission of QoS tagged usage records for the purpose of displaying to a user interface to inform a user of status of QoS related charging, accounting, service plan and/or billing. In some embodiments, the user interface is a device user interface. In some embodiments, the user interface is an web site user interface.

In some embodiments, a service controller (e.g., or other network element providing similar functions for this purpose) is used for collection, aggregation, and/or transmission of QoS tagged usage records for the purpose of synchronizing a user interface to inform a user of status of QoS related charging, accounting, service plan, and/or billing.

FIG. **9** is a functional diagram illustrating a device based service processor **115** and a service controller **122** in accordance with some embodiments. For example, this provides relatively full featured device based service processor implementation and service controller implementation. As shown, this corresponds to a networking configuration in which the service controller **122** is connected to the Internet, data, and/or phone network **120** and not directly to the access network **1610**. As shown, a data plane (e.g., service traffic plane) communication path is shown in solid line connections and control plane (e.g., service control plane) communication path is shown in dashed line connections. As will be apparent, the division in functionality between one device agent and another is based on, for example, design choices, networking environments, devices and/or services/applications, and various different combinations can be used in various different implementations. For example, the functional lines can be re-drawn in any way that the product designers see fit. As shown, this includes certain divisions and functional breakouts for device agents as an illustrative implementation, although other, potentially more complex, embodiments can include different divisions and functional breakouts for device agent functionality specifications, for example, in order to manage development specification and testing complexity and workflow. In addition, the placement of the agents that operate, interact with or monitor the data path can be moved or re-ordered in various embodiments. For example, the functional elements shown in FIG. **9** are described below with respect to FIGS. **10** and **11**.

As shown in FIG. **9**, service processor **115** includes a service control device link **1691**. For example, as device based service control techniques involving supervision across a network become more sophisticated, it becomes increasingly important to have an efficient and flexible control plane communication link between the device agents and the network elements communicating with, controlling, monitoring, or verifying service policy. In some embodiments, the service control device link **1691** provides the device side of a system for transmission and reception of service agent to/from network element functions. In some embodiments, the traffic efficiency of this link is enhanced by buffering and framing multiple agent messages in the transmissions. In some embodiments, the traffic efficiency is further improved by controlling the transmission frequency or linking the transmission frequency to the rate of service usage or traffic usage. In some embodiments, one or more levels of security or encryption are used to make the link

19

robust to discovery, eavesdropping or compromise. In some embodiments, the service control device link **1691** also provides the communications link and heartbeat timing for the agent heartbeat function. As discussed below, various embodiments disclosed herein for the service control device link **1691** provide an efficient and secure solution for transmitting and receiving service policy implementation, control, monitoring and verification information with other network elements.

As shown in FIG. 9, the service controller **122** includes a service control server link **1638**. In some embodiments, device based service control techniques involving supervision across a network (e.g., on the control plane) are more sophisticated, and for such it is increasingly important to have an efficient and flexible control plane communication link between the device agents (e.g., of the service processor **115**) and the network elements (e.g., of the service controller **122**) communicating with, controlling, monitoring, or verifying service policy. For example, the communication link between the service control server link **1638** of service controller **122** and the service control device link **1691** of the service processor **115** can provide an efficient and flexible control plane communication link, a service control link **1653** as shown in FIG. 9, and, in some embodiments, this control plane communication link provides for a secure (e.g., encrypted) communications link for providing secure, bidirectional communications between the service processor **115** and the service controller **122**. In some embodiments, the service control server link **1638** provides the network side of a system for transmission and reception of service agent to/from network element functions. In some embodiments, the traffic efficiency of this link is enhanced by buffering and framing multiple agent messages in the transmissions (e.g., thereby reducing network chatter). In some embodiments, the traffic efficiency is further improved by controlling the transmission frequency and/or linking the transmission frequency to the rate of service usage or traffic usage. In some embodiments, one or more levels of security and/or encryption are used to secure the link against potential discovery, eavesdropping or compromise of communications on the link. In some embodiments, the service control server link **1638** also provides the communications link and heartbeat timing for the agent heartbeat function.

In some embodiments, the service control server link **1638** provides for securing, signing, encrypting and/or otherwise protecting the communications before sending such communications over the service control link **1653**. For example, the service control server link **1638** can send to the transport layer or directly to the link layer for transmission. In another example, the service control server link **1638** further secures the communications with transport layer encryption, such as TCP TLS SSH version 1 or 2 or another secure transport layer protocol. As another example, the service control server link **1638** can encrypt at the link layer, such as using IPSEC, various possible VPN services, other forms of IP layer encryption and/or another link layer encryption technique.

As shown in FIG. 9, the service controller **122** includes an access control integrity server **1654**. In some embodiments, the access control integrity server **1654** collects device information on service policy, service usage, agent configuration and/or agent behavior. For example, the access control integrity server **1654** can cross check this information to identify integrity breaches in the service policy implementation and control system. In another example, the access

20

control integrity server **1654** can initiate action when a service policy violation or a system integrity breach is suspected.

In some embodiments, the access control integrity server **1654** (and/or some other agent of service controller **122**) acts on access control integrity agent **1694** reports and error conditions. Many of the access control integrity agent **1654** checks can be accomplished by the server. For example, the access control integrity agent **1654** checks include one or more of the following: service usage measure against usage range consistent with policies (e.g., usage measure from the network and/or from the device); configuration of agents; operation of the agents; and/or dynamic agent download.

In some embodiments, the access control integrity server **1654** (and/or some other agent of service controller **122**) verifies device service policy implementations by comparing various service usage measures (e.g., based on network monitored information, such as by using IPDRs or CDRs, and/or local service usage monitoring information) against expected service usage behavior given the policies that are intended to be in place. For example, device service policy implementations can include measuring total data passed, data passed in a period of time, IP addresses, data per IP address, and/or other measures such as location, downloads, email accessed, URLs, and comparing such measures expected service usage behavior given the policies that are intended to be in place.

In some embodiments, the access control integrity server **1654** (and/or some other agent of service controller **122**) verifies device service policy, and the verification error conditions that can indicate a mismatch in service measure and service policy include one or more of the following: unauthorized network access (e.g., access beyond ambient service policy limits); unauthorized network speed (e.g., average speed beyond service policy limit); network data amount does not match policy limit (e.g., device not stop at limit without re-up/revising service policy); unauthorized network address; unauthorized service usage (e.g., VOIP, email, and/or web browsing); unauthorized application usage (e.g., email, VOIP, email, and/or web); service usage rate too high for plan, and policy controller not controlling/throttling it down; and/or any other mismatch in service measure and service policy. Accordingly, in some embodiments, the access control integrity server **1654** (and/or some other agent of service controller **122**) provides a policy/service control integrity service to continually (e.g., periodically and/or based on trigger events) verify that the service control of the device has not been compromised and/or is not behaving out of policy.

As shown in FIG. 9, service controller **122** includes a service history server **1650**. In some embodiments, the service history server **1650** collects and records service usage or service activity reports from the Access Network AAA Server **1621** and the Service Monitor Agent **1696**. For example, although service usage history from the network elements can in certain embodiments be less detailed than service history from the device, the service history from the network can provide a valuable source for verification of device service policy implementation, because, for example, it is extremely difficult for a device error or compromise event on the device to compromise the network based equipment and software. For example, service history reports from the device can include various service tracking information, as similarly described above. In some embodiments, the service history server **1650** provides the service history on request to other servers and/or one or more agents. In some embodiments, the service history server

21

1650 provides the service usage history to the device service history 1618. In some embodiments, for purposes of facilitating the activation tracking service functions (described below), the service history server 1650 maintains a history of which networks the device has connected to. For example, this network activity summary can include a summary of the networks accessed, activity versus time per connection, and/or traffic versus time per connection. As another example, this activity summary can further be analyzed or reported to estimate the type of service plan associated with the traffic activity for the purpose of bill sharing reconciliation.

As shown in FIG. 9, service controller 122 includes a policy management server 1652. In some embodiments, the policy management server 1652 transmits policies to the service processor 115 via the service control link 1653. In some embodiments, the policy management server 1652 manages policy settings on the device (e.g., various policy settings as described herein with respect to various embodiments) in accordance with a device service profile. In some embodiments, the policy management server 1652 sets instantaneous policies on policy implementation agents (e.g., policy implementation agent 1690). For example, the policy management server 1652 can issue policy settings, monitor service usage and, if necessary, modify policy settings. For example, in the case of a user who prefers for the network to manage their service usage costs, or in the case of any adaptive policy management needs, the policy management server 1652 can maintain a relatively high frequency of communication with the device to collect traffic and/or service measures and issue new policy settings. In this example, device monitored service measures and any user service policy preference changes are reported, periodically and/or based on various triggers/events/requests, to the policy management server 1652. In this example, user privacy settings generally require secure communication with the network (e.g., a secure service control link 1653), such as with the policy management server 1652, to ensure that various aspects of user privacy are properly maintained during such configuration requests/policy settings transmitted over the network. For example, information can be compartmentalized to service policy management and not communicated to other databases used for CRM for maintaining user privacy.

In some embodiments, the policy management server 1652 provides adaptive policy management on the device. For example, the policy management server 1652 can issue policy settings and objectives and rely on the device based policy management (e.g., service processor 115) for some or all of the policy adaptation. This approach can require less interaction with the device thereby reducing network chatter on service control link 1653 for purposes of device policy management (e.g., network chatter is reduced relative to various server/network based policy management approaches described above). This approach can also provide robust user privacy embodiments by allowing the user to configure the device policy for user privacy preferences/settings so that, for example, sensitive information (e.g., geo-location data, website history) is not communicated to the network without the user's approval. In some embodiments, the policy management server 1652 adjusts service policy based on time of day. In some embodiments, the policy management server 1652 receives, requests or otherwise obtains a measure of network availability and adjusts traffic shaping policy and/or other policy settings based on available network capacity.

22

As shown in FIG. 9, service controller 122 includes a network traffic analysis server 1656. In some embodiments, the network traffic analysis server 1656 collects/receives service usage history for devices and/or groups of devices and analyzes the service usage. In some embodiments, the network traffic analysis server 1656 presents service usage statistics in various formats to identify improvements in network service quality and/or service profitability. In other embodiments, the network traffic analysis server 1656 estimates the service quality and/or service usage for the network under variable settings on potential service policy. In other embodiments, the network traffic analysis server 1656 identifies actual or potential service behaviors by one or more devices that are causing problems for overall network service quality or service cost.

As shown in FIG. 9, service controller 122 includes a beta test server 1658. In some embodiments, the beta test server 1658 publishes candidate service plan policy settings to one or more devices. In some embodiments, the beta test server 1658 provides summary reports of network service usage or user feedback information for one or more candidate service plan policy settings. In some embodiments, the beta test server 1658 provides a mechanism to compare the beta test results for different candidate service plan policy settings or select the optimum candidates for further policy settings optimization.

As shown in FIG. 9, service controller 122 includes a service download control server 1660. In some embodiments, the service download control server 1660 provides a download function to install and/or update service software elements (e.g., the service processor 115 and/or agents/components of the service processor 115) on the device, as described herein.

As shown in FIG. 9 service controller 122 includes a billing event server 1662. In some embodiments, the billing event server 1662 collects billing events, provides service plan information to the service processor 115, provides service usage updates to the service processor 115, serves as interface between device and central billing server 1619, and/or provides trusted third party function for certain ecommerce billing transactions.

As shown in FIG. 9, the Access Network AAA server 1621 is in network communication with the access network 1610. In some embodiments, the Access Network AAA server 1621 provides the necessary access network AAA services (e.g., access control and authorization functions for the device access layer) to allow the devices onto the central provider access network and the service provider network. In some embodiments, another layer of access control is required for the device to gain access to other networks, such as the Internet, a corporate network and/or a machine to machine network. This additional layer of access control can be implemented, for example, by the service processor 115 on the device. In some embodiments, the Access Network AAA server 1621 also provides the ability to suspend service for a device and resume service for a device based on communications received from the service controller 122. In some embodiments, the Access Network AAA server 1621 also provides the ability to direct routing for device traffic to a quarantine network or to restrict or limit network access when a device quarantine condition is invoked. In some embodiments, the Access Network AAA server 1621 also records and reports device network service usage (e.g., device network service usage can be reported to device service history 1618).

As shown in FIG. 9, the device service history 1618 is in network communication with the access network 1610. In

some embodiments, the device service history **1618** provides service usage data records used for various purposes in various embodiments. In some embodiments, the device service history **1618** is used to assist in verifying service policy implementation. In some embodiments, the device service history **1618** is used to verify service monitoring. In some embodiments, the device service history **1618** is used to verify billing records and/or billing policy implementation. In some embodiments, the device service history **1618** is used to synchronize and/or verify the local service usage counter.

As shown in FIG. 9, the central provider billing server **1619** is in network communication with the access network **1610**. In some embodiments, the central provider billing server **1619** provides a mediation function for central provider billing events. For example, the central provider billing server **1619** can accept service plan changes. In some embodiments, the central provider billing server **1619** provides updates on device service usage, service plan limits and/or service policies. In some embodiments, the central provider billing server **1619** collects billing events, formulates bills, bills service users, provides certain billing event data and service plan information to the service controller **122** and/or device **100**.

As shown in FIG. 9, in some embodiments, modem selection and control **1811** selects the access network connection and is in communication with the modem firewall **1655**, and modem drivers **1831**, **1815**, **1814**, **1813**, **1812** convert data traffic into modem bus traffic for one or more modems and are in communication with the modem selection and control **1811**. In some embodiments, different profiles are selected based on the selected network connection (e.g., different service profiles/policies for WWAN, WLAN, WPAN, Ethernet and/or DSL network connections), which is also referred to herein as multimode profile setting. For example, service profile settings can be based on the actual access network (e.g., home DSL/cable or work network) behind the Wi-Fi not the fact that it is Wi-Fi (or any other network, such as DSL/cable, satellite, or T-1), which is viewed as different than accessing a Wi-Fi network at the coffee shop. For example, in a Wi-Fi hotspot situation in which there are a significant number of users on a DSL or T-1 backhaul, the service controller can sit in a service provider cloud or an MVNO cloud, the service controls can be provided by a VSP capability offered by the service provider or the service controller can be owned by the hotspot service provider that uses the service controller on their own without any association with an access network service provider. For example, the service processors can be controlled by the service controller to divide up the available bandwidth at the hotspot according to QoS or user sharing rules (e.g., with some users having higher differentiated priority (potentially for higher service payments) than other users). As another example, ambient services (as similarly described herein) can be provided for the hotspot for verified service processors.

In some embodiments, the service processor **115** and service controller **122** are capable of assigning multiple service profiles associated with multiple service plans that the user chooses individually or in combination as a package. For example, a device **100** starts with ambient services that include free transaction services wherein the user pays for transactions or events rather than the basic service (e.g., a news service, eReader, PND service, pay as you go session Internet) in which each service is supported with a bill by account capability to correctly account for any subsidized partner billing to provide the transaction services (e.g.,

Barnes and Noble may pay for the eReader service and offer a revenue share to the service provider for any book or magazine transactions purchased from the device **100**). In some embodiments, the bill by account service can also track the transactions and, in some embodiments, advertisements for the purpose of revenue sharing, all using the service monitoring capabilities disclosed herein. After initiating services with the free ambient service discussed above, the user may later choose a post-pay monthly Internet, email and SMS service. In this case, the service controller **122** would obtain from the billing system **123** in the case of network based billing (or in some embodiments, the service controller **122** billing event server **1622** in the case of device based billing) the billing plan code for the new Internet, email and SMS service. In some embodiments, this code is cross referenced in a database (e.g., the policy management server **1652**) to find the appropriate service profile for the new service in combination with the initial ambient service. The new superset service profile is then applied so that the user maintains free access to the ambient services, and the billing partners continue to subsidize those services, the user also gets access to Internet services and may choose the service control profile (e.g., from one of the embodiments disclosed herein). The superset profile is the profile that provides the combined capabilities of two or more service profiles when the profiles are applied to the same device **100** service processor. In some embodiments, the device **100** (service processor **115**) can determine the superset profile rather than the service controller **122** when more than one "stackable" service is selected by the user or otherwise applied to the device. The flexibility of the service processor **115** and service controller **122** embodiments described herein allow for a large variety of service profiles to be defined and applied individually or as a superset to achieve the desired device **100** service features.

As shown in FIG. 9, an agent communication bus **1630** represents a functional description for providing communication for the various service processor **115** agents and functions. In some embodiments, as represented in the functional diagram illustrated in FIG. 9, the architecture of the bus is generally multipoint to multipoint so that any agent can communicate with any other agent, the service controller or in some cases other components of the device, such as user interface **1697** and/or modem components. As described below, the architecture can also be point to point for certain agents or communication transactions, or point to multipoint within the agent framework so that all agent communication can be concentrated, or secured, or controlled, or restricted, or logged or reported. In some embodiments, the agent communication bus is secured, signed, encrypted, hidden, partitioned and/or otherwise protected from unauthorized monitoring or usage. In some embodiments, an application interface agent (not shown) is used to literally tag or virtually tag application layer traffic so that the policy implementation agent(s) **1690** has the necessary information to implement selected traffic shaping solutions. In some embodiments, an application interface agent (not shown) is in communication with various applications, including a TCP application **1604**, an IP application **1605**, and a voice application **1602**.

In some embodiments, device assisted services (DAS) techniques for providing an activity map for classifying or categorizing service usage activities to associate various monitored activities (e.g., by URL, by network domain, by website, by network traffic type, by application or application type, and/or any other service usage activity categorization/classification) with associated IP addresses are pro-

25

vided. In some embodiments, a policy control agent (not shown), service monitor agent **1696**, or another agent or function (or combinations thereof) of the service processor **115** provides a DAS activity map. In some embodiments, a policy control agent, service monitor agent, or another agent or function (or combinations thereof) of the service processor provides an activity map for classifying or categorizing service usage activities to associate various monitored activities (e.g., by Uniform Resource Locator (URL), by network domain, by website, by network traffic type, by application or application type, and/or any other service usage activity classification/categorization) with associated IP addresses. In some embodiments, a policy control agent, service monitor agent, or another agent or function (or combinations thereof) of the service processor determines the associated IP addresses for monitored service usage activities using various techniques to snoop the DNS request(s) (e.g., by performing such snooping techniques on the device **100** the associated IP addresses can be determined without the need for a network request for a reverse DNS lookup). In some embodiments, a policy control agent, service monitor agent, or another agent or function (or combinations thereof) of the service processor records and reports IP addresses or includes a DNS lookup function to report IP addresses or IP addresses and associated URLs for monitored service usage activities. For example, a policy control agent, service monitor agent, or another agent or function (or combinations thereof) of the service processor can determine the associated IP addresses for monitored service usage activities using various techniques to perform a DNS lookup function (e.g., using a local DNS cache on the monitored device **100**). In some embodiments, one or more of these techniques are used to dynamically build and maintain a DAS activity map that maps, for example, URLs to IP addresses, applications to IP addresses, content types to IP addresses, and/or any other categorization/classification to IP addresses as applicable. In some embodiments, the DAS activity map is used for various DAS traffic control and/or throttling techniques as described herein with respect to various embodiments. In some embodiments, the DAS activity map is used to provide the user various UI related information and notification techniques related to service usage as described herein with respect to various embodiments. In some embodiments, the DAS activity map is used to provide service usage monitoring, prediction/estimation of future service usage, service usage billing (e.g., bill by account and/or any other service usage/billing categorization techniques), DAS techniques for ambient services usage monitoring, DAS techniques for generating micro-CDRs (e.g., also referred to as service usage partition, service usage recording partition, service charging bucket, device generated CDRs, such as in the case where the device and not a network component are generating the usage records, ambient usage records, specialized service usage records, or other terms to indicate a service usage data record generated to provide a more refined or detailed breakdown of service usage for the device), and/or any of the various other DAS related techniques as described herein with respect to various embodiments.

In some embodiments, all or a portion of the service processor **115** functions disclosed herein are implemented in software. In some embodiments, all or a portion of the service processor **115** functions are implemented in hardware. In some embodiments, all or substantially all of the service processor **115** functionality (as discussed herein) is implemented and stored in software that can be performed on (e.g., executed by) various components in device **100**. In

26

some embodiments, it is advantageous to store or implement certain portions or all of service processor **115** in protected or secure memory so that other undesired programs (and/or unauthorized users) have difficulty accessing the functions or software in service processor **115**. In some embodiments, service processor **115**, at least in part, is implemented in and/or stored on secure non-volatile memory (e.g., non volatile memory can be secure non-volatile memory) that is not accessible without pass keys and/or other security mechanisms. In some embodiments, the ability to load at least a portion of service processor **115** software into protected non-volatile memory also requires a secure key and/or signature and/or requires that the service processor **115** software components being loaded into non-volatile memory are also securely encrypted and appropriately signed by an authority that is trusted by a secure software downloader function, such as service downloader. In some embodiments, a secure software download embodiment also uses a secure non-volatile memory. Those of ordinary skill in the art will also appreciate that all memory can be on-chip, off-chip, on-board and/or off-board.

FIG. **10** provides a table summarizing various service processor **115** functional elements in accordance with some embodiments. Many of these agents are similarly described above, and the table shown in FIG. **10** is not intended to be an exhaustive summary of these agents, nor an exhaustive description of all functions that the agents perform or are described herein, but rather FIG. **10** is provided as a summary aid in understanding the basic functions of each agent in accordance with some embodiments, and how the agents interact with one another, with the service controller server elements, and/or with other network functions in certain embodiments to form a reliable device based service delivery solution and/or platform.

FIG. **11** provides a table summarizing various service controller **122** functional elements in accordance with some embodiments. Many of these agents/elements are similarly described above, and the table shown in FIG. **11** is not intended to be an exhaustive summary of these server elements, nor an exhaustive description of all functions that the elements perform or are described herein, but rather FIG. **11** is provided as a summary aid in understanding the basic functions of each element in accordance with some embodiments, and how the elements interact with one another, certain network elements, and/or the service processor agents in certain embodiments to form a reliable device based service delivery solution and/or platform.

FIG. **12** illustrates a device stack providing various service usage measurement from various points in the networking stack for a service monitor agent, a billing agent, and an access control integrity agent to assist in verifying the service usage measures and billing reports in accordance with some embodiments. As shown in FIG. **12**, several service agents take part in data path operations to achieve various data path improvements, and, for example, several other service agents can manage the policy settings for the data path service, implement billing for the data path service, manage one or more modem selection and settings for access network connection, interface with the user and/or provide service policy implementation verification. Additionally, in some embodiments, several agents perform functions to assist in verifying that the service control or monitoring policies intended to be in place are properly implemented, the service control or monitoring policies are being properly adhered to, that the service processor or one or more service agents are operating properly, to prevent unintended errors in policy implementation or control, and/

or to prevent tampering with the service policies or control. As shown, the service measurement points labeled I through VI represent various service measurement points for service monitor agent 1696 and/or other agents to perform various service monitoring activities. Each of these measurement points can have a useful purpose in various embodiments described herein. For example, each of the traffic measurement points that is employed in a given design can be used by a monitoring agent to track application layer traffic through the communication stack to assist policy implementation functions, such as the policy implementation agent 1690, or in some embodiments, the modem firewall agent 1655 or the application interface agent, in making a determination regarding the traffic parameters or type once the traffic is farther down in the communication stack where it is sometimes difficult or impossible to make a complete determination of traffic parameters. The particular locations for the measurement points provided in these figures are intended as instructional examples, and other measurement points can be used for different embodiments, as will be apparent to one of ordinary skill in the art in view of the embodiments described herein. Generally, in some embodiments, one or more measurement points within the device can be used to assist in service control verification and/or device or service troubleshooting.

In some embodiments, the service monitor agent and/or other agents implement virtual traffic tagging by tracking or tracing packet flows through the various communication stack formatting, processing and encryption steps, and providing the virtual tag information to the various agents that monitor, control, shape, throttle or otherwise observe, manipulate or modify the traffic. This tagging approach is referred to herein as virtual tagging, because there is not a literal data flow, traffic flow or packet tag that is attached to flows or packets, and the book-keeping to tag the packet is done through tracking or tracing the flow or packet through the stack instead. In some embodiments, the application interface and/or other agents identify a traffic flow, associate it with a service usage activity and cause a literal tag to be attached to the traffic or packets associated with the activity. This tagging approach is referred to herein as literal tagging. There are various advantages with both the virtual tagging and the literal tagging approaches. For example, it can be preferable in some embodiments, to reduce the inter-agent communication required to track or trace a packet through the stack processing by assigning a literal tag so that each flow or packet has its own activity association embedded in the data. As another example, it can be preferable in some embodiments, to re-use portions of standard communication stack software or components, enhancing the verifiable traffic control or service control capabilities of the standard stack by inserting additional processing steps associated with the various service agents and monitoring points rather than re-writing the entire stack to correctly process literal tagging information, and in such cases, a virtual tagging scheme may be desired. As yet another example, some standard communication stacks provide for unused, unspecified or otherwise available bit fields in a packet frame or flow, and these unused, unspecified or otherwise available bit fields can be used to literally tag traffic without the need to re-write all of the standard communication stack software, with only the portions of the stack that are added to enhance the verifiable traffic control or service control capabilities of the standard stack needing to decode and use the literal tagging information encapsulated in the available bit fields. In the case of literal tagging, in some embodiments, the tags are removed prior to passing the packets or flows to the

network or to the applications utilizing the stack. In some embodiments, the manner in which the virtual or literal tagging is implemented can be developed into a communication standard specification so that various device or service product developers can independently develop the communication stack and/or service processor hardware and/or software in a manner that is compatible with the service controller specifications and the products of other device or service product developers.

It will be appreciated that although the implementation/use of any or all of the measurement points illustrated in FIG. 12 is not required to have an effective implementation, such as was similarly shown with respect to various embodiments described herein, various embodiments can benefit from these and/or similar measurement points. It will also be appreciated that the exact measurement points can be moved to different locations in the traffic processing stack, just as the various embodiments described herein can have the agents affecting policy implementation moved to different points in the traffic processing stack while still maintaining effective operation. In some embodiments, one or more measurement points are provided deeper in the modem stack where, for example, it is more difficult to circumvent and can be more difficult to access for tampering purposes if the modem is designed with the proper software and/or hardware security to protect the integrity of the modem stack and measurement point(s).

Referring to FIG. 12, describing the device communications stack from the bottom to the top of the stack as shown, the device communications stack provides a communication layer for each of the modems of the device at the bottom of the device communications stack. Example measurement point VI resides within or just above the modem driver layer. For example, the modem driver performs modem bus communications, data protocol translations, modem control and configuration to interface the networking stack traffic to the modem. As shown, measurement point VI is common to all modem drivers and modems, and it is advantageous for certain embodiments to differentiate the traffic or service activity taking place through one modem from that of one or more of the other modems. In some embodiments, measurement point VI, or another measurement point, is located over, within or below one or more of the individual modem drivers. The respective modem buses for each modem reside between example measurement points V and VI. In the next higher layer, a modem selection & control layer for multi-mode device based communication is provided. In some embodiments, this layer is controlled by a network decision policy that selects the most desirable network modem for some or all of the data traffic, and when the most desirable network is not available the policy reverts to the next most desirable network until a connection is established provided that one of the networks is available. In some embodiments, certain network traffic, such as verification, control, redundant or secure traffic, is routed to one of the networks even when some or all of the data traffic is routed to another network. This dual routing capability provides for a variety of enhanced security, enhanced reliability or enhanced manageability devices, services or applications. In the next higher layer, a modem firewall is provided. For example, the modem firewall provides for traditional firewall functions, but unlike traditional firewalls, in order to rely on the firewall for verifiable service usage control, such as access control and security protection from unwanted networking traffic or applications, the various service verification techniques and agents described herein are added to the firewall function to verify compliance with service policy and pre-

vent tampering of the service controls. In some embodiments, the modem firewall is implemented farther up the stack, possibly in combination with other layers as indicated in other Figures. In some embodiments, a dedicated firewall function or layer is provided that is independent of the other processing layers, such as the policy implementation layer, the packet forwarding layer and/or the application layer. In some embodiments, the modem firewall is implemented farther down the stack, such as within the modem drivers, below the modem drivers, or in the modem itself. Example measurement point IV resides between the modem firewall layer and an IP queuing and routing layer. As shown, an IP queuing and routing layer is separate from the policy implementation layer where the policy implementation agent implements a portion of the traffic control and/or service usage control policies. As described herein, in some embodiments, these functions are separated so that a standard network stack function can be used for IP queuing and routing, and the modifications necessary to implement the policy implementation agent functions can be provided in a new layer inserted into the standard stack. In some embodiments, the IP queuing and routing layer is combined with the traffic or service usage control layer. For example, a combined routing and policy implementation layer embodiment can also be used with the other embodiments, such as shown in FIG. 12. Measurement point III resides between the IP queuing and routing layer and a policy implementation agent layer. Measurement point II resides between the policy implementation agent layer and the transport layer, including TCP, UDP, and other IP as shown. The session layer resides above the transport layer, which is shown as a socket assignment and session management (e.g., basic TCP setup, TLS/SSL) layer. The network services API (e.g., HTTP, HTTPS, FTP (File Transfer Protocol), SMTP (Simple Mail Transfer Protocol), POP3, DNS) resides above the session layer. Measurement point I resides between the network services API layer and an application layer, shown as application service interface agent in the device communications stack of FIG. 12.

As shown in FIG. 12, the application service interface layer is above the standard networking stack API and, in some embodiments, its function is to monitor and in some cases intercept and process the traffic between the applications and the standard networking stack API. In some embodiments, the application service interface layer identifies application traffic flows before the application traffic flows are more difficult or practically impossible to identify farther down in the stack. In some embodiments, the application service interface layer in this way assists application layer tagging in both the virtual and literal tagging cases. In the case of upstream traffic, the application layer tagging is straight forward, because the traffic originates at the application layer. In some downstream embodiments, where the traffic or service activity classification relies on traffic attributes that are readily obtainable, such as source address or URL, application socket address, IP destination address, time of day or any other readily obtained parameter, the traffic type can be identified and tagged for processing by the firewall agent or another agent as it initially arrives. In other embodiments, as described herein, in the downstream case, the solution is generally more sophisticated when a traffic parameter that is needed to classify the manner in which the traffic flow is to be controlled or throttled is not readily available at the lower levels of the stack, such as association with an aspect of an application, type of content, something contained within TLS, IPSEC or other secure format, or other information associated with the traffic. Accordingly, in

some embodiments, the networking stack identifies the traffic flow before it is fully characterized, categorized or associated with a service activity, and then passes the traffic through to the application interface layer where the final classification is completed. In such embodiments, the application interface layer then communicates the traffic flow ID with the proper classification so that after an initial short traffic burst or time period the policy implementation agents can properly control the traffic. In some embodiments, there is also a policy for tagging and setting service control policies for traffic that cannot be fully identified with all sources of tagging including application layer tagging.

As shown in FIG. 12, a service monitor agent, which is also in communication with the agent communication bus 1630, communicates with various layers of the device communications stack. For example, the service monitor agent, performs monitoring at each of measurement points I through VI, receiving information including application information, service usage and other service related information, and assignment information. An access control integrity agent is in communication with the service monitor agent via the agent communications bus 1630, as also shown.

FIG. 13 illustrates an embodiment similar to FIG. 12 in which some of the service processor is implemented on the modem and some of the service processor is implemented on the device application processor in accordance with some embodiments. In some embodiments, a portion of the service processor is implemented on the modem (e.g., on modem module hardware or modem chipset) and a portion of the service processor is implemented on the device application processor subsystem. It will be apparent to one of ordinary skill in the art that variations of the embodiment depicted in FIG. 13 are possible where more or less of the service processor functionality is moved onto the modem subsystem or onto the device application processor subsystem. For example, such embodiments similar to that depicted in FIG. 13 can be motivated by the advantages of including some or all of the service processor network communication stack processing and/or some or all of the other service agent functions on the modem subsystem (e.g., and such an approach can be applied to one or more modems). For example, the service processor can be distributed as a standard feature set contained in a modem chipset hardware or software package or modem module hardware or software package, and such a configuration can provide for easier adoption or development by device OEMs, a higher level of differentiation for the chipset or modem module manufacturer, higher levels of performance or service usage control implementation integrity or security, specification or interoperability standardization, and/or other benefits.

Referring to FIG. 13, describing the device communications stack from the bottom to the top of the stack as shown, the device communications stack provides a communication layer for modem MAC/PHY layer at the bottom of the device communications stack. Measurement point IV resides above the modem MAC/PHY layer. The modem firewall layer resides between measurement points IV and III. In the next higher layer, the policy implementation agent is provided, in which the policy implementation agent is implemented on the modem (e.g., on modem hardware). Measurement point II resides between the policy implementation agent and the modem driver layer, which is then shown below a modem bus layer. The next higher layer is shown as the IP queuing and routing layer, followed by the transport layer, including TCP, UDP, and other IP as shown.

31

The session layer resides above the transport layer, which is shown as a socket assignment and session management (e.g., basic TCP setup, TLS/SSL) layer. The network services API (e.g., HTTP, HTTPS, FTP (File Transfer Protocol), SMTP (Simple Mail Transfer Protocol), POP3, DNS) resides above the session layer. Measurement point I resides between the network services API layer and an application layer, shown as application service interface agent in the device communications stack of FIG. 13.

FIGS. 14A through 14E illustrate various embodiments of intermediate networking devices that include a service processor for the purpose of verifiable service usage measurement, reporting, and billing reports in accordance with some embodiments. For example, FIGS. 14A through 14E illustrate various extended modem alternatives for access network connection through an intermediate modem or networking device combination that has a connection (e.g., LAN connection) to one or more devices 100.

In some embodiments, device 100 includes a 3G and/or 4G network access connection in combination with the Wi-Fi LAN connection to the device 100. For example, the intermediate device or networking device combination can be a device that simply translates the Wi-Fi data to the WWAN access network without implementing any portion of the service processor 115 as shown in FIG. 14A. In some embodiments, an intermediate device or networking device combination includes a more sophisticated implementation including a networking stack and some embodiments, a processor, as is the case for example if the intermediate networking device or networking device combination includes a router function, in which case the service processor 115 can be implemented in part or entirely on the intermediate modem or networking device combination. The intermediate modem or networking device combination can also be a multi-user device in which more than one user is gaining access to the 3G or 4G access network via the Wi-Fi LAN connection. In the case of such a multi-user network, the access network connection can include several managed service links using multiple instantiations of service processor 115, each instantiation, for example, being implemented in whole or in part on device 100 with the intermediate modem or networking device combination only providing the translation services from the Wi-Fi LAN to the WWAN access network.

Referring now to FIGS. 14B through D, in some embodiments, the service processors 115 are implemented in part or in whole on the intermediate modem or networking device combination. In the case where the service processor 115 is implemented in part or in whole on the intermediate modem or networking device combination, the service processor 115 can be implemented for each device or each user in the network so that there are multiple managed service provider accounts all gaining access through the same intermediate modem or networking device combination. In some embodiments, the functions of service processor 115 are implemented on an aggregate account that includes the WWAN access network traffic for all of the users or devices connected to the Wi-Fi LAN serviced by the intermediate modem or networking device combination. In some embodiments, the central provider can also provide an aggregated account service plan, such as a family plan, a corporate user group plan and/or an instant hotspot plan. In the case where there is one account for the intermediate modem or networking device combination, the intermediate modem or networking device combination can implement a local division of services to one or more devices 100 or users in which the services are controlled or managed by the intermediate

32

modem or networking device combination or the device 100, but the management is not subject to service provider control and is auxiliary to the service management or service policy implementation performed by service processors 115. In some embodiments, another service model can also be supported in which there is an aggregate service provider plan associated with one intermediate modem or networking device combination, or a group of intermediate modems or networking device combinations but where each user or device still has its own service plan that is a sub-plan under the aggregate plan so that each user or device has independent service policy implementation with a unique instantiation of service processor 115 rather than aggregate service policy implementation across multiple users in the group with a single instantiation of service processor 115.

As shown in FIG. 14B, in some embodiments, device 100 includes a Wi-Fi modem, a Wi-Fi modem combined with a 3G and/or 4G WWAN modem on intermediate modem or networking device combination 1510, and the intermediate modem or networking device combination forwards WWAN access network traffic to and from device 100 via the Wi-Fi link. For example, the service processor 115 can be implemented in its entirety on device 100 and the service provider account can be associated exclusively with one device. Similarly, as shown in FIG. 14C, such an implementation can be provided using a different access modem and access network, such as a 2G and/or 3G WWAN, DSL wire line, cable DOCSIS wire line or fiber wire line configuration in place of the 3G and/or 4G access network connection to the intermediate modem or networking device combination 1510. In addition, various other embodiments similarly use DSL as shown in FIG. 14D, USB, Ethernet, Bluetooth, or another LAN or point to point connection from device 100 to the intermediate modem or networking device combination 1510, or a femto cell modem and DSL/cable/T1/other combination as shown in FIG. 14E.

FIG. 15 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a proxy server(s) 270 in accordance with some embodiments. As shown, FIG. 2 includes a proxy server(s) 270 in communication with a 4G/3G/2G wireless network operated by, for example, a central provider. For example, the proxy server(s) 270 can be used to implement and/or assist in providing various techniques described herein, such as service usage measurement and/or other techniques as described herein.

FIG. 16 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a service provider network in accordance with some embodiments, involving one or more of service controllers, service processors and/or proxy server/routers. This diagram illustrates the connections and flows for the provisioning system, new subscriber activation system and order system with the equipment functions having similar or identical embodiments as described in the context of FIG. 4 except in this case one or more proxy server/routers 270 are included for embodiments as described herein.

While the specific illustrations in FIGS. 16 through 18 and associated descriptions refer to a the wireless 3G/4G network, it will be apparent to one of ordinary skill in the art that these techniques can similarly be applied to other types of access networks including, for example, Wi-Fi, cable, DSL, fiber, satellite, and/or other networks. It will also now be apparent to one of ordinary skill in the art that in the context of the proxy server/router embodiments illustrated in FIGS. 16 through 18 that many other embodiments are

possible for the CDR (or FDR, start/stop records, interim accounting records, IPDRs, micro-CDR or service charging bucket accounting) recording, reporting, aggregating, mediating and/or synchronizing usage records within the various network elements and/or device elements.

FIG. 17 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a service provider network in accordance with some embodiments, involving one or more of service controllers, service processors and/or proxy server/routers. This diagram illustrates the connections and flows for the charging system with the equipment functions having similar or identical embodiments as described in the context of FIGS. 4 through 8 except in this case one or more proxy server/routers 270 are included for embodiments as described herein. The charging flows are illustrated by three line types: solid lines for device generated CDRs and proxy server/router generated CDRs, dashed lines for network equipment generated CDRs, and double dot-dash line types for CDRs that have been mediated by a service controller 120 and/or mediation system 118.

FIG. 18 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including a service provider network in accordance with some embodiments, involving one or more of service controllers and/or proxy server/routers. This diagram illustrates the connections and flows for the charging system with the equipment functions having similar or identical embodiments as described in the context of FIG. 17 except in this case there is no service processor CDR feed and only CDR feeds from the network equipment and the proxy server/routers. The charging flows are illustrated by three line types: solid lines for proxy server/router generated CDRs, dashed lines for network equipment generated CDRs, and double dot-dash line types for CDRs that have been mediated by a service controller 120 and/or mediation system 118.

In some embodiments, the proxy server is used for one or more of the functions of: generating, collecting, aggregating, mediating and/or reporting service usage charging information. In some embodiments, the proxy server/router is used for one or more of collecting, aggregating, mediating and/or reporting service usage charging information in which a client device service processor assists in routing traffic to and/or from the proxy server. In some embodiments, the proxy server is used for one or more of the functions of: generating, collecting, aggregating, mediating and/or reporting service usage charging information and the policies controlling the device service processor and the proxy servers are coordinated and distributed by a service controller. In some embodiments, the proxy server/router is used for one or more of: collecting, aggregating, mediating and/or reporting service usage charging information where a client device service processor assists in routing traffic to and/or from the proxy servers and the service processors are coordinated and distributed by a service controller. In some embodiments, in which the proxy server is used in conjunction with a device service processor (e.g., or other client function), the service processor detects one or more types of ambient service activity or other special service activity and routes the traffic to the appropriate proxy server destination for that activity.

In some embodiments, one or more proxy servers or routers are used for one or more of: generating, collecting, aggregating, mediating and/or reporting service usage charging information in which the proxy server communi-

cates with a client device UI for the purpose of informing the user of service status and/or receiving user feedback on service status.

In some embodiments, one or more proxy servers or routers are used for controlling service usage in accordance with a set of policies for specialized services that are programmed by a service design UI (VSP Interface 4910). In some embodiments, one or more proxy servers or routers are used for controlling service usage in accordance with a set of policies for specialized services that are programmed by a network element with a service design UI, and the policies controlling the device service processor and the proxy server are coordinated and distributed by a service controller.

In some embodiments, one or more proxy servers or routers are used for one or more of generating, collecting, aggregating, mediating and/or reporting service usage charging information in which a client device service processor assists in routing traffic to and/or from the proxy server and the policies controlling the device service processor and the proxy server are coordinated and distributed by a service controller.

In some embodiments, the proxy servers can be distributed and managed across diverse geographical separations so that the device traffic to the proxy server/routers does not need to traverse long distances before being connected through the proxy server/routers to the Internet, data, or phone network 120. For example, this geographic diversity or redundancy can be accomplished in a number of ways including using a redundant server or router architecture with a load balancer to determine the best hop to the nearest equipment location, coupled with a distributed data base to persist the device service state so that if a proxy server/router element goes down the device service may be effectively resumed by a different proxy server/router in the same data center or in a geographically diverse data center. In some embodiments, the device service state is persisted in this way by a distributed or redundant diversity data base. In some embodiments, the device service state is held by the service controller network, which in some embodiments, are also distributed and/or redundant. In some embodiments, the device service state is persisted by other network equipment elements. In some embodiments, the device service state is persisted by the device client software (e.g., service processor).

In some embodiments, it may not be possible to accurately identify every network service access attempt or service usage (e.g., or traffic access) as belonging to a given service usage partition (e.g., a given ambient service usage, background network chatter usage, user service plan usage, emergency service usage, and/or other type of service usage). As used herein, the terms service usage partition, service usage recording partition, service charging bucket, and micro-CDRs are used interchangeably. Accordingly, it is desirable to provide a service charging bucket for traffic that is allowed and not definitively identified as belonging to a known service charging bucket. This allows for techniques to employ an "allow but verify" approach to traffic that is likely to be legitimately associated with an ambient service or a user service or a network service that is intended to be allowed, but is not definitively identified as being associated with an allowed service.

As an example, there may be a web site access associated with an ambient service that does not have a reference identifier or other traffic parameter that allows the service processor to associate it with the correct ambient service. In this case, a set of rules can be applied to determine if it is likely that the web site access is a legitimate access given the

35

access control policies that are in place, and if it is the access can be allowed and the traffic usage either recorded in the ambient service charging bucket that it is suspected to be associated with, or the traffic usage can be charged to a network chatter service usage bucket, or the traffic usage can be charged to the user service usage bucket, or the traffic usage may be recorded in a “not classified but allowed” service charging bucket. In some embodiments, in which such traffic is charged to the “not classified but allowed” service usage charging bucket, additional verification measures are employed to ensure that the amount of traffic that is not classified but allowed does not grow too large or become a back-door for service usage errors. For example, the access control policy rules for allowing unclassified traffic can be relatively loose as long as the amount of service usage charges accumulating in the not classified charging bucket remains within certain bounds, and/or the rate of service usage charged to the not classified bucket remains within certain bounds, but if the not classified traffic becomes large or the rate of not classified traffic growth becomes large then the rules governing when to allow not classified traffic can be tightened.

As another example, a browser application can access a web site that is known to be an ambient service website, and that web site might serve back a series of traffic flows, some of which are associated with the ambient service website through URL identifiers that are known to be part of the website, and other traffic can be associated with the ambient service website by virtue of a referring website tag or header, and some traffic can be returned to the same application with a relatively close time proximity to the other traffic as being identified as ambient traffic. In this example, as long as the not classified traffic service charging bucket does not exceed a given pre-set policy limit on its size, and/or does not grow faster than a given pre-set policy rate, and/or is received within a certain pre-set policy period of time difference from the time that other ambient service charging bucket traffic is received, then the not classified traffic is continued to be allowed. However, if the not classified traffic amount or rate of growth exceeds the pre-set policy limits, or if the period of time between when verified ambient service traffic is received and the not classified traffic is received exceeds policy limits, then the not classified traffic can be blocked or other action can be taken to further analyze the not classified traffic.

In some embodiments, it is important to provide a hierarchy of service usage charging rules for the various service usage partitions on a device. As an example, for a given service plan there can be two ambient service charging buckets, a network chatter (e.g., or network overhead) service charging bucket, and a user service plan service charging bucket and it is desirable to make sure that no ambient services or network overhead service or unclassified service is charged to the user service plan, and it is also desirable to ensure that all known ambient service traffic is charged to the appropriate ambient service partner, and it is desirable to ensure that no network overhead service or unclassified service is charged to ambient service partners. In such situations, a service charging bucket hierarchy can be provided as follows: determine if a traffic flow (e.g., or socket) is associated with network overhead, and if so allow it and charge that service bucket, then determine if a traffic flow (or socket) is associated with ambient service #1, and if so allow it and charge that service bucket, then determine if a traffic flow (or socket) is associated with ambient service #2, and if so allow it and charge that service bucket, then determine if a traffic flow (or socket) is associated with not

36

classified traffic, and if so allow it and charge that service bucket, then if the traffic is not associated with any of the above service charging buckets allow it and charge it to the user service plan charging bucket. In another example, if the user has not yet chosen to pay for a user service plan, then the same hierarchical access control and service charging policy can be used except the final step would be: then if the traffic is not associated with any of the above service charging buckets block the traffic. Hierarchical service charging bucket identification such as depicted in these examples can be a crucial aspect of a robust access control policy and/or service charging policy system. Many other access control policy hierarchies and service charging bucket policy hierarchies will now be apparent to one of ordinary skill in the art.

In some embodiments, the not classified traffic is charged according to service charging rules that rely on the most likely candidate service charging bucket for the traffic. As another example, if the not classified traffic is being delivered to the same application as other known ambient service traffic and the time difference between delivery of the known ambient service traffic and the not classified traffic is small, then the not classified traffic can be charged to the ambient service in accordance with a pre-set charging policy rule specifying these conditions. Other embodiments that will now be apparent to one of ordinary skill in the art. For example, another charging rule for not classified traffic could be to perform a pro-rata allocation of the not classified traffic to all of the other service charging buckets with the pro-rata allocation being based on the percentage of the total traffic used by the device for each service charging bucket. As another example, the not classified traffic can be charged to a subset of the service charging buckets for the device (e.g., all ambient services plus the network overhead service) in accordance with the pro-rata share for each service included in the pro-rata split.

In some embodiments, the user service plan agreement is structured so that the user acknowledges that ambient services in which the access connection to the service is sponsored, paid for, and/or partially subsidized by an entity other than the user are a benefit to the user, and/or the user acknowledges that there is no inherent right to free ambient services, and that the service usage accounting system may not always properly characterize usage for a sponsored or subsidized ambient service (e.g., or some other specialized service) in the correct accounting service charging bucket, and, thus, the user service plan account can be charged and/or billed with some of this traffic. By having the user acknowledge a service use agreement of this form then some ambient traffic can be charged to the user service plan account, including, for example, allowed but not classified traffic, excess ambient service usage beyond pre-set policy limits, ambient service usage during busy network periods or on congested network resources, and/or other criteria/measures. In some embodiments, the user might be notified that they are being charged for service activities that are sometimes subsidized or free to the user. As discussed above, it is important to ensure that a not classified service charging bucket does not become a back door for service charging errors or hacking. It will now be apparent to one of ordinary skill in the art that the not classified service usage charges can be verified in a variety of manners, including, for example, observing the size of the not classified service charging bucket as compared to other service usage charges on the device (e.g., total device service usage, ambient service usage, user bucket service usage, and/or other cri-

teria/measures), capping the not classified bucket, and/or capping the rate of growth of the not classified bucket.

In some embodiments, it is important to verify not only that the total device service usage amount is correct, but that the service usage is being reported in the proper service charging buckets. For example, if the service processor software can be hacked so that it correctly reports the total service usage, but reports user service plan traffic under one or more ambient service buckets, then simply verifying that the total amount of service usage is correct will not be sufficient to prevent the device from obtaining free user service that can be charged to ambient service partners. There are a variety of direct and indirect embodiments to accomplish this verification of service charging bucket divisions. For example, in direct verification embodiments, one or more alternative measures of service usage are employed to cross-check the accuracy of the service charging bucket divisions. In indirect embodiments one of two classes of verification are employed: the size and rate of growth for service charging buckets is analyzed and compared to a pre-set group of policies to detect and/or modify service charging bucket growth that is out of policy; and/or the proper operation of the service processor elements involved in service charging bucket partitioning is verified.

Various embodiments involving direct verification of service charging bucket usage and/or accounting include the use of network based service usage measures such as CDRs, IPDRs, flow data records (e.g., FDRs—detailed reports of service usage for each service flow, such as network socket connection, opened and used to transmit data to or from the device), accounting records, interim accounting records or other similar usage records to verify that the device is within service policy and/or the device based service usage reports are accurate. Use of such network generated service usage records to directly verify service charging and/or proper service usage policy adherence are described herein. When network address destination and/or source information is available in these records, as described herein, this can be used in some embodiments, to verify the service charging bucket accounting provided by the device service processor. In some embodiments, some types of service usage records include real-time data but not necessarily all of the useful information needed to help verify service charging bucket accounting, while other types of service usage records provide more detail (e.g., IP address for destination and source) but do not always arrive in real-time. For example, in some embodiments, FDRs are created each time a new service flow (e.g., network socket connection) is opened and then closed. At the time the service flow is closed, a (e.g., possibly time stamped) data usage record indicating source address, destination address and amount of data transmitted is created and sent to a charging aggregation function in the network. The charging aggregation function can then forward the FDRs to the service controller for verification or direct accounting of service charging bucket accounting. By comparing the FDR addresses with known ambient service traffic address associations, the partitioning of service charging buckets between one or more ambient services and other services such as a user service plan service charging bucket may be verified. However, in some cases it can be a long period of time for an FDR to be generated when a device service flow (e.g., socket) remains open for a long period of time, as in the case for example with a long file download, a peer to peer connection with a socket keep alive, or a proxy server service with a socket keep alive. In such cases, it can be disadvantageous to have large amounts of data to be transferred without an FDR to confirm device service pro-

cessor based reports, and in some cases this can provide an opportunity for service processor service reporting hacks. This can be remedied in a variety of ways by using other network reported service usage information to augment the FDR information. For example, start and stop accounting records can sometimes be obtained in some embodiments, from a network element such as a service gateway or the AAA servers (e.g., or other network equipment elements depending on the network architecture). Although start and stop records do not possess the detail of service usage information that FDRs, CDRs, IPDRs, interim accounting records or other service usage records possess, they do inform the service controller that a device is either connected to the network or has stopped connecting. If a device is connected to the network and is not transmitting device usage reports or heartbeats, then the service controller is alerted that an error or hacking condition is likely. As another example of how two or more types of network reported service usage information may be used to create a better real time or near real-time check on device service usage, if both FDRs and start/stop accounting records are available, the service controller can send a stop-then-resume service command to the device (e.g., or alternatively send a stop then resume service command to a network equipment element), which will cause the device to terminate all open service flows before re-initiating them, and once the service flows are stopped then the FDR flow records will be completed and transmitted for any service flows that were in process but unreported when the stop service command was issued. This will cause any long term open socket file transfers to be reported in the FDR flow records thus plugging the potential back door hole in the FDR service usage accounting verification method.

As another example showing how multiple types of network generated service usage accounting records may be used to complement each other and strengthen the verification of service charging bucket accounting partitions, interim data records can be used with FDRs. Interim data records are available in accordance with some embodiments, in which the interim data records are generated on a regularly scheduled basis by a network element (e.g., gateway, base station, HLR, AAA, and/or other network element/function). Interim data records are typically near real time records that report the aggregate traffic usage for the device as of a point in time, but often do not include traffic address information or other traffic details. In embodiments in which both interim accounting records and FDRs are available, when the interim accounting records are indicating service usage that is not being reported in the FDR stream this is evidence that a device has one or more long term socket connections that are open and are not terminating. In this case, the service controller can verify that the device based usage reports are properly accounting for the total amount of service usage reported by the interim accounting records, and/or the service controller can force an FDR report for the open sockets by issuing a stop-resume service command as similarly discussed above.

As described herein, other embodiments involving direct verification of service charging bucket accounting can be provided. One example is to route ambient service traffic to a proxy server or router programmed to support only the network access allowed for the ambient service and to account for the ambient service usage. Additional proxy servers or routers can be similarly programmed for each ambient service that is part of the device service plan, and in some embodiments, another proxy server or router is programmed to support traffic control and account for the user service plan service access. By comparing the service

usage accounting for each of these proxy servers or routers, the device generated service charging bucket accounting can be directly verified. In some embodiments, the usage accounting provided by the proxy servers or routers is used directly for service usage accounting.

In some embodiments, ambient service partner feedback is used to verify service charging bucket accounting. For example, web servers used by ambient service partners to provide ambient services can identify a user device based on header information embedded in the HTML traffic, and then account for either the service used by the device during the ambient service sessions or account for the number of transactions the user completes. If service usage is recorded, then it can be reported to the service controller and be used directly to verify ambient service charging bucket accounting. If transactions are all that are recorded, then this can be reported to the service controller and the amount of ambient service used by the device can be compared with the number of transactions completed to determine if the ambient service usage is reasonable or should be throttled or blocked. It will now be apparent to one of ordinary skill in the art that other embodiments can be provided that employ more than one type of network generated service usage records to verify service usage accounting and/or verify service charging bucket accounting.

Other embodiments involving indirect methods for verifying or controlling service charging bucket accounting include monitoring the size and/or growth rate of ambient service usage. In some embodiments, the access control policy rules call for restricting a given ambient service access when the amount of service usage charges accumulating in the ambient service charging bucket exceed a pre-set policy limit, and/or when the rate of service usage for the ambient service exceeds a pre-set policy limit. For example, once these limits are reached, the ambient service can be throttled back for a period of time, blocked for a period of time, or charged to the user service plan charging bucket. In some embodiments, before these actions are taken the user UI can be used to notify the user of the service policy enforcement action. In some embodiments, indirect verification of service charging bucket accounting includes the various techniques described herein for verifying proper operation of the service processor agent software and/or protecting the service processor agent software from errors, manipulation, or hacking.

In some embodiments, the device service processor directs traffic destined for a given ambient service to a proxy server or router programmed to support that ambient service, and any traffic control policies and/or access control policies for the ambient service are implemented in the proxy server or router. For example, in such embodiments the proxy server or router can be programmed to only allow access to one or more ambient services that are authorized by the device service plan, with the proxy server or router controlling device access so that other network destinations cannot be reached. Continuing this example embodiment, the proxy server or router can account for the ambient service usage in an ambient service charging bucket as discussed elsewhere. In such proxy server or router ambient service control embodiments, the same traffic association techniques described elsewhere that allow incoming traffic associated with an ambient service website or other service to be identified, allowed or blocked, potentially throttled, and accounted for in a service charging bucket can be implemented in the proxy server or router programming. Such proxy server or router embodiments can also implement user service plan service charging buckets, user service plan

traffic controls, and user service plan access control as discussed herein. In some embodiments, the proxy server or router analyzes the HTML traffic content of the traffic flows as described herein to perform such associations, traffic control and/or service usage accounting. Similarly, in some embodiments, a proxy server or router can provide the "surf-out" capabilities described herein by performing the same surf-out traffic associations (e.g., HTML branch reference associations and/or other branch associations) described herein. It will now be apparent to one of ordinary skill in the art that many of the adaptive ambient service control and service usage charging functions described herein for a service processor can be readily implemented with a proxy server or router that is appropriately programmed.

In some embodiments, routing of device traffic for one or more ambient services and/or user service plan services to a proxy server or router is accomplished by the device service processor using the device service processor traffic control embodiments described herein. In some embodiments, routing of device traffic for one or more ambient services and/or user service plan services to a proxy server or router is accomplished by dedicated network equipment such as the gateways (e.g., SGSN, GGSN, PDSN, or PDN), home agents, HLRs or base stations, with the network equipment being provisioned by a service controller (e.g., or other interchangeable network element with similar functions for this purpose) to direct the device traffic to the proxy server or router. In some embodiments, the ambient service traffic or the user service plan traffic is controlled by the proxy server according to a service plan policy set supplied by the service controller (e.g., or equivalent network function for this purpose). The traffic control service policy thus implemented by the proxy server can control traffic based on one or more of the following: period of time, network address, service type, content type, application type, QoS class, time of day, network busy state, bandwidth, and data usage.

In some embodiments, a proxy server or router is used to verify accounting for a given service, for example, an ambient service. In some embodiments, this is accomplished by the device service processor directing the desired service flows to a proxy server or router programmed to handle the desired service flows, with the proxy server or router being programmed to only allow access to valid network destinations allowed by the access control policies for the desired service, and the proxy server or router also being programmed to account for the traffic usage for the desired services. In some embodiments, the proxy service usage accounting may then be used to verify device based service usage accounting reported by the service processor. In some embodiments, the accounting thus reported by the proxy server or router can be used directly to account for service usage, such as ambient service usage or user service plan service usage.

In some embodiments, in which a proxy server is used for device service usage accounting, the proxy server maintains a link to the device service notification UI via a secure communication link, such as the heartbeat device link described herein. For example, the proxy server or router can keep track of device service usage versus service plan usage caps/limits and notify the user device UI through the device communication link (e.g., heartbeat link) between the service controller and the device. In some embodiments, the proxy server/router communicates with a device UI in a variety of ways, such as follows: UI connection through a device link (e.g., heartbeat link), through a device link connected to a service controller (e.g., or other network

element with similar function for this purpose), presenting a proxy web page to the device, providing a pop-up page to the device, and/or installing a special portal mini-browser on the device that communicates with the proxy server/router. In some embodiments, the UI connection to the proxy server/router is used as a user notification channel to communicate

usage notification information, service plan choices, or any of the multiple services UI embodiments described herein. In some embodiments, for the proxy server/router techniques for implementing service traffic/access controls and/or service charging bucket accounting, it is desirable to have the same information that is available to the service processor on the device, including, for example, application associated with the traffic, network busy state, QoS level, or other information about the service activity that is available at the device. For example, such information can be used to help determine traffic control rules and/or special services credit is due (e.g., ambient services credit). In some embodiments, information available on the device can be communicated to the proxy server/router and associated with traffic flows or service usage activities in a variety of ways. For example, side information can be transmitted to the proxy server/router that associates a traffic flow or service activity flow with information available on the device but not readily available in the traffic flow or service activity flow itself. In some embodiments, such side information may be communicated over a dedicated control channel (e.g., the device control link or heartbeat link), or in a standard network connection that in some embodiments, can be secure (e.g., TLS/SSL, or a secure tunnel). In some embodiments, the side information available on the device can be communicated to the proxy server/router via embedded information in data (e.g., header and/or stuffing special fields in the communications packets). In some embodiments, the side information available on the device can be communicated to the proxy server/router by associating a given secure link or tunnel with the side information. In some embodiments, the side information is collected in a device agent or device API agent that monitors traffic flows, collects the side information for those traffic flows, and transmits the information associated with a given flow to a proxy server/router. It will now be apparent to one of ordinary skill in the art that other techniques can be used to communicate side information available on the device to a proxy server/router.

For example, just as the hierarchy of charging rules can be important for implementations in which the service processor is creating the service charging bucket accounting, it can also be important in implementations that use a proxy server or router for service charging bucket accounting. Accordingly, various embodiments described herein for creating a hierarchy of service usage charging rules can be applied to proxy server or proxy router embodiments. It will be apparent to one of ordinary skill in the art that the service charging bucket embodiments and traffic control and access control embodiments described herein for allowed but not classified buckets apply equally to the proxy server/router embodiments. For example, pre-defined service policy rules can be programmed into the proxy server/router to control the traffic flows and/or place usage limits or access limits on an ambient service, or a user service plan service. It will also now be apparent to one of ordinary skill in the art that the embodiments described herein disclosing an initial allowed service access list, temporarily allowing additional service activities until they are determined to be allowed or not allowed, expanding the allowed service activity list, maintaining a not allowed service activity list and expanding the not allowed service activity list also apply equally to proxy

server/router embodiments. Similarly, it will now be apparent to one of ordinary skill in the art that the proxy/server router embodiments can be employed to directly generate the service charging bucket (or micro-CDR) usage reports used to provide further detail and/or billing capabilities for service usage. In some embodiments, in which the device service processor directs traffic to a proxy server/router, there are advantageous design feature embodiments available that can reduce the need to provision network to detect and force specialized device service traffic to the appropriate proxy server/router. For example, this can be done by creating a "usage credit" system for the services supported by the proxy server/router. Total service usage is counted on the one hand by the device service processor, or by other network equipment, or by both. Credit on the other hand for ambient service or other specialized access service usage that is not charged to the user is then provided for services that the device directs through the proxy server/router destination (e.g., URL or route hop) supporting the particular ambient service or other specialized access service. If the device correctly directs traffic to the proxy server/router, then the counting and/or access rules are correctly implemented by the proxy server/router. The service can be thus controlled and/or accounted for. When the service is accounted for, the proxy server/router reports the service charging bucket accounting back to the service controller (e.g., or other network equipment responsible for service charging bucket/micro CDR mediation) and the user service plan service charging bucket account can be credited for the services. Traffic that reaches the proxy server/router is controlled by the access rules and/or traffic control rules and/or QoS control rules of the proxy server/router programming, so there is no question regarding the type of service that is supported with the service charging buckets that are reported to mediation functions (e.g., mediation functions can be performed by one or more of service controller, usage mediation, billing, AAA, and/or HLR/home agent). As the proxy server/router is in the network and can be physically secured and protected from hacking, there is high confidence that the service control and/or charging rules intended for ambient services or some other specialized service are properly implemented and that the proxy server/router connection is being used for the intended service and not some other unintended hacked service. If the device is somehow hacked or otherwise in error so that the traffic is not directed through the appropriate proxy server/router, then the proxy server/router does not log the traffic in micro CDRs/buckets and no specialized service usage credit is sent to the mediation functions, so there is no usage credit deducted from the device user service plan service usage totals. Thus, the user pays for the services when the device is hacked to avoid the proxy server/router. The user account service agreement can specify that if the user tampers with software and traffic is not routed to servers then credit will not be provided and user plan will be charged.

In some proxy server/router embodiments, the usage credit is sometimes recorded by the proxy server/router detecting which device is performing the access. Device identification can be accomplished in a variety of ways including a header/tag inserted into the traffic by the device, a route in the network specified for that device, a secure link (e.g., TLS/SSL, IP Sec, or other secure tunnel), a unique device IP address or other credential (e.g., where proxy server/router has access to an active IP address look up function), a unique proxy server/router address and/or socket for the device.

In some embodiments, the coordination of the device service controller traffic control elements with a proxy server/router can make it simpler to locate, install, provision and operate the proxy servers. The proxy server/routers do not need to be located “in line” with the access network because it is the device’s responsibility to make sure the traffic is routed to the servers/routers or else there is not credit and the user account is charged. In some embodiments, this makes it unnecessary or reduces the need to force device traffic routes in carrier network. In some embodiments, the proxy server/routers can be located in carrier network or on the Internet. If the proxy server/routers are on Internet, then traffic can be authenticated in a firewall before being passed to server/routers to enhance security to attack.

In some embodiments, the service charging bucket recording software in the proxy server/router can be programmed into an ambient service partners network equipment directly thus eliminating the need for special apparatus. The ambient service partner’s equipment (e.g., a web server, load balancer or router) can recognize the device using one of the techniques described above, aggregate the device service charging bucket accounting, and periodically send the usage accounting to the service controller or other network service usage mediation function.

Programming and/or provisioning the types of ambient services, user service plan services and/or specialized services disclosed in various embodiments described herein can be a complex process. In some embodiments, a simplified central network user or MVNO user programming interface, also referred to herein as a Virtual Service Provider Work Station 4910 or interchangeably as a service design interface or VSP Interface, is used to program the necessary policy settings for such services is desirable. For example, a service design interface (VSP Interface 4910) is provided that organizes and/or categorizes the various policy settings that are required to set up an ambient service (e.g., or other service) including one or more of the following: a policy list of service activities that are allowed under the ambient service (e.g., or other service), access control policies, rules for implementing and/or adapting an allowed list of network destinations, rules for implementing and/or adapting a blocked list of network destinations, service charging bucket policies, user notification policies, service control, and/or service charging bucket verification policies, actions to be taken upon verification errors. In some embodiments, the required information for one or more of these policy sets is formatted into a UI that organizes and simplifies the programming of the policies. In some embodiments, the UI is partly graphical to help the user understand the information and what settings need to be defined in order to define the service. In some embodiments, the UI is created with an XML interface. In some embodiments, the UI is offered via a secure web connection. In some embodiments, a basic service policy for an ambient service (e.g., or another service) is created that includes one or more of the above service policy settings, and then this service policy set becomes a list or an object that can be replicated and used in multiple service plan policy set definitions (e.g., “dragged and dropped” in a graphical UI). In some embodiments, the resulting set of policies created in this service design interface (VSP Interface 4910) are then distributed to the necessary policy control elements in the network and/or on the device that act in coordination to implement the service policy set for a given device group. For example, if a service processor is used in conjunction with a service controller,

programmed on the service controller and the device service processor into the service controller, and the service controller loads the service controller policy settings subset into the service controller components that control the policies and loads the device policy settings subset to the devices that belong to that device group. In embodiments in which a proxy server/router is used to help control and account for services, in some embodiments, the service design interface (VSP Interface 4910) loads the service policy settings subsets that need to be programmed on the proxy server/router into the proxy server/router. In embodiments where other network equipment (e.g., gateways, base stations, service usage recording/aggregation/feed equipment, AAA, home agent/HLR, mediation system, and/or billing system) need to be provisioned or programmed, in some embodiments, the service design interface (VSP Interface 4910) also loads the appropriate device group policy subsets to each of the equipment elements. Accordingly, various techniques can be used as described herein to greatly simplify the complex task of translating a service policy set or service plan into all the myriad equipment and/or device settings, programming, and/or provisioning commands required to correctly implement the service. It will now be apparent to one of ordinary skill in the art that several of these techniques can similarly be used for the VSP service design interface (VSP Interface 4910). In some embodiments, a service processor is used in conjunction with one or more proxy servers or routers for one or more of QoS charging, QoS provisioning and/or QoS control. In some embodiments, one or more proxy servers or routers are used in conjunction with a service controller (e.g., or other network function with similar functions for this purpose) for one or more of QoS charging, QoS provisioning, and/or QoS control. In some embodiments, a service processor is used in conjunction with one or more proxy servers or routers and a service controller (e.g., or other network function with similar functions for this purpose) for one or more of QoS charging, QoS provisioning, and/or QoS control. In some embodiments, one or more proxy servers or routers are used in conjunction with a service controller (e.g., or other network function with similar functions for this purpose) for one or more of QoS charging, QoS provisioning, and/or QoS control, in which the proxy server or router policies for this QoS service implementation are programmed by a service design UI (VSP Interface 4910).

FIG. 19 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments. The description of the network equipment element functions is generally identical to the embodiments depicted in other figures, except that one or more service controllers 120 and/or proxy servers/routers 270 and/or service design interfaces (VSP Interface 4910) are shared between the two networks as described herein. For example, the network equipment, charging record formats, provisioning systems can be similar in the two networks, or may be completely different since in the various embodiments the service controller 120, service processor 115, and/or proxy sever router 270 are used to provide unified roaming services, or in some embodiments, to provide common network service features across the different networks.

FIG. 20 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments, involving one or more of service controllers and/or service processors. FIG. 20 is similar to FIG. 19 except that FIG. 20 illustrates

that various types of access network technology and equipment can be used on any number of the central provider networks (e.g., 2G/3G/4G cellular wireless plus WiFi is the example in FIG. 19 while 2G/3G/4G cellular wireless plus DSL and cable is the example in FIG. 20). FIGS. 19 and 20 do not show the access network connections to the central provider #2 core network, but that the second central provider network can have all of or some of the access equipment elements that the first central provider network possesses, or the second central provider network can possess different access network technology and equipment as described herein would be apparent to one of ordinary skill in the art. For example, if any aspect of the 2G/3G/4G technology is different for the two networks, then multi-mode wireless modules can be used in the device modems to allow for access connection using one technology on the first central provider network, and access connection using a second technology on the second central provider network. As many of the techniques and embodiments described herein allow for network service policy implementation at layers above the modem physical layer, modem MAC layer and the access network access control and authorization layers, then any number of multi-mode modem technologies can be employed to bridge connect to either of the two networks while enhancing roaming services or providing unified network services in one or more of the areas of service traffic control, user notification interfaces, charging policies and/or systems, QoS services, instant activation services and/or billing services. It will be apparent to one of ordinary skill in the art that while the discussion herein is for service provider (e.g., central provider, MVNO, VSP, etc) networks, many of the embodiments can similarly be applied to private networks such as, for example, enterprise networks, enterprise WAN solutions and/or remote office solutions, government networks, emergency networks, and/or networks involving intermediate networking devices.

FIG. 21 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments. For example, FIG. 21 includes elements that can be identical or very similar to those described in the context of FIGS. 5 through 8, except that in FIG. 21 there are two central provider networks in which the CDR reporting and service controls are unified through one or more common service controllers 120 and/or service processors 115. As shown in FIG. 21, the charging flows are illustrated by three line types: solid lines for device generated CDRs, dashed lines for network equipment generated CDRs, and double dot-dash line types for CDRs that have been mediated by a service controller 120 and/or mediation system 118.

In the context of the multiple service provider network embodiments illustrated in FIGS. 19 through 23, many of the device assisted service techniques disclosed herein that involve a service processor and/or a service controller, including, for example, service usage charging, QoS enabled service usage charging, service usage notification, service usage control, and service usage user feedback UI that apply to the case of improving roaming services or unifying different service provider networks with a common service may be employed. In addition, in some embodiments, the application of verifiable device assisted services (e.g., verifiable service processor service policy adherence) can be accomplished both on the first service provider network and the second service provider network by virtue of the fact that both networks are supplying network generated service

usage information to the one or more service controllers and/or proxy servers or routers.

Accordingly, this allows a number of embodiments for one or not of the service providers, including, for example: common service usage and/or recording platform that can provide unified service usage reporting and/or billing across multiple networks, common network service controls and/or traffic controls across multiple networks, common user information sharing across multiple networks, and common QoS controls across multiple networks, common service design services across multiple networks. It is also possible to unify networks with different access technologies into a common network service. For example, by creating different network interface “shim” software layers, it is possible to interface a common set of service controller capabilities, service processor capabilities, proxy server and/or router capabilities and/or service design interface capabilities into different networks that have different equipment and even different communication protocols for provisioning equipment, CDR collection and aggregation equipment, traffic control equipment, AAA, home agent/HLR, billing systems, QoS reservation and control equipment, and/or other capabilities/protocols. The equipment software shims translate the different equipment communication protocols for each network into the common protocols supported in the service controller, service processor, proxy server and/or router, and/or service design center apparatus.

In some embodiments, the techniques described herein to unify network services across multiple networks can be used to provide consistent service control, notification, billing across networks locally or globally without the need to back-haul the data traffic through wholesale pipes from one network to another as is the case with conventional network technologies. For example, a subscriber can be acquired and/or billed for by network #1 and then be serviced with a uniform service experience and deep service notification, control, charging capabilities on both network #1 and network #2. In some embodiments, a subscriber can be acquired and billed for by network #2 and then be serviced with a uniform service experience and deep service notification, control, charging capabilities on both network #1 and network #2. Service plans, settings, notification, carrier branding, etc. can all be the same or different for subscribers originated on one network versus the other. Additionally, in some embodiments, one or more of the service controller, proxy servers or routers and/or other apparatus that handles traffic including the mediation systems can be distributed and/or redundant with diversity as discussed elsewhere to reduce the need to back haul device service traffic or service control traffic. In some embodiments, it is possible for both carrier networks to possess a common service design network element and UI despite the fact that the different networks can have very different provisioning, AAA, traffic handling, charging, billing, etc equipment.

FIG. 22 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments, involving one or more of service controllers, service processors and/or proxy server/routers. FIG. 22 includes elements that can be identical or very similar to those described in the context of FIGS. 5 through 8, 16 through 18, and 21 except that in FIG. 21 there are two central provider networks where the CDR reporting and service controls are unified through one or more common service controllers 120, service processors 115 and/or proxy server/routers 270. As shown in FIG. 22: the charging flows are illustrated using the following three

47

line types: solid lines for device generated CDRs and/or proxy server/router generated CDRs, dashed lines for network equipment generated CDRs, and double dot-dash line types for CDRs that have been mediated by a service controller 120 and/or mediation system 118.

FIG. 23 illustrates a wireless network architecture for providing device assisted CDR creation, aggregation, mediation and billing including two service provider networks in accordance with some embodiments, involving one or more of service controllers and/or proxy server/routers. FIG. 23 includes elements that may be identical or very similar to those described in the context of FIG. 22, except in this case there is no service processor CDR feed and only CDR feeds from the network equipment and the proxy server/routers. As shown in FIG. 23, the charging flows are illustrated using the following three line types: solid lines for device generated CDRs and/or proxy server/router generated CDRs, dashed lines for network equipment generated CDRs, and double dot-dash line types for CDRs that have been mediated by a service controller 120 and/or mediation system 118.

In some embodiments, one or more proxy servers or routers are used to create a common network service between two or more different networks. In some embodiments, one or more proxy servers or routers are used in conjunction with device client (DAS service processor) used to create a common network service between two or more different networks. In some embodiments, one or more proxy servers or routers are used to create a common network service between two or more different networks in which the proxy server is programmed with a centralized service design interface. In some embodiments, one or more proxy servers or routers are used to create a common service usage recording and/or accounting system across different networks. In some embodiments, one or more proxy servers or routers are used in conjunction with device client (e.g., DAS service processor) used to create a common network service between two or more different networks, in which the proxy server and the device client have coordinated policy settings and/or are both controlled by a service controller element in a network.

In some embodiments, one or more proxy servers or routers are used in conjunction with a device client (e.g., DAS service processor) to create a common network service between two or more different networks in which the proxy router and the device client have coordinated policy settings and/or are both controlled by a service controller element in a network.

In some embodiments, one or more proxy servers or routers are used to create a common network service between two or more different networks in which the proxy server is programmed with a centralized service design interface and the is controlled by a service controller element in a network.

In some embodiments, one or more proxy servers or routers are used to create a common network service between two or more different networks in which the proxy server is programmed with a centralized service design interface and one or more of the service policies are is controlled by a service controller element in a network. In some embodiments, one or more proxy servers or routers are used to create a coordinated QoS signaling, reservation, provisioning, control, charging, and/or user interface across different service provider networks.

Those of ordinary skill in the art will appreciate that various other rules can be provided for the rules engine as described herein. Those of ordinary skill in the art will also

48

appreciate that the functions described herein can be implemented using various other network architectures and network implementations (e.g., using various other networking protocols and corresponding network equipment and techniques).

Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There are many alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

INCORPORATION BY REFERENCE

The following U.S. patent applications are incorporated herein by reference for all purposes: application Ser. No. 12/695,980, entitled ENHANCED ROAMING SERVICES AND CONVERGED CARRIER NETWORKS WITH DEVICE ASSISTED SERVICES AND A PROXY, filed Jan. 28, 2010; application Ser. No. 12/380,778, entitled VERIFIABLE DEVICE ASSISTED SERVICE USAGE BILLING WITH INTEGRATED ACCOUNTING, MEDIATION ACCOUNTING, AND MULTI-ACCOUNT, filed on Mar. 2, 2009; application Ser. No. 12/380,780, entitled AUTOMATED DEVICE PROVISIONING AND ACTIVATION, filed Mar. 2, 2009; application Ser. No. 12/695,019, entitled DEVICE ASSISTED CDR CREATION, AGGREGATION, MEDIATION AND BILLING, filed on Jan. 27, 2010; application Ser. No. 12/695,021, entitled QUALITY OF SERVICE FOR DEVICE ASSISTED SERVICES, filed on Jan. 27, 2010; provisional Patent Application No. 61/206,354, entitled SERVICES POLICY COMMUNICATION SYSTEM AND METHOD, filed Jan. 28, 2009; provisional Patent Application No. 61/206,944, entitled SERVICES POLICY COMMUNICATION SYSTEM AND METHOD, filed Feb. 4, 2009; provisional Application No. 61/207,393, entitled SERVICES POLICY COMMUNICATION SYSTEM AND METHOD, filed Feb. 10, 2009; provisional Patent Application No. 61/207,739, entitled SERVICES POLICY COMMUNICATION SYSTEM AND METHOD, filed on Feb. 13, 2009; provisional Patent Application No. 61/270,353, entitled DEVICE ASSISTED CDR CREATION, AGGREGATION, MEDIATION AND BILLING, filed on Jul. 6, 2009; provisional Application No. 61/264,126, entitled DEVICE ASSISTED SERVICES ACTIVITY MAP, filed Nov. 24, 2009; provisional Application No. 61/252,151, entitled SECURITY TECHNIQUES FOR DEVICE ASSISTED SERVICES, filed Oct. 15, 2009; and provisional Application No. 61/252,153, entitled DEVICE GROUP PARTITIONS AND SETTLEMENT PLATFORM, filed Oct. 15, 2009.

What is claimed is:

1. A method performed by a wireless device communicatively coupled to a network system by a wireless access network, the network system including a network element corresponding to a network element destination, the method comprising:

receiving, from the network system, a device policy comprising a service usage activity classification and information identifying the network element destination;

detecting, using the device policy, wireless device traffic associated with a plurality of service activities;

identifying, using the device policy, one or more service activities of the plurality of service activities as being included in the service usage activity classification, the service usage activity classification comprising side

49

- information associated with the one or more identified service activities and collected on the one or more identified service activities; and
 directing or routing, using the device policy, the side information associated with the one or more identified service activities over the wireless access network based on the information identifying the network element destination. 5
2. The method of claim 1, wherein the plurality of service activities comprise one or more transactions.
3. The method of claim 1, wherein the side information comprises a service usage history. 10
4. The method of claim 1, wherein the side information comprises an identifier for a usage activity tag.
5. The method of claim 1, wherein identifying the one or more service usage activities comprises compartmentalizing usage according to pre-assigned device activity transaction codes. 15
6. The method of claim 1, wherein the network element comprises a service controller.
7. The method of claim 1, wherein the side information comprises device-assisted service measures for the one or more identified service activities. 20
8. The method of claim 6, wherein the service controller is reachable through the wireless access network through an Internet connection.
9. The method of claim 1, wherein the side information comprises service policy device information directed to an access control integrity server. 25
10. The method of claim 1, wherein the side information comprises service usage device information directed to an access control integrity server. 30
11. A wireless device configured to be communicatively coupled to a network system by a wireless access network, the network system including a network element corresponding to a network element destination, the wireless device comprising: 35
- a processor configured to:
 - receive, from the network system, a device policy comprising a service usage activity classification and information identifying the network element destination;

50

- detect, using the device policy, wireless device traffic associated with a plurality of service activities;
 - identify, using the device policy, one or more service activities of the plurality of service activities as being included in the service usage activity classification, the service usage activity classification comprising side information associated with the one or more identified service activities and collected on the one or more identified service activities; and
 - direct or rout, using the device policy, the side information associated with the one or more identified service activities over the wireless access network based on the information identifying the network element destination.
12. The wireless device of claim 11, wherein the plurality of service activities comprise one or more transactions.
13. The wireless device of claim 11, wherein the side information comprises a service usage history.
14. The wireless device of claim 11, wherein the side information comprises an identifier for a usage activity tag.
15. The wireless device of claim 11, wherein identifying the one or more service usage activities comprises compartmentalizing usage according to pre-assigned device activity transaction codes. 25
16. The wireless device of claim 11, wherein the network element comprises a service controller.
17. The wireless device of claim 11, wherein the side information comprises device-assisted service measures for the one or more identified service activities. 30
18. The wireless device of claim 16, wherein the service controller is reachable through the wireless access network through an Internet connection.
19. The wireless device of claim 11, wherein the side information comprises service policy device information directed to an access control integrity server.
20. The wireless device of claim 11, wherein the side information comprises service usage device information directed to an access control integrity server.

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